

Notes

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Part I

Analysis in One Variables

Chapter 1 Continuous Function on $\mathbb{R}$, Limits of Function

1.1 Sequence

Topology of $\mathbb{N}$ given by $(\mathbb{N}, <)$.

1.1.1 Cluster Points

Definition 1.1.1.1 Let $(a_n) \in \text{Func}(\mathbb{N}, \mathbb{R})$ or $\text{Func}(\mathbb{N}, \mathbb{C})$. z is a **cluster point** of (a_n) if

$$\forall \epsilon > 0 \forall N_0 \exists n > N_0 \wedge a_n \in B_\epsilon(z)$$

Equivalently, for all $B_\epsilon(z) \cap \{a_n | n \geq N_0\} \neq \emptyset, \forall N_0$.

1.1.2 Convergence

(Definition in general in topology)

Definition 1.1.2.1 Given $(a_n) \in \text{Func}(\mathbb{N}, \mathbb{R})$ or $\text{Func}(\mathbb{N}, \mathbb{C})$. Then

$$a_n \rightarrow z := (\forall \epsilon \exists N_0 (A_{N_0} \subseteq B_\epsilon(z)))$$

where $A_{N_0} := \{a_n | n \geq N_0\}$.

Equivalently, $\forall \epsilon \exists N_0 (n \geq N_0 \rightarrow |a_n - z| < \epsilon)$. Take the notion of

$$\lim_{n \rightarrow \infty} a_n = z$$

(In this section) $P((a_n), A) := \forall \epsilon \exists N_0 (n > N_0 \rightarrow a_n \in B_\epsilon(A))$

◇ **Lemma 1.1.2.2** (Uniqueness of Convergence in $\mathbb{R}, \mathbb{C}$) If $P((a_n), A), P(a_n, B)$, then $A = B$.

◦ **Proof** Via Hausdorffness. □

Operations

♠ **Proposition 1.1.2.3** (Operations limits of $\mathbb{R}$ or $\mathbb{C}$) Suppose the following limits exists:

$$\lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} (a_n \cdot b_n) = (\lim_{n \rightarrow \infty} a_n) \cdot (\lim_{n \rightarrow \infty} b_n)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}, \lim_{n \rightarrow \infty} b_n \neq 0$$

◦ **Proof**

- Suppose $A = \lim_{n \rightarrow \infty} a_n, B = \lim_{n \rightarrow \infty} b_n$. Now selection $N = \max\{N_1, N_2\}$, $|a_n - A| < \epsilon/2, |b_n - B| < \epsilon/2$, then $|(a_n + b_n) - (A + B)| < \epsilon$. With uniqueness.

- Similarly.
-

□

◇ **Lemma 1.1.2.4** (Majorance of Sequence) Suppose $(a_n), (b_n), (c_n)$ be sequence. Then, we have

$$a_n \leq b_n \leq c_n, n \geq N_0 \Rightarrow \lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} b_n \leq \lim_{n \rightarrow \infty} c_n$$

◦ **Proof** It is easy to verify: Suppose $a_n \leq b_n, n \geq N_0$. If $\lim_{n \rightarrow \infty} a_n > \lim_{n \rightarrow \infty} b_n$, then

$$b_n < B + \epsilon < A - \epsilon < a_n, \epsilon = (B - A)/2$$

Contradiction.

□

◇ **Lemma 1.1.2.5** (The Sequence with Limit is Bounded) Suppose $\lim_{n \rightarrow \infty} a_n = A$, then $\{a_n\}$ is bounded.

◦ **Proof** Omitted.

□

Average

♠ **Proposition 1.1.2.6** Suppose $\lim_{n \rightarrow \infty} a_n = A$, then

$$\lim_{n \rightarrow \infty} \frac{a_0 + \cdots + a_n}{n+1} = A$$

◦ **Proof** Suppose $n > N_1, |a_n - A| < \epsilon$. Now, select some N_2 such that

$$\frac{(a_0 - A) + \cdots + (a_{N_1} - A)}{n+1} < \epsilon, n > N_2$$

so, $N = \max\{N_1, N_2\}$,

$$\left| \frac{(a_0 - A) + \cdots + (a_N - A)}{N+1} \right| \leq \left| \frac{a_0 + \cdots + a_{N_1}}{N+1} \right| + \left| \frac{a_{N_1+1} + \cdots}{N+1} \right| < 2\epsilon$$

□

1.1.3 Monotone Sequence

Definition 1.1.3.1 Let (a_n) be a sequence

- (a_n) is **strictly increasing** if $a_{n+1} > a_n$;
- (a_n) is **increasing** if $a_{n+1} \geq a_n$.

Similarly for decreasing.

♠ **Proposition 1.1.3.2** If (a_n) is increasing and has a upper bound, then

$$\lim_{n \rightarrow \infty} a_n = \sup\{a_n | n \in \mathbb{N}\} \in \mathbb{R}$$

◦ **Proof** Firstly the limit exists: Because of completeness of $\mathbb{R}$, $\sup\{a_n | n \in \mathbb{N}\} \in \mathbb{R}$. Then omitted. $\square$

1.1.4 Subsequence

Definition 1.1.4.1

- We firstly define the **strictly increasing function** of index

$$\sigma : \mathbb{N} \rightarrow \mathbb{N}$$

such that: $\sigma(n) < \sigma(n+1), n \in \mathbb{N}$.

- Let $(a_n) \in \text{Func}(\mathbb{N}, S)$ a subsequence is $(a_{\sigma(n)})$.

► **Theorem** (Bolzano-Weistrasss) Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

◦ **Proof** Furthermore, this theorem has a equivalent explanation: $\mathbb{R}$ is secondly countable, and there exists at least one cluster point.

Such point could be given by

$$\sup\{a_n | n \in \mathbb{N}\}$$

And the elements by countable basis. $\square$

Furthermore, we could extend ∞ to the topology by compactification, so bounded condition could be removed.

Definition 1.1.4.2 Define:

- $\overline{\lim} a_n := \lim_{N \rightarrow \infty} \sup_{n \leq N} a_n$;
- $\underline{\lim} a_n := \lim_{N \rightarrow \infty} \inf_{n \in \mathbb{N}} a_n$;

♠ **Proposition 1.1.4.3** We have

$$\lim_{n \rightarrow \infty} \sup(a_n) \geq \lim_{n \rightarrow \infty} \inf(a_n)$$

◦ **Proof** Omitted. $\square$

1.1.5 Cauchy Sequence

Definition 1.1.5.1 A sequence (a_n) is **Cauchy sequence** if

$$\forall \epsilon \exists N (N \in \mathbb{N} \wedge \forall n \forall m (n > N \wedge m \geq 0 \rightarrow |a_{m+n} - a_n| < \epsilon))$$

♠ **Proposition 1.1.5.2**

Cauchy Sequence $\Leftrightarrow$ Convergent.

◦ **Proof**

- $\Rightarrow$ If the sequence is Cauchy, then take monotone $\overline{\lim} a_n$ and $\underline{\lim} a_n$. We have
- $\Leftarrow |a_{n+m} - a_n| \leq |a_{n+m} - A| + |a_n - A| \leq \epsilon$.

□

1.1.6 Series

Definition 1.1.6.1 A **series** is the limit of a formal sum

$$S_n := \sum_{k=0}^n a_k \quad S := \lim_{n \rightarrow \infty} S_n$$

Convergence

Definition 1.1.6.2 Series $\sum_{k=0}^{\infty} a_k$ is **convergent** if the limit exists.

♠ **Proposition 1.1.6.3** For convergent series $\sum_{k=0}^{\infty} a_k, \sum_{k=0}^{\infty} b_k$, we have

$$\begin{aligned} \sum_{k=0}^{\infty} (a_k + b_k) &= \sum_{k=0}^{\infty} a_k + \sum_{k=0}^{\infty} b_k \\ \sum_{k=0}^{\infty} (ca_k) &= c \sum_{k=0}^{\infty} a_k \end{aligned}$$

◦ **Proof** Omitted.

□

♠ **Proposition 1.1.6.4** (Cauchy's Criterion) $\sum_{k=0}^{\infty} a_k$ is convergent iff

$$\forall \epsilon > 0 \exists N (\forall n \geq N \forall p > 0 (|\sum_{k=n}^{n+p} a_k| < \epsilon))$$

◦ **Proof** Omitted.

□

Positive Series

Definition 1.1.6.5 Series $\sum_{k=0}^{\infty} a_k$ is **positive** iff $a_k \geq 0$.

♠ **Proposition 1.1.6.6** Positive series $\sum_{k=0}^{\infty} a_k$ is convergent $\Leftrightarrow \sum_{k=0}^{\infty} a_k \leq M$ for some M .

◦ **Proof** Omitted. □

♠ **Proposition 1.1.6.7** If $a_k \geq b_k \geq 0$, then

$$\sum_{k=0}^{\infty} a_k \text{ is convergent} \Rightarrow \sum_{k=0}^{\infty} b_k \text{ is convergent}$$

◦ **Proof** Omitted. □

Absolute Convergence

Definition 1.1.6.8 $\sum_{k=0}^{\infty} a_k$ is **absolutely convergent** if $\sum_{k=0}^{\infty} |a_k|$ is convergent.

D'Alembert's Criterion

◇ **Lemma 1.1.6.9** (D'Alembert) Suppose $\sum_{k=0}^{\infty} a_k$ is positive series. If there exists some $N \in \mathbb{N}$,

- $r < 1$ such that for all $n \geq N$, $a_n > 0$

$$\frac{a_{n+1}}{a_n} \leq r$$

Then, $\sum_{k=0}^{\infty} a_k$ is convergent.

- such that for all $n \geq N$,

$$\frac{a_{n+1}}{a_n} \geq 1$$

Then, $\sum_{k=0}^{\infty} a_k$ is divergent.

◦ **Proof**

-
-

□

Remark

Cauchy's Root Criterion

◇ **Lemma 1.1.6.10** (Cauchy) Suppose $\sum_{k=0}^{\infty} a_k$ is positive series. If there exists some N ,

- and $r < 1$ such that for all $n \geq N$

$$\sqrt[n]{a_n} < r$$

- such that $n \geq N$

$$\sqrt[n]{a_n} \geq 1$$

Then, $\sum_{k=0}^{\infty} a_k$ is divergent.

◦ **Proof**

-
-

□

Cauchy's

Kummer/Raabe/Bertrand's Criterion

Abel's Criterion

Dirichlet's Criterion

Lebnitz's Criterion

1.1.7 Power Series

Definition 1.1.7.1 A **power series** is a formal sum of

$$f(z) := \sum_{k=0}^{\infty} a_k z^k$$

► **Theorem** (Cauchy-Hadamard) For $\sum_{k=0}^{\infty} a_n x^n$, define

$$\rho := \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{a_n}}$$

Then, the series is absolutely convergent in $|z| < \rho$, divergent in $|z| > \rho$.

◦ **Proof ...**

□

Exponential

Define:

$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

$$\exp(x) := \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Geometric Sequence

Definition 1.1.7.2 Geometric Sequence

$$G(z) = \sum_{i=0}^{\infty} z^i = \lim_{n \rightarrow \infty} G_n(z) \quad G_n(z) = \sum_{i=0}^n z^i$$

♠ **Proposition 1.1.7.3**

-
- $G(z) = \frac{1}{1-z}, |z| < 1$

◦ **Proof** With $G(z) = \lim_{n \rightarrow \infty} \frac{1-z^{n+1}}{1-z} = \frac{1}{1-z}$. □

Newton's Method**1.1.8 Riemann's Sequence Theorem**▶ **Theorem** (Riemann)

◦ **Proof** ... □

1.1.9 Fubini's Theorem, Tonelli's Theorem▶ **Theorem** (Fubini)

◦ **Proof** ... □

▶ **Theorem** (Tonelli)

◦ **Proof** ... □

1.1.10 L'Hospital's Law**1.1.11 Infinite Product****1.2 Limit of Functions****1.3 Continuous Functions on $\mathbb{R}$**

Definition 1.3.0.1 $f : U \rightarrow \mathbb{R}$ is **continuous** at $a \in U$ if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

1.3.1 Intermediate Value Theorem

◇ **Lemma 1.3.1.1** If $f \in C([a, b], \mathbb{R})$ with $f(c) > 0$, $c \in [a, b]$. Then, there exists $\delta > 0$ such that

$$\forall x \in (c - \delta, c + \delta) \cap [a, b] (f(x) > 0)$$

◦ **Proof ...**

□

► **Theorem** (Intermediate Value Theorem) Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous with

$$f(a) > c$$

◦ **Proof ...**

□

1.3.2 Extreme Value Theorem

Lesbegue Value

1.3.3 title

Chapter 2 Derivative

2.1 Derivatives

Definition 2.1.0.1 Define:

Chapter 3 Integration

3.1 Antiderivative

Definition 3.1.0.1 (Indefinite Function) Suppose

$$f : [a, b] \rightarrow \mathbb{R}$$

Then, $F : [a, b] \rightarrow \mathbb{R}$ is an antiderivative of f if

- F is continuous;
- $F'(x) = f(x), x \in (a, b)$;

◇ **Lemma 3.1.0.2** Suppose F, G be the antiderivative, then we have

$$F - G = C$$

◦ **Proof** Notice that

$$\frac{d}{dx}(F - G) = F' - G' = 0 \Rightarrow F - G = C$$

□

◇ **Lemma 3.1.0.3** (Linearity) Suppose F_1, F_2 be antiderivative of f_1, f_2 , then

$$(\alpha F_1 + \beta F_2 + c)' = \alpha f_1 + \beta f_2$$

◦ **Proof** Omitted

□

◇ **Lemma 3.1.0.4** (Chain's Rule) Suppose $g : D \rightarrow \mathbb{R}, h : I \rightarrow D$ with antiderivative G of g . Then, suppose

$$f = (g \circ h)h' : I \rightarrow \mathbb{R}$$

We have

$$F = G \circ h$$

◦ **Proof** Omitted.

□

3.1.1 Partial Integration

♠ **Proposition 3.1.1.1** ...

◦ **Proof** ...

□

★ Corollary 3.1.1.2 ...

◦ Proof ...

□

3.2 Riemannian Integration

3.2.1 title

3.3 Theorems of Integration in One Variable

3.3.1 Newton-Lebnitz Theorem

► Theorem ...

◦ Proof ...

□

3.3.2 Integrations Intermediate Value Theorem

3.4 Indefinite Integration

3.5 Summe and Integration

◇ Lemma 3.5.0.1 We have: For $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$

$$\sum_{i=1}^n f(n) \text{ is convergent} \Leftrightarrow \int_1^\infty f(x) dx \text{ is convergent}$$

◦ Proof ...

□

Chapter 4 Disccusion of Series and Infinite Product, in the Category of Analysis in One Variable

Chapter 5 ODE

5.1 title

Part II

Analysis in Multivariables, and the Differential Manifold

Part III

Topological Manifolds

Part IV

Differential Topology

Part V

Geometric Topology

Part VI

Smooth Manifold

Chapter 6 Smooth Manifold

6.1 Definitions

Definition 6.1.0.1 An n -dimensional **smooth manifold** is a topological manifold $(M, \{(U_i, \varphi_i)\})$ with **smooth transition function**

$$\varphi_i \circ \varphi_j^{-1} \in C_{diff}^\infty(\hat{U}_j, \hat{U}_i)$$

And for every smooth charts, there exists a **maximal chart** which contains the smooth charts. Such atlas is called the **smooth structure** of M .

◇ **Lemma 6.1.0.2** Let M be a topological manifold:

- Every smooth atlas is contained in a maximal smooth atlas;
- Two smooth atlases for M determines the same maximal smooth atlas iff their union is smooth.

◦ **Proof**

- Let $\mathcal{A}$ be the collection of charts that are smooth compatible with every chart in $\bar{\mathcal{A}}$.
 $\bar{\mathcal{A}}$ is a smooth chart. Firstly, $\mathcal{A} \subseteq \bar{\mathcal{A}}$. For two charts $(U, \varphi), (V, \psi) \in \bar{\mathcal{A}}$ where $U \cap V \neq \emptyset$, select some $p \in U \cap V$ and a chart (W, θ) where $p \in U \cap V \cap W$. Omitted.
- Omitted.

□

► **Theorem** (Milnor, 1956) S^7 has more than one the smooth structures.

► **Theorem** (Exotic $\mathbb{R}^4$) $\mathbb{R}^4$ has infinite many of smooth structures.

6.1.1 Smooth Functions

Definition 6.1.1.1 A function $f : M \rightarrow \mathbb{R}$ is **smooth function** if for every coordinate (U_i, φ_i)

$$f \circ \varphi_i^{-1} : \hat{U}_i \rightarrow \mathbb{R}$$

is smooth.

6.1.2 Construction of Smooth Manifold

◇ **Lemma 6.1.2.1** (Smooth Manifold Construction Lemma) Let M be a set, and suppose we are given a collection $\{U_\alpha\}$ of subse of M , together with injective maps $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$. With following properties:

- For each α , $\varphi_\alpha(U_\alpha)$ is an open subset of $\mathbb{R}^n$;
- $\varphi_\alpha(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$;
- Whenever $U_\alpha \cap U_\beta \neq \emptyset$, $\varphi_\alpha \circ \varphi_\beta^{-1}$ is a diffeomorphism;
- Countably many U_α cover M ;

- Hausdorffness;

◦ **Proof** Omitted. □

6.2 Smooth Functions and Smooth Maps

6.2.1 Definitions

Definition 6.2.1.1

- A function $f : M \rightarrow \mathbb{R}^k$ is **smooth** if for every $p \in M$, there exists a smooth chart (U, φ) and the composition $f \circ \varphi^{-1} : \hat{U} \rightarrow \mathbb{R}^k$ is smooth;
Such $f \circ \varphi^{-1}$ is called the **coordinate representation** of f ;
- Furthermore, $F : M \rightarrow N$ is smooth if for some charts $(U, \varphi), (V, \psi)$,

$$\varphi \circ F \circ \psi^{-1} : \psi(U \cap V) \rightarrow \varphi(U \cap V)$$

is smooth

It is easy to verify:

- $C^\infty(M)$ is commutative $\mathbb{R}$ -algebra;
- For every charts, the representation map is smooth;

◇ **Lemma 6.2.1.2** (Smoothness is Local) Let M, N be two smooth manifolds, $F : M \rightarrow N$ be a map. If every point $p \in M$, restriction $F|_U$ is smooth, then F is smooth; Conversely also holds.

◦ **Proof** Omitted; □

◇ **Lemma 6.2.1.3** (Pasting) If given $F_\alpha : U_\alpha \rightarrow N$ with $F_\alpha|_{U_\alpha \cap U_\beta} = F_\beta|_{U_\alpha \cap U_\beta}, \cup U_\alpha = M$. Then there exists $F : M \rightarrow N$ such that

$$F|_{U_\alpha} = F_\alpha$$

◦ **Proof** ... □

◇ **Lemma 6.2.1.4** Every smooth map between smooth manifolds is continuous.

◦ **Proof** Omitted. □

6.2.2 Diffeomorphisms

Definition 6.2.2.1 ...

6.3 Smooth Covering Maps

6.4 Proper Maps

Definition 6.4.0.1 $F : M \rightarrow N$ is **proper** if the inverse image of any compact $K \subseteq N$ is compact.

◇ **Lemma 6.4.0.2** Suppose M is a compact space and N is a Hausdorff space, then every continuous map $F : M \rightarrow N$ is proper.

◦ **Proof ...**

□

6.5 Partition Unity

Chapter 7 Tangent Spaces

7.1 Tangent Vectors

7.1.1 Derivative on Opens

Definition 7.1.1.1 ...

7.1.2 Formel Derivations

7.1.3 Tangent Vectors on a Manifold

♣ **Example 7.1.3.1** (Tangent Space of Vector Space) Let $V \cong \mathbb{R}^n$ be a vector space. We have

$$T_w V = V; w \in V$$

7.1.4 Pushforwards

7.2 Coordinate Computations

7.3 Tangent Vectors to Curves, Fibres

7.3.1 Curves

7.3.2 Fibres

Chapter 8 Vector Fields

8.1 Tangent Bundles

Definition 8.1.0.1 Define: For manifold M , the **tangent space**

$$TM = \coprod_{p \in M} T_p M$$

Then:

♠ **Proposition 8.1.0.2** (Smooth Structure of TM , Smoothness of Bundle Map)

◦ **Proof ...**

□

8.1.1 Functoriality

♠ **Proposition 8.1.1.1** $T : Man^1 \rightarrow VBun$ is a functor.

◦ **Proof**

- Firstly, T is well-defined;
- Secondly, $T(g \circ f) = T(g) \circ T(f)$. $f : M \rightarrow N, g : N \rightarrow L$,

$$\begin{aligned} T_p(g \circ f)(X)(h) &= X(h \circ (g \circ f)) \\ &= T_p(f)(X)(h \circ g) \\ &= T_p(g) \circ T_p(f)(X) \end{aligned}$$

where $X \in T_p M$.

□

Especially

♠ **Proposition 8.1.1.2** If $f : M \rightarrow N$ is smooth map, so is $T(f)$.

◦ **Proof** Consider followin diagram

$$\begin{array}{ccc} TM & \xrightarrow{T(f)} & TN \\ \downarrow & & \downarrow \\ M & \xrightarrow{f} & N \end{array}$$

Take charts

□

8.1.2 Pfaffian Form

Definition 8.1.2.1 Let M be a differentiable manifold and V a finite dimensional vector space. A **V -valued Pfaffian form**, also called 1-form on M is a smooth map

$$\omega : TM \rightarrow V$$

where $\omega_p : T_p M \rightarrow V$ is linear.

The set of all Pfaffian forms on M

$$\Omega^1(M, V)$$

Especially, when $V = \mathbb{R}$, denoted by $\Omega_1(M)$.

Definition 8.1.2.2 Define the product

$$\begin{aligned} C^\infty(M) \times \Omega^1(M, V) &\rightarrow \Omega^1(M, V) \\ f \cdot \omega &:= (f \circ \pi_{TM}) \cdot \omega \end{aligned}$$

♣ **Example 8.1.2.3** ...

Pullback

Definition 8.1.2.4 Let $f : N \rightarrow M$. Then, the **pullback**

$$\begin{aligned} f^* : \Omega^1(M, V) &\rightarrow \Omega^1(N, V) \\ \omega &\mapsto \omega \circ T f \end{aligned}$$

8.2 Lie Algebra of Vector Fields

◇ **Lemma 8.2.0.1** If $U \subseteq M$ is an open subset and $f \in C^\infty(U, V)$ is a smooth function on U with values in V , where V is a finite dimensional vector space, $X \in \mathcal{V}(M)$.

Then we obtain for each a smooth function

$$\mathcal{L}_X f := df \circ X|_U \in C^\infty(U, V)$$

◦ **Proof** Omitted. □

Definition 8.2.0.2 Such $\mathcal{L}_X(f)$ is called **Lie derivative** of f with respect to X .

8.2.1 Lie Bracket

◇ **Lemma 8.2.1.1** Let $U \subseteq \mathbb{R}^n$ be an open subset. Then we obtain a Lie bracket on the space $C^\infty(U, \mathbb{R}^n)$ by

$$[X, Y](p) := dY(p)X(p) - dX(p)Y(p); p \in U$$

The map

$$C^\infty(U, \mathbb{R}^n) \rightarrow \text{End}(C^\infty(U, \mathbb{R})), X \mapsto \mathcal{L}_X$$

where $\mathcal{L}_X(f)(p) = df(p)X(p)$. is an injective homomorphism of Lie algebras. i.e. $\mathcal{L}_{[X, Y]} = [\mathcal{L}_X, \mathcal{L}_Y]$

◦ **Proof** ... □

8.3 Vector Field as Derivations

Chapter 9 The Cotangent Bundle

9.1 Tangent Covectors on Manifolds

Definition 9.1.0.1 Let M be a manifold. For each $p \in M$, we define the **cotangent space** at p ,

$$T_p^* M := (T_p M)^\vee$$

9.2 Differentials

Definition 9.2.0.1 ...

Chapter 10 Submersions, Immersions and Embeddings

Chapter 11 Submanifolds

11.1 Embedded Submanifold

♠ **Proposition 11.1.0.1** Suppose $F : M \rightarrow N$ smooth, $q \in N$ regular value. Then, for $p \in f^{-1}(q)$,

$$T_p(f^{-1}(q)) = \text{Ker}(d_p F)$$

11.2 Immersed Submanifold

Definition 11.2.0.1 Immersed submanifold: Image of injective immersion.

♣ **Example 11.2.0.2** $\mathbb{R} \rightarrow T, x \mapsto (\exp(2\pi ix), \exp(2\pi irx))$.

Fact If f is closed injective immersion then $\text{Im}(f)$ is an embedded submanifold.

Chapter 12 Differential Forms

12.1 Differential Form

12.1.1 title

Here is some of review of

♠ **Proposition 12.1.1.1** We have

$$\begin{cases} \dim \Lambda(V) = 2^d \\ \dim \Lambda_k(V) = \binom{d}{k}, 0 \leq k \leq d \end{cases}, \dim(V) = d, \Lambda_k = \{0\}; k \geq d$$

◦ **Proof ...**

□

Univeresal Mapping Property

♠ **Proposition 12.1.1.2** Denote $\varphi : V \times \cdots \times V \rightarrow \Lambda_k(V), (v_1, \dots, v_k) \mapsto v_1 \wedge \cdots \wedge v_k$. Then, for arbitrary alternating multilinear map h of $V \times \cdots \times V$ into the vector space W . There exists a unique map $\tilde{h} : \Lambda_k(V) \rightarrow W$ such that

$$\tilde{h} \circ \varphi = h$$

With the following diagram commutes

$$\begin{array}{ccc} & \Lambda_k(V) & \\ \varphi \uparrow & \searrow & \\ V \times \cdots \times V & \xrightarrow{h} & W \end{array}$$

◦ **Proof ...**

□

★ **Corollary 12.1.1.3** Suppose that $W = F$. Then, we could take the notion of

$$\Lambda_k(V)^* \cong A_k(V)$$

◦ **Proof ...**

□

Pairing

Definition 12.1.1.4 Let V and W be real finite dimensional vector spaces, a **pairing** of V and W is a bilinear map

$$(-, -) : V \times W \rightarrow \mathbb{R}$$

A **pairing** is called **non-singular** if whenever $w \neq 0$ in W , there exists

Linear Transformation of $\Lambda(V)$

Take the notion

$$\text{End}(\Lambda(V))$$

for the endomorphisms of $\Lambda(V)$

12.1.2 Tensor Fields and Differential Forms

12.1.3 The Lie Derivative

12.2 Orientation

12.2.1 Orientation of Vector Spaces

Definition 12.2.1.1 Let V be an N -dimensional real inner product space. We extend the inner product from V to all of $\Lambda(V)$

$$\langle w_1 \wedge \cdots \wedge w_p, v_1 \wedge \cdots \wedge v_p \rangle := \det \langle w_i, v_j \rangle$$

The bilinear map can be extended linearly to $\Lambda_p(V)$.

♠ **Proposition 12.2.1.2** If $e_1, \dots, e_n$ is an orthonormal basis of V , then the corresponding basis of $\Lambda(V)$ is an orthonormal basis for $\Lambda(V)$.

◦ **Proof ...**

□

Definition 12.2.1.3 (Consistently Oriented) Assume that $(v_1, \dots, v_n)$ and $(w_1, \dots, w_n)$ be two bases of V with $\dim(V) = n$, then we have the transform matrix

$$v^T = Bw^T$$

Then, they are **consistently oriented** if $\det(B) > 0$.

12.2.2 Orientation of Manifold

Definition 12.2.2.1 Let M be a manifold. Then a frame

$$(E_1, \dots, E_n)$$

is **(positively) oriented** if

12.2.3 The Orientation Covering

12.2.4 The Orientation of Hypersurface

12.2.5 The Orientation of

Chapter 13 Integration on Manifolds

Definition 13.0.0.1 Let M be a manifold. A **partition of unity** is a collection of smooth maps $(\varphi_i : M \rightarrow \mathbb{R})$

- there exists $\varphi_i : U_i \rightarrow V_i$ such that

$$\text{supp}(\varphi_i) \subseteq U_i$$

- For each $p \in M$, only finitely many of $\varphi_i(p) \neq 0$.

Definition 13.0.0.2 Let M be an oriented n -dim smooth manifold, ω a diff n -form, local integrable. Choose a partition of unity (φ_i) , define

$$\int \omega = \sum_i \int h_i^{-1*}(\varphi_i \circ \omega)$$

Chapter 14 De Rham Cohomology

Chapter 15 The de Rham Theorems

Chapter 16 Integral Curves, and Flows

16.1 Integral Curves

Definition 16.1.0.1 ...

Chapter 17 Integral Manifolds and Foliations

Chapter 18 The Hodge Theorem

Part VII

Bundles

Chapter 19 Bundles

19.1 Bundles

What is bundle?

19.1.1 Definitions

Definition 19.1.1.1

- In most general case, a **bundle** over an object B in a category C is simply an object E of C equipped with a morphism in C from E to B

$$p : E \rightarrow B$$

And let $(x : U \rightarrow B)$ be a generalized element of B . (92.2.1.1), the **fibre** E_x of the bundle over x is the pullback is the pullback x^*E

$$\begin{array}{ccc} x^*E = U \times_B E & \longrightarrow & E \\ \downarrow & & \downarrow p \\ U & \xrightarrow{x} & B \end{array}$$

Here, $U = 1$.

Such bundles over B is a category. Called **slide category** C/B of C over B .

- The morphism $E_1 \rightarrow E_2$ is a diagram such that the following diagram commutes

$$\begin{array}{ccc} E_1 & \xrightarrow{\quad} & E_2 \\ & \searrow & \swarrow \\ & B & \end{array}$$

Some of Bundles are listed as following:

- Fibre Bundle;
- Micro Bundle;
- Principle Bundle;
- Vector Bundle;
- Bundle with Connection;

Cross Sections

Definition 19.1.1.2 A **cross section** of a bundle (E, p, B) is a

Product Bundle

Stiefel Variety

Definition 19.1.1.3 ...

Grasmannian Variety

Definition 19.1.1.4 ...

19.1.2 Morphisms of Bundles

19.1.3 Product and Fibre Products

19.1.4 Restriction and Induced Bundles

19.1.5 Local Properties

19.2 Connections

Chapter 20 Fibre Bundles

20.1 Fibre Bundles

Definition 20.1.0.1 Let $\eta : E \rightarrow B$ be a bundle. Then, a **fibre bundle** is (η, E, B, F) with standard fiber F , such that for every general element $1 \rightarrow B$, the pullback of E along x is isomorphic to F .
i.e. $E_x \cong F$.

Chapter 21 Associated Bundles

Chapter 22 Projective Modules and Vector Bundles

22.1 Free module, GL_n and Stably Free Modules

22.1.1 The Module over Rings

Review:

- Vector Space(over division ring);
 - Base;
 - IBP property;
- ◇ **Lemma 22.1.1.1** If there exists a ring map $f : R \rightarrow F$, where F is a field or division ring, then R has IBP property.

- **Proof** Consider the following map

$$M \rightarrow M \otimes_R F = V \quad m \mapsto m \otimes_R 1$$

Then, $\sum \lambda_i m_i = 0 \rightarrow \lambda_i = 0 \Leftrightarrow \sum \lambda_i (m_i \otimes_R 1) = 0 \rightarrow \lambda_i = 0$.

Especially, if commutative, we could take $F = R/m$. □

Thus, $\dim(V) = \dim(M)$ is well-defined.

Definition 22.1.1.2 In general, for the ring R , take the notation of

$$GL_n(R)$$

for the group of invertible matrices. Especially, $n = 1$, $GL_1(R) = R^\times$.

◇ **Lemma 22.1.1.3** Any finite-dimensional algebra over a field/division ring must satisfies *IBP*, because the rank of free R -module M is an invariant:

$$\text{rank}(M) = \dim_F(M)/\dim_F(R)$$

- **Proof ...** □

22.1.2 Stably Free

Definition 22.1.2.1 An R -module P is called **stably free** of rank $n - m$ if

$$P \oplus R^m \cong R^n$$

for some m, n .

Conversely:

♣ **Example 22.1.2.2** ...

22.1.3 Stably Free Property

Definition 22.1.3.1 ...

22.1.4 Some of Noncommutative Ring Theory

♠ **Proposition 22.1.4.1** (P.M.Cohn) Consider the following conditions on a ring R :

- R satisfies IBP;
- For all m, n , if $R^m \cong R^n \oplus P$, then $m \geq n$;
- For all n , if $R^n \cong R^n \oplus P$, then $P = 0$.

◦ **Proof** ...

□

22.2 Projective Modules

Given in noncommutative ring theory/Homological Algebra

22.3 Picard Group of Commutative Rings

Given in noncommutative ring theory.

22.4 Topological Vector Bundle, and Chern Classes

We will work on field $F = \mathbb{R}, \mathbb{C}$ or $\mathbb{H}$.

22.4.1 Families of Vector Spaces

Definition 22.4.1.1

- Let X be a topological space. A family of vector spaces is a topological space E , together with map

$$\eta : E \rightarrow X$$

with a finite dimensional vector space structure on each $E_x = \eta^{-1}(\{x\})$.

The structure of E_x is compatible with the topological structure of E . i.e.

$$F \times E \rightarrow E \quad E \times_X E \rightarrow E$$

are continuous.

- A **homomorphism** from $\eta : E \rightarrow X$ to $\varphi : F \rightarrow X$ is:
 - A continuous map $f : E \rightarrow F, \eta = \varphi f$;
 - $f_x : E_x \rightarrow F_x$ is a linear map of vector spaces;

If $Y \subseteq X$, then define

$$E|Y = (E|Y, Y, \eta|_{\eta^{-1}(Y)})$$

With restriction map $\eta|_{\eta^{-1}(Y)} : \eta^{-1}(Y) \rightarrow Y$.

In general:

Definition 22.4.1.2 If $f : Y \rightarrow X$ is any continuous map, then the **pullback** of $\eta : E \rightarrow X$ is

$$f^*\eta : f^*E \rightarrow Y$$

where $f^*E = Y \times_X E$

$$\begin{array}{ccc} f^*E & \longrightarrow & E \\ \downarrow f^*\eta & & \downarrow \eta \\ Y & \xrightarrow{f} & X \end{array}$$

Vector Bundles

♣ **Example 22.4.1.3** (Trivial Bundle) If V is an n -dimensional vector space. The projection from $T^n = X \times V$ to X is a **constant family**

$$\eta : X \times V \rightarrow X$$

Such bundle is called **trivial vector bundle**.

Definition 22.4.1.4 (Vector Bundle) A **vector bundle** is a family of vector spaces such that:

- Local Trivial: For every point $x \in X$, $\eta|_U$ is a trivial.

$$\begin{array}{ccc} \eta^{-1}(U) & \xrightarrow{\quad} & T = U \times V \\ & \searrow & \swarrow \\ & U & \end{array}$$

So sometimes a vector bundle is called a **locally trivial** family of vectors.

Definition 22.4.1.5 (Category of vector bundles) Define the category $Vect_F(X)$, with the elements:

- $Ob(Vect_F(X)) = \{\text{Vector Bundles on } X\}$
- $Mor(Vect_F(X)) = \{\text{Morphisms between elements in } Ob(Vect_F(X))\}$

And a functor is given by: $f^* : Vect(Y) \rightarrow Vect(X)$.

Product

Definition 22.4.1.6 Let $\eta : E \rightarrow X, \varphi : F \rightarrow X$ be two vector bundle. Then, the **product of vector bundles** is defined as $(E \times F, X, \eta \times \varphi)$

$$\eta \times \varphi : E \times F \rightarrow X \times X$$

This gives a bilinear functor

$$\boxtimes : Vect(X) \times Vect(X) \rightarrow Vect(X \times X)$$

Tensor Product

Definition 22.4.1.7 Let $\eta : E \rightarrow X, \varphi : F \rightarrow X$ be two vector bundles. Then,

$$E \otimes F := \coprod_{x \in X} E_x \otimes F_x$$

with $\eta \otimes \varphi : E \otimes F \rightarrow X, e \otimes f \mapsto x; e \in E_x, f \in F_x$.

This is also a bilinear functor

$$\otimes : Vect(X) \times Vect(X) \rightarrow Vect(X)$$

Whitney Sum

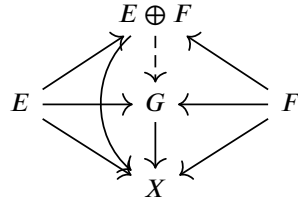
Definition 22.4.1.8 The **Whitney sum** $E \oplus F$ of two vector bundle $\eta : E \rightarrow X, \varphi : F \rightarrow X$ is the family of all vector spaces $E_x \oplus F_x$, which can be seen as a topological subspace of $E \times F$:

$$(E_x \times F_x)_{x \in X} \hookrightarrow E \times F$$

With the new vector bundle construction induced by $\eta \times \varphi$:

$$\eta \oplus \varphi := \eta \times \varphi|_{E \oplus F}$$

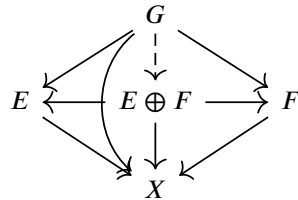
Whitney sum is the categoricial coporudct in $Vect(X)$.



Given $E \oplus F \rightarrow G : E_x \times F_x \ni (e, f) \mapsto i_1(e) + i_2(f)$ which is well-defined.

And commutes: Via the product structure.

Especially: $E \oplus F$ is also the categoricial coproduct:



$G \rightarrow E \oplus F, g \mapsto p_1(g) + p_2(g)$. Omitted.

★ **Corollary 22.4.1.9** $Vect(X)$ is an additional category.

◦ **Proof** Via checking definition. □

♣ **Example 22.4.1.10** (Mobius Bundles) The Mobius bundle E^1 over S

$$\pi : E^1 \rightarrow S$$

With strip $E_x \cong \mathbb{R}, x \in S^1$.

E^1 can be seen as the pasting of $[0, 1] \times \mathbb{R}$ via

$$(0, v) \sim (1, -v)$$

♣ **Example 22.4.1.11** (Whitney Sum of Tangent Bundle of Product Manifolds) For the product of manifolds $M_1 \times M_2$, we have

$$T_{(x,y)}(M \times N) \cong T_x M \oplus T_y N$$

Thus, we have

$$T(M \times N) = TM \oplus TN \rightarrow M \times N$$

New Vector Bundles from Olds:

- Restriction;
- Induced;
- Cartesian Product
- Whitney Sum;

Dimension of Vector Bundles

Notice that: $\dim(E) : X \rightarrow \mathbb{N}, x \mapsto \dim E_x$ is locally constant.

Definition 22.4.1.12 An n -dimensional vector space is a bundle E such that $\dim(E) = n$ is a constant function. Especially, on connected X , the function is constant.

•

Definition 22.4.1.13 ...

Definition 22.4.1.14 Define:

$$V_{n,F}(X) :=$$

Subbundles

Definition 22.4.1.15

- A **subbundle** of a vector space $\eta : E \rightarrow X$ is a subspace F of E which is a vector bundle under induced structure.
i.e. F_x is a vector subspace of E_x , and family $F \rightarrow X$ is locally trivial.
- The **quotient bundle** E/F is defined as the union of vector spaces

$$(E_x/F_x)$$

given quotient topology (by η).

Short exact sequence

$$0 \longrightarrow F \longrightarrow E \longrightarrow E/F \longrightarrow 0$$

in $VB(X)$.

Definition 22.4.1.16 A vector bundle $E \rightarrow X$ is said to be of **finite type** if there is a finite covering $U_1, \dots, U_n$ such that $E|_{U_i}$ is a trivial bundle.

♠ **Proposition 22.4.1.17**

-
-

◦ **Proof** ...

□

► **Theorem** (Subbundle Theorem) Let $E \rightarrow X$ be a vector bundle on paracompact space X :

-
-

◦ **Proof** With the conclusion 21.4.1, there exists a metric on the bundle.

□

Sections

Definition 22.4.1.18

- Let $\eta : E \rightarrow X$ be a vector space. A **global section** of η is a continuous map $s : X \rightarrow E$ such that

$$\eta \circ s = 1_X$$

- It is nowhere zero if $s(x) \neq 0, \forall x \in X$.

In general, we could define the **local section**

$$s : U \rightarrow E \quad \eta \circ s = id_U$$

Vector Bundle under Riemannian Metrics

Review:

- Riemannian Metrics;
- Hermitian Metrics;

Metric on $E_x \Rightarrow$ Metric on E .

A fundamental result is:

► **Theorem** If X is a paracompact space, then there exists a metric on E , when $F = \mathbb{R}, \mathbb{C}, \mathbb{H}$.

◦ **Proof ...**

□

Patching Vector Bundles

Transition of vector bundles: Consider two locally trivial maps

...

Definition 22.4.1.19 (Tensor Product of Bundles)

Definition 22.4.1.20 (Determinant Bundle)

Cancellation Theorems

► **Theorem** (Real Cancellation Theorem)

◦ **Proof ...**

□

Homotopy Invariance Theory

Classifying Vector Bundles

► **Theorem** (Classifying Theorem) Let X be paracompact space. Then,

$$VB_n(X) \cong [X, Grass_n]$$

◦ **Proof ...**

□

Definition 22.4.1.21 (Classifying Spaces)

► **Theorem** (Classifying Theorem for Line Bundles)

...

◦ **Proof ...**

□

22.4.2 Serre-Swan Theorem

♠ **Proposition 22.4.2.1** (Global Section) Let $\eta : E \rightarrow X$ be a vector bundle. Let $\Gamma(E)$ be the set of global sections of E .

Then, $\Gamma(E)$ is a module over $C^0(X)$ of continuous functions on X .

Especially, if E is a trivial bundle of dimension n , then $\Gamma(E)$ is a free $C^0(X)$ -module of rank n .

◦ **Proof** Module axiom:

- Sum:
- Scalar Product:

□

★ **Corollary 22.4.2.2** If X is paracompact, then $\Gamma(E)$ is a locally free $C^0(X)$ -module.
 $\Gamma(E)$ is a finitely generated projective module if X is compact or E is of finite type.

◦ **Proof** ...

□

► **Theorem** (Swan) Let X be a compact Hausdorff space, and write R for $C^0(X)$.
 Then, $\Gamma : VB(X) \rightarrow P(R)$ of finitely generated projective modules is a functor. The homomorphism

$$\Gamma_{VB(X)}(E, F)$$

This implies, there exists an equivalence of categories $VB(X) \cong P(C^0(X))$.

◦ **Proof** ...

□

Remark The Serre's theorem will be given in AG.

Here:

- Construction:
- Affines:

Projective Flag Modules

Splitting Principle

Pontrijagin Classes

22.4.3 Characteristic Classes

Abstract The determinant line bundle $\det(E)$ of a vector bundle E yields to a cohomology class.

-
-

Axioms for Stiefel-Whitney Classes

† **Axiom** ...

Axioms for Chern Classes

† Axiom ...

Pontrjagin Classes

Definition 22.4.3.1 (Complex Conjugate Bundle)

22.4.4 Others

Complexification of Real Vector Bundles

22.5 Algebraic Vector Bundles

Part VIII

Lie Groups

Part IX

Riemann Manifold

Chapter 23 Riemannian Metrics

23.1 Definitions

Definition 23.1.0.1

- Let M be a smooth manifold. A **Riemannian metric** on M is a 2-tensor field $g \in \mathcal{T}^2 M$, whose value

$$g_p : T_p M \times T_p M \rightarrow \mathbb{R}$$

which is positive definite in the sense of $g_p(v, v) \geq 0$.

- A **Riemannian manifold** is a pair (M, g) .

Remark Riemannian metric vs Metric Space

♣ **Example 23.1.0.2** (Euclidean Metric)

Isometries

Local Representation

Definition 23.1.0.3 (Local Representation of Riemannian Metric)

♠ **Proposition 23.1.0.4** (Existence of Orthonormal Frame)

◦ **Proof ...**

□

Definition 23.1.0.5 (Unit Tangent Bundle) Define the **unit tangent bundles**

$$UTM := \{(p, v) \mid |v|_g = 1\}$$

♠ **Proposition 23.1.0.6** If (M, g) is a Riemannian-Manifold with or without boundary, its unit tangent bundle UTM is a smooth, properly embedded codimension-1 submanifold with boundary in TM , with

$$\pi : UTM \rightarrow M \quad \partial(UTM) = \pi^{-1}(\partial M)$$

Then, UTM is connected $\Leftrightarrow M$ is connected. UTM is compact $\Leftrightarrow M$ is compact.

◦ **Proof ...**

□

23.1.1 Riemannian Submanifold

◇ **Lemma 23.1.1.1** Suppose $(\tilde{M}, \tilde{g})$ is a Riemannian manifold with or without boundary. M is a smooth manifold with or without boundary. And

$$F : M \rightarrow \tilde{M}$$

is a smooth map. The smooth 2-tensor field (pullback)

$$g = F^* \tilde{g}$$

is a Riemannian metric iff F is an immersion.

◦ **Proof ...**

□

23.2 Models of Riemannian Manifolds

23.2.1 Symmetries

23.2.2 Covering Space of Riemannian Manifold

Eucilidean Space

Sphere

Hyperbolic Space

Killing-Hopf Theorem

► **Theorem** Suppose (M, g) be compelete, simply connected Riemannian n -manifold with constant sectional curvature, $n \geq 2$.

Then, M is isometric to one of the model spaces $\mathbb{R}^n, \mathbb{S}^n(R), \mathbb{H}^n(R)$.

◦ **Proof ...**

□

★ **Corollary 23.2.2.1** ...

◦ **Proof ...**

□

Space Form

Chapter 24 Riemannian Connections

24.1 Affine Connections

Definition 24.1.0.1 Define the **affine connections**

$$\begin{aligned} \nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) &\rightarrow \mathfrak{X}(M) \\ (X, Y) &\mapsto \end{aligned}$$

With following properties:

- $(C^\infty(M)\text{-Linear}) \nabla_{fX+gY}Z = f\nabla_XZ + g\nabla_YZ;$
- $(\mathbb{R}\text{-Linear}) \nabla_X(aY + bZ) = a\nabla_X(Y) + b\nabla_X(Z)$
- $(\text{Lebnitz}) \nabla_X(fY) = f\nabla_XY + X(f)Y$

where $X, Y, Z \in \mathfrak{X}(M), f, g \in C^\infty(M)$.

♠ **Proposition 24.1.0.2** Let M be a differentiable manifold with an affine connection ∇ . There exists a unique correspondence which associates to a vector field V along the differentiable curve

$$c : I \rightarrow M$$

another vector field along c , called the **covariant derivative** of V along c , such that:

-
-
-

◦ **Proof ...**

□

24.1.1 Parallel Vector Field

24.2 Riemannian Metrics, Levi-Civita Theorems

Chapter 25 Geodesics

25.1 The Geodesics Flow

25.2 Minimalizing Properties

Chapter 26 Curvatures

26.1 Curvatures

26.2 Sectional Curvature

26.3 Ricci Curvature and Scalar

26.4 Tensors on Riemannian Manifolds

26.5 Manifolds of Nonpositive Curvature

26.5.1 Cartan-Hadamard Manifold

► *Theorem* (Cartan-Hadamard)

◦ **Proof ...** □

Definition 26.5.1.1 ...

♠ **Proposition 26.5.1.2** ...

◦ **Proof ...** □

♠ **Proposition 26.5.1.3** ...

◦ **Proof ...** □

26.5.2 Cartan's Fixed Point Theorem

26.5.3 Cartan's Torsion Theorem

Chapter 27 Riemannian Submanifold

Chapter 28 Gauss-Bonnet Theorem

Chapter 29 Jacobi Field

Chapter 30 Comparison Theory

Chapter 31 Curvature and Topology

Chapter 32 Isomorphic Immersions

Chapter 33 Complete Manifold, Hopf-Rinow and Hadamard Theorems

Chapter 34 Spaces of Constant Curvature

Chapter 35 Variations of Energy

Chapter 36 The Morse Index Theorem

Chapter 37 The Fundamental Group of Manifolds of Negative Curvatures

Chapter 38 The Sphere Theorem

Chapter 39 The Laplacian

This will be given in geometric analysis.

Part X

Foundation of Differential Geometry

Chapter 40 Differentiable Manifolds

40.1 Differentiable Manifolds

40.1.1 Pseudo Group Axioms

Definition 40.1.1.1 A pseudogroup of transformations on a topological space S is a set Γ of transformations satisfying the following axioms:

- Each $f \in \Gamma$ is a homeomorphism of an open set of S onto another open set of S .

$$f_{\cong} : U \rightarrow V; U, V \in \text{Open}(S)$$

- If $f \in \Gamma$, then the restriction of f to an arbitrary open subset of domain of f is in Γ . (And it is still a homeomorphism)

$$f \in \Gamma \Rightarrow f|U \in \Gamma$$

- Let $U = \cup U_i$. Then,

$$f|U_i \in \Gamma \Rightarrow f \in \Gamma$$

- For every open U ,

$$id_U \in \Gamma$$

-

$$f \in \Gamma \Rightarrow f^{-1} \in \Gamma$$

- $f \in \Gamma$ is a homeomorphism of U onto V , $f' \in \Gamma$ is a homeomorphism of U' onto V' , then

$$\Gamma \ni f' \circ f : f^{-1}(V \cap U') \rightarrow f'(V \cap U')$$

♣ **Example 40.1.1.2** (Pseudogroup $\Gamma^r(\mathbb{R}^n)$) Define:

$$\Gamma^r(\mathbb{R}^n) := \{f \in C^r(U, V) | U, V \in \text{Open}(\mathbb{R}^N)\}$$

There is an inclusion

$$\Gamma^{r+1}(\mathbb{R}^n) \hookrightarrow \Gamma^r(\mathbb{R}^n)$$

♣ **Example 40.1.1.3** Furthermore, Pseudogroup

$$\mathcal{O}(\Omega)$$

40.1.2 Atlas

Definition 40.1.2.1 An **atlas** of a topological space M is a family of **charts** (U_i, φ_i) which satisfies:

- Each U_i is an open set of M , and

$$\bigcup U_i = M$$

- $\varphi_i : U_i \rightarrow S$ is a homeomorphism onto its image;
- If $U_i \cap U_j \neq \emptyset$, then

$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \rightarrow \varphi_j(U_i \cap U_j)$$

is an element of Γ .

Definition 40.1.2.2 An **complete atlas** of M compatible with Γ is an atlas of M compatible with Γ .

Then, every atlas is contained in a unique complete atlas.

Remark Suppose A, A' be two atlases. Then, if they are **compatible**

...

40.1.3 Differentiable Manifold

Definition 40.1.3.1 ...

40.1.4 Differentiable Mapping

40.1.5 Differentiable Curve, Tangent Vectors

Differentiable Curve

Definition 40.1.5.1 ...

Tangent Vectors

Definition 40.1.5.2 Let $\mathfrak{F}(p), p \in M$ be the equivalence class of differentiable functions at p ,

$$\mathfrak{F}(p) := \{(U, f) | f \in C^1(U, \mathbb{R}), p \in U \subseteq X \text{ open}\} / \sim \quad (U, f) \sim (V, g) \Leftrightarrow f|_{U \cap V} = g|_{U \cap V}$$

$\mathfrak{p}$ is an $\mathbb{R}$ -linear space.

Then, let $x : [a, b] \rightarrow M$ be a C^1 -differentiable function where

$$x(t_0) = p$$

Then, define

$$X : \mathfrak{F}(p) \rightarrow \mathbb{R}$$

$$[(U, f)] \mapsto \left(\frac{df(x(t))}{dt} \right) |_{t_0}$$

In other word, Xf is derivative of f in the direction of the curve $x(t)$ at $t = t_0$.
The function X satisfies the following properties:

-
-

Lie Bracket

40.2 Tensor Algebra, Tensor Fields

Given in Smooth manifold.

40.3 Lie Groups

Given in Lie theory.

40.4 Fibre Bundles

40.4.1 Differentiable Principle Fibre Bundle

Definition 40.4.1.1 A differentiable principle fibre bundle over M with group G consists of $P(M, G, \pi)$, where:

- G acts freely on P on the right:

$$P \times G \rightarrow P \quad (u, a) \mapsto ua = R_a u \in P$$

- M is quotient space of P , and the canonical projection

$$\pi : P \rightarrow M$$

is differentiable;

- P is locally trivial:

We call:

-

Fibre of Differentiable Principle Fibre Bundle

40.4.2 Fundamental Vector Field

Definition 40.4.2.1 Given a principle fiber bundle $P(M, G, \pi)$, the action of G on P induces a homomorphism σ of the Lie algebra $\mathfrak{g}$ of G into the Lie algebra $\mathfrak{X}(P)$ of vector fields on P .

$$\mathfrak{g} = L(G) \rightarrow \mathfrak{X}(P)$$

Constructions:

- Exponential Maps:

- Consider the action

$$\Phi_u : G \rightarrow P \quad g \mapsto u \cdot g$$

Then,

$$\sigma : \mathfrak{g} = T_e G \rightarrow \mathfrak{X}(P) \sigma(A)_u = (d\Phi_u)_e(A)$$

Take the following notation

$$A^* := \sigma A$$

Furthermore, we have

♠ **Proposition 40.4.2.2**

$$\mathfrak{g} \rightarrow T_u P \quad A \mapsto (A^*)_u$$

is linear isomorphism.

◦ **Proof**

-
-
-

□

♠ **Proposition 40.4.2.3** Let A^* be fundmanetal vector field, then there exists correspondence

$$\begin{aligned} A \in \mathfrak{g} &\Rightarrow A^* \in \mathfrak{X}(P) \\ (ad(a^{-1}))A \in \mathfrak{g} &\Rightarrow (R_a)_* A^* \end{aligned}$$

◦ **Proof ...**

□

The mapping σ is injective, but not surjective. But we could define a Lie-subalgebra of $\mathfrak{X}(P)$, so that $\sigma : L(G) \rightarrow \mathfrak{X}(P)$ is bijective.

Definition 40.4.2.4 ...

♠ **Proposition 40.4.2.5** Let M be a manifold. $\{U_\alpha\}$ is an open covering of M and G a Lie group.

Given mapping

◦ **Proof ...**

□

Chapter 41 Theory of Connections

41.1 Connections in a Principle Vector Bundle

Remark Firstly, the fibre through u is

$$E_{\pi(u)}$$

Definition 41.1.0.1 Let (P, M, G, π) be a principle fibre bundle over a manifold M . For each $u \in P$, $T_u(P)$ be the tangent space of P at u , and G_u be the subspace of $T_u(P)$ consisting of vectors tangent to fibre through u .

$$G_u := T_u(E_{\pi(u)}) = \text{Ker}(\pi_u)$$

- $T_u(P) = G_u \oplus Q_u$; G_u is the vertical subspace, Q_u is the horizontal subspace;
Take the following notation vX, hX .

- $Q_{ua} = (R_a)^* Q_u$; $u \in P, a \in G$
- Q depends differentiably on u ;

A **connection** is a distribution $u \mapsto Q_u$.

Remark 1 The tangent space is the formal tangent space, not rely on the open subset and the tangent line via limit.

41.1.1 Connection Form

◇ **Lemma 41.1.1.1** $\sigma_u : \mathfrak{g} \rightarrow T_u(P)$ is injective, $u \in P$, with induced linear isomorphism

$$\sigma_u : \mathfrak{g} \rightarrow V_u(P) \quad A \mapsto (A^*)_u$$

◦ **Proof ...**

□

Definition 41.1.1.2 Given a connection Γ in P . We define the 1-form P with the value in Lie algebra of G as follows: $\omega \in \Omega^1(P, \mathfrak{g})$,

$$\omega_p : T_p(P) \rightarrow \mathfrak{g} \quad X \mapsto A$$

such that $(A^*)_u = vX$.

Remark 1 $\mathfrak{g} \rightarrow G_u, A \mapsto (A^*)_u$ is a linear isomorphism.

♠ **Proposition 41.1.1.3** The connection form ω of a connection satisfies the following conditions:

- $\omega(A^*) = A$;
- $(R_a)^* \omega = \text{ad}(a^{-1})\omega$, that is

...

for every $a \in G$ and every vector field X on P .

◦ **Proof ...** □

♠ **Proposition 41.1.1.4** ...

◦ **Proof ...** □

♠ **Proposition 41.1.1.5** ...

41.2 Existence and Extension of Connections

► **Theorem ...**

◦ **Proof ...** □

41.3 Parallelism

41.3.1 Horizontal Lift

Definition 41.3.1.1 Let $\tau = x(t)$ be a piecewise differentiable curve of C^1 in M . A **horizontal lift** or **lift** is a **horizontal curve**

$$\tau^* = u(t)$$

such that $x(t) = \pi(u(t))$.

Horizontal: Piecewise differentiable, the tangent vectors are horizontal.

♠ **Proposition 41.3.1.2** (Existence, Uniqueness)

◦ **Proof ...** □

41.3.2 Parallel Displacement

41.4 Holonomy Groups

Another Reference: Riemannian Holonomy Groups and Calibrated Geometry by Dominic D. Joyce

Definition 41.4.0.1 For point $x \in M$, denote

41.5 Curvature Form, and Structure Equation

Definition 41.5.0.1 Let $P(M, G, \pi)$ be a PFB and representation on G on V , with $\rho(a) \in \text{End}(V), a \in G$. $\rho(ab) = \rho(a)\rho(b)$.

A **pseudotensorial form** of degree on P of type (ρ, V) is a V -valued r -form φ such that

$$R_a^* \varphi = \rho(a^{-1}) \cdot \varphi; a \in G$$

Such a form is called **tensorial form** if it is horizontal in the sense of

$$\varphi(X_1, \dots, X_i, \dots, X_r) = 0; X_i \in V_i P$$

♠ **Proposition 41.5.0.2**

-
-
-

◦ **Proof ...**

□

Especially:

Definition 41.5.0.3

- The connection form ω is a pseudotensorial 1-form of type $ad(G)$;
- $D\omega$ is called **tensorial 2-form** of type $ad(G)$, denoted by Ω , is called the **curvature form** of ω .

41.5.1 Structure Equation

► **Theorem** (Structure Equation, E. Cartan) Let ω be a connection form and Ω be its curvature form, then

$$\Omega(X, Y) = d\omega(X, Y) + \frac{1}{2}[\omega(X), \omega(Y)]$$

◦ **Proof ...**

□

♠ **Proposition 41.5.1.1 ...**

◦ **Proof ...**

□

41.5.2 Bianchi's Identity

► **Theorem** (Bianchi)

$$D\Omega = 0$$

◦ **Proof ...**

□

41.6 Mappings of Connections

41.7 Reduction Theorem

41.8 Holonomy Theorem

41.9 Flat Connections

41.10 Local, and Infinitesimal Holonomy Groups

41.11 Invariant Connections

Chapter 42 Linear Affine Connections

42.1 Connections in Vector Bundles

42.1.1 Fibre Metrics

42.2 Linear Connections

Chapter 43 Curvature and Space Forms

Chapter 44 Transformations

Chapter 45 Submanifolds

Chapter 46 Variation of Length Integral

Chapter 47 Homogeneous Spaces

Chapter 48 Symmetric Spaces

Part XI

Morse Theory

Part XII

Floer Theory

Part XIII

Spin Geometry

Part XIV

Fourier Analysis

Part XV

Real Analysis

Part XVI

Functional Analysis

Part XVII

ODE

Part XVIII

PDE

Part XIX

Complex Analysis in One Variable

Part XX

Complex Analysis in Multivariables

Part XXI

Riemann Surface

Part XXII

Complex Geometry

Part XXIII

Linear Algebra

Chapter 49 Linear Spaces

49.1 Linear Space

◇ Lemma 49.1.0.1

◦ **Proof ...**

□

49.2 Basis

49.2.1 Steinitz's Exchange Lemma

◇ **Lemma 49.2.1.1** (Steinitz) Let U, V be two finite subsets of V . Let U be a set of linear independent vectors. and W spans V . Then,

- $\text{card}(W) \leq \text{card}(U)$;
- There is subset $W' \subseteq W$ with $|W'| = |U|$ and $U \cup W'$ spans V .

◦ **Proof ...**

□

S.Maclane develops this lemma for matroid:

49.3 Dimension

Part XXIV

Appendix: Linear Algebra over Division Ring

Part XXV

Fundamental of Group Theory

Part XXVI

Ring, Modules and Associative Algebras

Chapter 50 Rings

Chapter 51 Modules

51.1 Tensor Product

Definition 51.1.0.1

◇ **Lemma 51.1.0.2** (Tensor Product commutes with Colimits) Let (M_i, μ_{ij}) be a system over a preordered set I , then

$$\operatorname{colim}(M_i \otimes N) \cong (\operatorname{Colim} M_i) \otimes N$$

◦ **Proof** □

◇ **Lemma 51.1.0.3** Let $M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \longrightarrow 0$ be an exact sequence of R -modules and homomorphisms, and N be an R -module, then the sequence

$$M_1 \otimes N \xrightarrow{f \otimes \operatorname{id}} M_2 \otimes N \xrightarrow{g \otimes \operatorname{id}} M_3 \otimes N \longrightarrow 0$$

is exact. In other word, $- \otimes N$ is right exact.

◦ **Proof** □

51.1.1 Finiteness

◇ Lemma 51.1.1.1

◦ **Proof** □

51.1.2 Tensor Algebra

51.2 Finite Case of Modules and Maps

51.2.1 Finite Modules, and Finitely presented modules

Definition 51.2.1.1 Let R be a ring and M is R -module:

- We say M is a **finite** R -module, or finitely generated module if there exists $n \in \mathbb{N}$ and $x_i \in M$, with exact sequence

$$\bigoplus R x_i \rightarrow M \rightarrow 0$$

- M is **finitely represented** if there exists $m, n \in \mathbb{N}$ and exact sequence

$$R^{\oplus m} \rightarrow R^{\oplus n} \rightarrow M \rightarrow 0$$

which implies:

- M is finitely generated;
- The relation between the generators are also finitely generated.

such choice of exact sequence is called a **presentation**.

◇ **Lemma 51.2.1.2** Let R be a ring, and

$$\alpha : R^{\oplus n} \rightarrow M \quad \beta : N \rightarrow M$$

be R -module maps. If $\text{Im}(\alpha) \subseteq \text{Im}(\beta)$, then there exists an R -module map $\gamma : R^{\oplus n} \rightarrow N$ such that $\alpha = \beta \circ \gamma$.

$$\begin{array}{ccc} N & \xrightarrow{\beta} & M \\ \uparrow \gamma & \nearrow \alpha & \\ R^{\oplus n} & & \end{array}$$

◦ **Proof** Let $\alpha(e_i) = \beta(n_i)$. Then, consider the mapping

$$\gamma : R^{\oplus n} \rightarrow N \quad (a_1, \dots, a_n) \mapsto \sum a_i n_i$$

□

◇ **Lemma 51.2.1.3** Let R be a ring, and $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of R -modules:

- If M_1 and M_3 are finite R -modules, then M_2 is a finite R -module;
- If M_1 and M_3 are finitely presented R -modules, then M_2 is finitely presented;
- If M_2 is a finite R -module, then M_3 is a finite R -module;
- If M_2 is finitely presented R -module and M_1 is a finite R -module, then M_3 is a finitely represented R -module;
- If M_3 is finitely presented R -module and M_2 is a finite R -module, then M_1 is a finite R -module.

◦ **Proof**

-
-
-
-
-

□

◇ **Lemma 51.2.1.4** Let R be a ring and let M be a finite R -module. There exists a filtration by finite R -module

$$\dots$$

such that each quotient M_i/M_{i-1} is isomorphic to R/I_i for some ideal I_i of R .

◦ **Proof** ...

□

◇ **Lemma 51.2.1.5** Let $R \rightarrow S$ be a map, and M be an S -module. If M is finite as an R -module, then M is finite as an S -module.

◦ **Proof** ...

□

51.2.2 Ring Maps of finite Type and of finite Representation

Chapter 52 Associative Algebra

Chapter 53 Polynomial Algebra

53.1 Discriminant

Chapter 54 Matrix Algebra

Part XXVII

Field and Galois Theory

Chapter 55 Fields and Galois Theory

55.1 Fields

Definition 55.1.0.1 A **field** is a five-tuple $(K, +, \times, 0, 1)$, where:

- $(K, +, 0)$ is an additive group;
- $(K^\times, \times, 0)$ is a multiplicative group;

We have know the construction of

$$Ring \rightarrow CRing \rightarrow ID \rightarrow Fields$$

♠ **Proposition 55.1.0.2** Let A be a domain. Then for every ring morphism $\phi : A \rightarrow K$ where K is a field, there exists factorization

$$\begin{array}{ccc} Quot(A) & \xrightarrow{\quad} & K \\ \uparrow & \nearrow & \\ A & & \end{array}$$

◦ **Proof** It is well-defined, induced by

$$\frac{a}{b} \mapsto \frac{\phi(a)}{\phi(b)}$$

Universal property omitted. □

Vector Spaces

◇ **Lemma 55.1.0.3** Every $V \in Vect_k$ is free.

◦ **Proof** Consider the Bases. (If infinite, rely on AC) □

★ **Corollary 55.1.0.4** Every exact sequence of modules over a field splits.

◦ **Proof** Each vector space is free, so is projective. And according to the proposition here: 93.2.0.5 □

Characteristic

Definition 55.1.0.5 Let F be a field, and a ring map

$$\phi : \mathbb{Z} \rightarrow F \quad 1 \mapsto 1$$

Then, the kernel $\ker(\phi) = (p)$, $\text{char}(F) := p$, is called the **characteristic** of F .

Under this notation:

Definition 55.1.0.6 Let F be a field:

- $\text{char}(F) = 0, \mathbb{Q} \subseteq F$
- $\text{char}(F) = p, F_p \subseteq F$

is called **prime field**, which is the smallest subfield of F .

55.1.1 Field Extensions

◇ Lemma 55.1.1.1

If F is a field and R is a nonzero ring, then, any ring homomorphism

$$\phi : F \rightarrow R$$

is injective.

◦ **Proof** Suppose $a \in \text{Ker}(\phi); a \neq 0$. Then, notice that

$$\phi(1) = \phi(aa^{-1}) = 0 \Rightarrow 1 = \phi(1) = 0$$

Contradiction. □

Definition 55.1.1.2 If F is a field contained in a field E , then E is said to be a **field extension** of F . We shall write

$$E/F$$

to indicate that E is an extension of F .

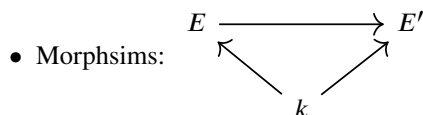
Category of Field Extension

With 53.1.1.1, we have known: The ring morphisms between fields must be injective. And: Those morphisms represent the field extensions.

For morphism $\phi : F \rightarrow K$, K can be seen as F -algebra.

Definition 55.1.1.3 Let k be a field. There is a **category of field extensions** of k .

- Objects: Morphisms $k \rightarrow E$;



Take the notation of morphisms $\text{Mor}_k(E, E')$.

Definition 55.1.1.4 (Tower of Extensions) A **tower of fields** $E_n/E_{n-1}/\dots/E_0$ consists of a sequence of extensions of fields $E_n/E_{n-1}, \dots, E_1/E_0$.

Definition 55.1.1.5

◇ **Lemma 55.1.1.6** ...

◦ **Proof** ...

□

Finite Extensions

Definition 55.1.1.7 Let F/E be an extension of fields. The dimension of F considered as an E -vector space is called **degree of extension** and is denoted as:

$$[E : F] := \dim_E(F)$$

Algebraic Extension

Definition 55.1.1.8 Consider a field extension F/E . An element $\alpha \in F$ is said to be **algebraic** over E if α is the root of some nonzero polynomial with coefficients in E .

If all elements of F are algebraic then F is said to be an **algebraic extension** of E .

Minimal Polynomials

Algebraic Closure

55.1.2 Separation Extension

Chapter 56 The Fundamental Theorem of Galois Theory

Chapter 57 Algebraic Closure

Chapter 58 Infinite Galois Extensions

Part XXVIII

Inverse Galois Theory

Open Problem (Inverse Galois Problem)

Whether or not every finite group G appears as the Galois group of some Galois extension of the rational number $\mathbb{Q}$. i.e.

$$\text{Gal}(K/\mathbb{Q}) = G$$

Equivalently, we should describe the property of the rational number field

$$\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$$

More genrally

Definition 58.0.0.1 Let G be a group, then G is **realizable** over K if there exists some L such that

$$\text{Gal}(L/K) = G$$

Part XXIX

Commutative Algebra

Chapter 59 Additional Text for Ring and Modules

59.1 Local Properties

59.1.1 Local Rings

59.1.2 Localization

59.2 Spectrum of Ring

Definition 59.2.0.1 Let R be CR. The spectrum is the topology given by:

$$\text{Spec}(R) := \{p \in R \mid p \text{ is prime}\}$$

59.2.1 Zariski Topology

Definition 59.2.1.1

59.2.2 Localization

59.2.3 Open and Closed Subsets of Spectra

Closed Points in $\text{Spec}(R)$

♠ **Proposition 59.2.3.1** Let $x = p \in \text{Spec}(R)$ be a point. Then

$$\{x\} \text{ is closed} \Leftrightarrow p \text{ is maximal ideal.}$$

◦ **Proof** $\{x\}$ is closed $\Leftrightarrow \{x\} = V(x) = \{p \in \text{Spec}(R) \mid x \subseteq p\} \Leftrightarrow x$ is a maximal ideal. □

59.2.4 Connected Component

59.2.5 Glueing Property

59.2.6 Glueing Functions

Goal: To show

$$\text{Spec}(R) = \cup_{i=1}^n D(f_i)$$

And given $h_i = h_j \in R_{f_i f_j}$, there exists $h \in R$ where $h = h_i \in R_{f_i}$.

- $D(f) \mapsto R_f$ defines a sheaf;
- If we think of the elements of R_f as the algebraic/regular functions on $D(f)$, then these glue as would continuous/differentiable functions on a topological/differentiable manifold.

◇ **Lemma 59.2.6.1** (Exactness of Sheaf of Modules)

59.2.7 Irrducible Component

59.2.8 Image

Images of Ring Maps of Finite Representation

◇ **Lemma 59.2.8.1** Let $U \subseteq \text{Spec}(R)$ be open. The followings are equivalent:

- U is **retrocompact** in R ;
- U is **quasi-comapct**
- U is finite union of standard opens;
- There exist a finitely generated ideal $I \subseteq R$ such that $X/V(I) = U$;

◦ **Proof**

- (3) $\Leftrightarrow$ (4) Omitted;
- (3) $\Rightarrow$ (4):
- (4) $\Rightarrow$ (1):
- (1) $\Rightarrow$ (3)

□

More on Images

59.3 Jacobson Radical of a Ring

59.3.1 Nakayama's Lemma

59.4 Additional Propositions for Prime Ideals

59.5 Nilpotent

Definition 59.5.0.1 Let R be a ring, $I \subseteq R$ be an ideal:

- I is **locally nilpotent** if for every $x \in I$, there exists n such that $x^n = 0$;
- I is **nilpotent** if there exists n such that $I^n = 0$;

◇ **Lemma 59.5.0.2** Let $R \rightarrow R'$ be a ring map, and $I \subseteq R$ be a locally nilpotent ideal, then IR' is a locally nilpotent ideal of R' .

◦ **Proof**

□

◇ **Lemma 59.5.0.3** Let R be a ring and $I \subseteq R$ be a locally nilpotent ideal, an element x of R is a unit iff image of x in R/I is a unit.

◦ **Proof**

□

◇ Lemma 59.5.0.4

◦ Proof

□

59.6 Jacobson Ring

Definition 59.6.0.1

Chapter 60 Chain Conditions

Definition 60.0.0.1

-
-

60.1 Noetherian Rings, Modules

60.1.1 Noetherian Ring

Definition 60.1.1.1

60.1.2 Noetherian Modules

Definition 60.1.2.1 Let R be a commutative ring, and M be $R - Mod$. Then, M is **Noetherian** if for any descending sequence of submodules

$$M_0 \supset M_1 \supset \cdots \supset \dots$$

There exists some $N \geq 0$ such that $M_N = M_{N+1} = \dots$ i.e. The submodules satisfy ACC.

♠ **Proposition 60.1.2.2** The following conditions are equivalent:

- M is Noetherian;
- For every subset $S \subseteq M$, there exists elements $m_1, \dots, m_r \in S$,

$$(S)_R = (m_1, \dots, m_r)_R$$

- Every submodule of M is finitely generated.

◦ **Proof**

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-
-

□

♠ **Proposition 60.1.2.3** The following conditions are equivalent:

- M is Noetherian;
- M is finitely generated.

◦ **Proof**

-
-

□

Polynomial Rings over Noetherian Rings

Given here:

60.2 Length

Definition 60.2.0.1 Let R be a ring. For any R -module M we define the **length** of M over R by the formula

$$\text{length}_R(M) := \sup\{\alpha \in \text{Ord} \mid M_0 = 0, M_\alpha = M, M_\gamma \subset M_{\gamma-1}\}$$

It is the supremum of lengths of chain of submodules. There exists a partial order relation for chains by extending elements. Then the length of M is the length of maximal chain.

60.2.1 Properties

◇ **Lemma 60.2.1.1** $\text{length}_R(M) < \infty \Leftrightarrow M$ is a finite R -module.

◦ **Proof ...**

□

Lengths of Modules of short Exact Sequence

◇ **Lemma 60.2.1.2** If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is a short exact sequence of modules over R , then

$$\text{length}_R(M) = \text{length}_R(M') + \text{length}_R(M'')$$

◦ **Proof** Via inequality: Because of

$$\text{length}(M)$$

And conversely: Take filtration $M_0 \subset \cdots \subset$

□

Local Rings

◇ **Lemma 60.2.1.3** Let R be a local ring with maximal ideal m . If M is an R -module and $m^n M \neq 0, n \geq 0$.

If M is an R -module and $m^n M \neq 0$ for all $N \geq 0$, then $\text{length}_R(M) = \infty$.

In other word: M has finite length if $m^n M = 0$ for some n .

◦ **Proof ...**

□

Ring Maps◇ **Lemma 60.2.1.4** ...◦ **Proof** ...

□

Length-Quotient of Fields**Localization**◇ **Lemma 60.2.1.5** ...◦ **Proof** ...

□

Finitely Generated Maximal Ideal◇ **Lemma 60.2.1.6** ...◦ **Proof** ...

□

60.2.2 Simple Modules

Definition 60.2.2.1 Let R be a ring. M be an R -module. M is **simple** if every submodule of M is either M or 0 .

◇ **Lemma 60.2.2.2** Let R be a ring and M be an R -module. The following are equivalent:

- M is simple;
- $\text{length}_R(M) = 1$;
- $M \cong R/m$ for some maximal ideal $m \subset R$.

◦ **Proof**

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-
-

□

60.3 Artinian Rings, Modules

Definition 60.3.0.1

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-

◇ **Lemma 60.3.0.2** Suppose R be is a finite dimensional algebra over a field. Then, R is artinian.

◦ **Proof ...**

□

60.4 K-Groups

Definition 60.4.0.1 Let R be a ring. We will introduce the two abelian groups associated to R :

-
- $K_0(R)$ with follwing properties
 1. For every finite projective R -module M , there is given an element $[M]$ in $K_0(R)$;
 2. For every short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$
 of finite projective R -module. Then we have $[M] = [M'] + [M'']$.
 3. The group $K_0(R)$ is generated by elements $[M]$;
 4. All relations in $K_0(R)$ are $\mathbb{Z}$ -linear combinations of the relations comming from exact sequences as above.

Chapter 61 Integrality

61.1 Integral Closure

Definition 61.1.0.1 Let S be a ring, and $R \subseteq S$ be a subring:

- Let $s \in S$, a monic polynomial

$$g = x^n + a_1 x^{n-1} + \cdots + a_n$$

with $g(s) = 0$ is called an **integral equation** for s over R .

- An element $s \in S$ is **integral** over R if there exists an integral equation for s over R .
- S is called **integral** over R if every element of S is integral over R .

More generally, we can consider the imbedding $R \hookrightarrow S$ and $s \in S$ is integral if there exists some polynomial $P \in R[X]$ with $P^\varphi(s) = 0$.

◇ **Lemma 61.1.0.2** (Nakayama's Lemma) Let $\varphi : R \rightarrow S$ be a ring map, and $y \in S$. If there exists a **finite** R -submodule M such that $1 \in M$, $yM \subseteq M$, then y is integral over R

- **Proof** Consider the map

$$M \rightarrow M \quad x \mapsto y \cdot x$$

Then, according to Nakayama's lemma, there exists a polynomial such that $P \in R[X]$, $P^\varphi(s) = 0$ □

◇ **Lemma 61.1.0.3** A finite ring map is integral.

- **Proof** Let $S = R$, then the map is finite. □

◇ **Lemma 61.1.0.4** Let $\varphi : R \rightarrow S$ be a ring map. Let $s_1, \dots, s_n$ be a finite set of elements in S . In this case, s_i is integral over R iff there exists an R -subalgebra $S' \subseteq S$ over R containing all of the s_i .

- **Proof** □

◇ **Lemma 61.1.0.5** Let $R \rightarrow S$ be a ring map. The following are equivalent:

- $R \rightarrow S$ is finite;
- $R \rightarrow S$ is integral and of finite type;
- There exists $x_1, \dots, x_n$ which generate S as an algebra over R such that each x_i is integral over R .

- **Proof** □

◇ **Lemma 61.1.0.6** Suppose $R \rightarrow S$ and $S \rightarrow T$ be integral ring maps. Then, $R \rightarrow T$ is integral.

◦ **Proof**

□

61.1.1 Integral Closure

◇ **Lemma 61.1.1.1** Let $R \rightarrow S$ be a ring homomorphism. Then, the collection

$$\bar{S} = \{s \in S \mid s \text{ is integral over } R\}$$

is an R -subalgebra of S .

◦ **Proof**

□

Definition 61.1.1.2

61.2 Normal Rings

Definition 61.2.0.1

61.3 Going Up, Going Down

Chapter 62 Associated Prime, Prime Decompositions

62.1 Supports and Annihilators

Definition 62.1.0.1 Let R be a ring and let M be an R -module. The support of M is the set

$$\text{supp}(M) := \{p \in \text{Spec}(R) \mid M_p \neq 0\}$$

62.2 Associated Prime

Definition 62.2.0.1 Let R be a ring. Let M be an R -module.

A prime p of R is **associated to** M if there exists an element $m \in M$ whose annihilator is p . i.e. $p = \{x \in R \mid xm = 0\}$ and p is prime.

Take the following notation of collections

$$\text{Ass}_R(M)$$

Remark Not every $\text{Ann}(m)$ is prime.

♣ **Example 62.2.0.2**

◇ **Lemma 62.2.0.3** Let R be a ring. M be an R -module. Then

$$\text{Ass}(M) \subseteq \text{supp}(M)$$

◦ **Proof**

□

62.2.1 Associated Prime under Some of Structures

Exact Sequence

◇ **Lemma 62.2.1.1**

◦ **Proof**

□

Filtrations

Noetherian Rings

Minimal Primes

Chapter 63 Seperable Extension, Perfect Fields

Chapter 64 Spectra

Chapter 65 Dimension

65.1 Dimension of Rings

65.1.1 Definitions: Chain, Length, Superemum, Height

Definition 65.1.1.1

- Let R be a ring. A **chain of prime ideals** is a sequence

$$p_0 \subseteq p_1 \subseteq \cdots \subseteq p_n$$

of prime ideals such that $p_i \neq p_{i+1}$;

- Define:

$$\text{length}((p_i)) := n$$

- (Krull Dimension) Let R be a ring. Then, the **superemum of integers** $n \geq 0$ such that R has a chain of prime ideals

$$p_0 \subseteq p_1 \subseteq \cdots \subseteq p_n; p_i \neq p_{i+1}$$

of length n . i.e. $\dim(R) = \sup(\text{length}(p_i))$;

- (Height) Define:

$$\text{ht}(p) := \dim(R_p)$$

◇ **Lemma 65.1.1.2** The **Krull dimension** of R is the superemum of the heights of its (maximal) primes.

- **Proof** $\dim(R) := \sup_{(p_i)} \text{length}(p_i)$, the superemum must be a chain to maximal ideal $m \in \text{Spec}_{\max}(R)$. Thus

$$\dim(R) = \sup_{(p_i), \text{end}(p_i) \in \text{Spec}_{\max}(R)} \text{length}(p_i)$$

□

◇ **Lemma 65.1.1.3** A Noetherian ring of dimension 0 is Artinian. Conversely, any artinian ring is Noetherian of dimension zero.

- **Proof**

□

◇ **Lemma 65.1.1.4** Let R be a Noetherian local ring. Then $\dim(R) = 0 \Leftrightarrow d(R) = 0$. **Remark** Notion $d(-)$: Here

- **Proof** ...

□

♠ **Proposition 65.1.1.5** Let R be a ring. The following are equivalent:

- R is Artinian;

- R is Noetherian and $\dim(R) = 0$;
- R has finite length as a module over itself;
- R is a finite product of Artinian local rings;
- R is Noetherian and $\text{Spec}(R)$ is a finite discrete topological space;
- R is a finite product of Noetherian local rings of dimension 0;
- R is a finite product of Noetherian local rings R_i with $d(R_i) = 0$;
- R is a finite product of Noetherian local rings R_i whose maximal ideals are nilpotent
- R is Noetherian, has finitely many maximal ideals and its Jacobson radical ideal is nilpotent;
- R is Noetherian and no strict inclusions among its primes;

◦ **Proof ...**

□

◇ **Lemma 65.1.1.6** Let R be a local Noetherian ring, the following are equivalent:

- $\dim(R) = 1$;
- $d(R) = 1$;
- there exists an $x \in m$, x not nilpotent such that $V(x) = \{m\}$;
- there exists an $x \in m$, x not nilpotent such that $m = \sqrt{(x)}$;
- there exists an ideal of definition generated by 1 element, and no ideal of definition is generated by 0 elements.

◦ **Proof ...**

□

♠ **Proposition 65.1.1.7** Let R be a local Noetherian ring, $d \geq 0$ be an integer, the followings are equivalent:

- $\dim(R) = d$;
- $d(R) = d$;
- there exists an ideal of definition generated by d elements, and no ideal of definition is generated by fewer than d elements;

◦ **Proof ...**

□

65.1.2 Noetherian Local Ring

Definition 65.1.2.1 Let (R, m) be a local ring. Define:

$$\dim_{\text{emb}}(R) := \dim_{\kappa(m)} m/m^2$$

◇ **Lemma 65.1.2.2** Let (R, m) be a Noetherian local ring, then

$$\dim(R) \leq \dim_{\kappa(m)} m/m^2$$

◦ **Proof** Via Nakayama's lemma, suppose $x_1, \dots, x_n$ generates $m/m^2 = M/IM$, $M = m$, $I = m$, and $m \subseteq \text{rad}(R) = \bigcap_{m' \in \text{Spec}_{\max}(R)} m' \Rightarrow x_1, \dots, x_n$ generates m , and $n = \dim_{\kappa(m)} m/m^2 \geq \dim(R)$ □

Definition 65.1.2.3 Let R be a Noetherian ring, $x \in R$:

- A **system of parameters** of R is a sequence of elements $x_1, \dots, x_d \in m$ which generates an ideal of definition of R ; 65.2.1.1
- If there exists $x_1, \dots, x_d \in m$ such that

$$m = (x_1, \dots, x_d)$$

then we call R a **regular local ring** and $x_1, \dots, x_d$ a **regular system of parameters**.

◇ **Lemma 65.1.2.4** Let (R, m) be a Noetherian local ring of dimension d

-
-

◦ **Proof ...** □

65.2 Application of Dimension Theory

◇ **Lemma 65.2.0.1** Let R be a Noetherian local domain of dimension ≥ 2 . A nonempty open subset $U \subseteq \text{Spec}(R)$ is infinite.

◦ **Proof ...** □

◇ **Lemma 65.2.0.2** A Noetherian ring with finitely many primes has dimension ≤ 1 .

◦ **Proof ...** □

◇ **Lemma 65.2.0.3** Let S be a nonzero finite type algebra over a field k . The following are equivalent:

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-
-
-
-
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-

◦ **Proof ...**

□

65.2.1 Noetherian Jacobson Rings

◇ **Lemma 65.2.1.1** (Noetherian Jacobson Rings)

- Any Noetherian domain R of dimension 1 with finitely many primes is Jacobson;
- Any Noetherian ring such that every prime p is either maximal or contained in infinitely many prime ideals is Jacobson.

◦ **Proof ...**

□

65.3 Support, and Dimension of Modules

◇ **Lemma 65.3.0.1** Let R be a Noetherian ring, and M be a finite R -module, there exists a filtration by R -modules

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$$

such that each quotient $M_i/M_{i-1} \cong R/p_i$ for some prime ideal p_i of R .

◦ **Proof ...**

□

Chapter 66 Associated Primes

66.1 Associated Primes

Definition 66.1.0.1 Let R be a ring, M be an R -module.

A prime p of R is associated to M if there exists an element $m \in M$ whose annihilator is p . The set of all such primes is denoted

$$\text{Ass}_R(M) \quad \text{or} \quad \text{Ass}(M) = \{p \in \text{Spec}(R) \mid \exists m \in M, p = \text{Ann}(m)\}$$

Remark Not every annihilator of principle element is prime. For example:

...

◇ **Lemma 66.1.0.2** Let R be a ring, M be an R -module, then

$$\text{Ass}(M) \subseteq \text{Supp}(M)$$

◦ **Proof** ...

□

66.2 Symbolic Powers

66.3 Relative Assassin

66.4 Weakly Associated Primes

66.5 Embedded Primes

Chapter 67 Hilbert-Sammuel Polynomials

67.1 Polynomials

67.1.1 Integral-Valued Polynomials

Definition 67.1.1.1 A binomial polynomial $k = 0, 1, \dots$ are:

- $Q_0(X) := 1$;
- $Q_1(X) := X$;
- $Q_k(X) := \binom{X}{k}$;

And difference operator

$$\Delta f(n) := f(n+1) - f(n)$$

Then:

$$\binom{X}{k+1} - \binom{X}{k} = \binom{X}{k-1} \Rightarrow \Delta Q_k = Q_{k-1}$$

♠ **Proposition 67.1.1.2** Let $f \in \mathbb{Q}[X]$ be a polynomial. The following properties are equivalent:

- f is $\mathbb{Z}$ -linear combination of binomial polynomial Q_k ;
- For all $n \in \mathbb{Z}$, $f(n) \in \mathbb{Z}$;
- $f(n) \in \mathbb{Z}$ for n large enough;
- Δf is $\mathbb{Z}$ -linear combination of binomial polynomials.

◦ **Proof** Clearly $(a) \Leftrightarrow (b)$; $(a) \Leftrightarrow (d)$.

- $(a) \Leftrightarrow (c) \Leftarrow$ We use induction on degree of f induced by $\deg(f) = \deg(P) - \deg(Q)$, $f = P/Q$.
 $\deg(f) = 0$ Omitted;
 Induction Steps: Apply (4).

□

Definition 67.1.1.3 If f satisfies one of the structure above, then f is called **integral-valued**. With following decomposition

$$f = \sum c_k Q_k$$

67.1.2 Polynomial-Like Functions

Definition 67.1.2.1 $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is a **polynomial-like** function if there exists some polynomial $P_f(X) \in \mathbb{Z}(X)$ such that

$$P_f(n) = f(n); n \gg 0$$

♠ **Proposition 67.1.2.2** The following conditions are equivalent:

- f is polynomial-like;
- Δf is polynomial-like;
- There exists $r \geq 0$ such that $\Delta^r f(n) = 0$ for all large enough n .

◦ **Proof** (1) $\Leftrightarrow$ (2) Omitted;

(1) $\Leftrightarrow$ (3)

- $\Rightarrow r = \deg(f)$;
- $\Leftarrow$ Integral r times.

□

A (small?) Approximation Problem

Exists $f \sim Q$? $f \sim P_f, Q \sim P_f \Rightarrow f \sim Q \Rightarrow Q \in \mathbb{Z}[X]$ failed.

67.1.3 The Hilbert Polynomials

title

Definition 67.1.3.1 Consider a graded ring $H = \bigoplus H_n$ having the following properties:

- H_0 is Artinian;
- The ring H is generated by H_0 and by a finite number $x_1, \dots, x_r \in H_1$.

Thus, H is the quotient of polynomial ring $H_0[X_1, \dots, X_r]$, with function

$$\chi(M, n) := l_{H_0}(M_n)$$

We have

♠ **Proposition 67.1.3.2** ...

◦ **Proof** ...

□

► **Theorem** (Hilbert) $\chi(M, n)$ is polynomial-like function.

◦ **Proof** ...

□

► **Theorem** Assume M_0 generates M as H -module, then:

- $\Delta^{r-1} Q(M) \leq l(M_0)$;
- The following properties are equivalent:
 - $\Delta^{r-1} Q(M) = l(M_0)$
 - $\chi(M, n) = l(M_0) \binom{n+r-1}{r-1}, n \geq 0$

– The natural map $M_0 \otimes_{H_0} H_0[X_1, \dots, X_r] \rightarrow M$ is an isomorphism.

•

◦ **Proof ...**

□

title

Definition 67.1.3.3 Let (R, m, κ) be a Noetherian local ring. Let M be a finite R -module, we define

- **Hilbert function** of M to be function

$$\varphi_M : n \mapsto \text{length}_R(m^n M / m^{n+1} M)$$

•

$$\chi_M : n \mapsto \text{length}_R(M / m^{n+1} M)$$

◇ **Lemma 67.1.3.4** It is easy to verify

$$\chi_M(n) = \sum_{i=0}^n \varphi_M(i)$$

◦ **Proof** By property of length.

□

67.1.4 The Samuel Polynomials

◇ **Lemma 67.1.4.1** Let A be a noetherian commutative ring, M be finitely generated A -modules, q is an ideal of A , then,

$$l(M/qM) < \infty \Leftrightarrow \text{Supp}(M) \cap V(q) \text{ are maximal ideals}$$

◦ **Proof ...**

□

★ **Corollary 67.1.4.2** ...

◦ **Proof ...**

□

Definition 67.1.4.3 ...

♠ **Proposition 67.1.4.4** ...

◦ **Proof ...**

□

Remark Psychology of Intuition of Hilbert's polynomial:

67.2 Noetherian Local Ring

Definition 67.2.0.1 Hilbert functions of module: Define

$$\varphi_M : n \mapsto \text{length}(m^n M / m^{n+1} M)$$

67.2.1 Ideal of Definition of R

Definition 67.2.1.1 Let (R, m) be local Noetherian ring. An ideal $I \subseteq R$ such that

$$\sqrt{I} = m$$

67.2.2 Hilbert Polynomial of M over R

Definition 67.2.2.1 ...

67.3 Poincare Series

Chapter 68 Multiplicities

68.1 Multiplicities of a Ring

68.2 Intersection Multiplicities of Two Modules

68.3 Tor-Formula

Chapter 69 Graded Rings

69.1 Graded Rings, Graded Module

69.1.1 Definitions

Definition 69.1.1.1

- Let S be a ring, endowed with a direct sum composition

$$S = \bigoplus_{d \geq 0} S_d$$

which satisfies $S_d \cdot S_e \subseteq S_{d+e}$.

- With this graded ring, define graded module

$$M = \bigoplus_{d \geq 0} M_d$$

which satisfies $S_d \cdot M_e \subseteq M_{d+e}$.

Remark We set $S_{-k} = 0, k \geq 0$.

- Especially, the **irrelevant ideal** is the ideal

$$S_+ = \bigoplus_{d > 0} S_d$$

Homogeneous Elements

Definition 69.1.1.2

- An element $x \in S$ is called **homogenous** if $x \in S_d$ for some $d \geq 0$;
- An element $x \in M$ is called **homogenous** if $x \in M_d$ for some $d \geq 0$;

Graded Ring/Module Homomorphism

Definition 69.1.1.3

- In general, let $M = \bigoplus M_d, M' = \bigoplus M'_d$ be two S -graded modules. The S -**graded map** is

$$\varphi : M \rightarrow M'$$

such that $\varphi(M_d) \subseteq M'_d$. Take the following notation:

$$GrHom_0(M, N) = \{f : M \rightarrow N \mid f \text{ is graded}\}$$

which is an S_0 -module. (The multiplication of the elements in S_0 does not change the homogeneous degree)

- Take the following notation

$$M(n)_d := M_{n+d}$$

which is called **twist** of M by n .

$$GrHom(M, N) := \bigoplus_{n \in \mathbb{Z}} GrHom_n(M, N) \quad GrHom_n(M, N) = GrHom_0(M, N(n))$$

Especially, there exists natural isomorphism

$$(M \oplus N)(n) = M(n) \oplus N(n) \quad (M \otimes_S N)(n) = M(n) \otimes_S N(n)$$

Sub-Graded Ring/Module, Quotients

As what we did for ring and module. Omitted.

69.1.2 Properties

◇ **Lemma 69.1.2.1** (Nakayama) Let S be a graded ring and M a graded S -module:

- $S^+ M = M$, M is finite $\Rightarrow M = 0$;
- $N, N' \subseteq M$ are graded submodule, $M = N + S_+ N'$, N' is finite, then $M = N$;
- If $N \rightarrow M$ is a map of graded modules,

$$N/S_+ N \rightarrow M/S_+ M$$

is surjective, M is finite, $N \rightarrow M$ is surjective;

- $x_1, \dots, x_n \in M$ are homogeneous, generate $M/S_+ M$ and M is finite. Then $x_1, \dots, x_n$ generates M .

◦ Proof

- By induction on the number of generators r : Choose the generators $y_1, \dots, y_r$ where y_i is homogeneous, $\deg(y_i) = d_i$. Now select $d_r = \min(d_i)$ by rearrangement Thus $y_r = 0$ by $d_r \geq d_r + 1$ if $y_r \neq 0$.
- Apply (1). $M/N = S_+ N'/S_+ N' \cap N \subseteq S_+(N'/N) \subseteq S_+(M/N)$ where $N'/N \subseteq M/N$ as a submodule. Thus we have $M/N = 0 \Rightarrow M = N$;
- Consider $M = \text{Im}(N \rightarrow M) + S_+ 0$ so $M = \text{Im}(N \rightarrow M)$.
- Consider the morphism

$$\oplus S(-d_i) \rightarrow M \quad (s_1, \dots, s_n) \mapsto \sum s_i x_i$$

□

Module $S^{(d)}$

Definition 69.1.2.2 Take the following notation

$$\begin{aligned} (S_n^{(d)}) &:= S_{nd} & (M_n^{(d)}) &:= M_{nd} \\ S^{(d)} &:= \bigoplus_{n \geq 0} S_{nd} & M^{(d)} &:= \bigoplus_{n \geq 0} M_{nd} \end{aligned}$$

◇ **Lemma 69.1.2.3** Let S be a graded ring, which is finitely generated over S_0 . Then for all sufficiently divisible d the algebra $S^{(d)}$ is generated in degree 1 over S_0 .

◦ Proof

□

Integral Closure

◇ **Lemma 69.1.2.4** Let $R \rightarrow S$ be a homomorphism of graded rings. $S' \subseteq S$ be the integral closure of R in S . Then

$$S' = \bigoplus_{d \geq 0} S'_d \cap S_d$$

i.e. S' is a graded R -subalgebra.

◦ **Proof**

□

Graded Free

Definition 69.1.2.5 A graded S -module M is called **graded-free** if it has a Basis made up of homogeneous elements. i.e. Homogeneous basis B with $S(B) = M$.

◇ **Lemma 69.1.2.6** (Serre, Local Algebra, P92) The followings are equivalent:

- M is graded-free;
- M is S -flat as a nongraded module;
- $Tor_1^A(M, k) = 0$;

◦ **Proof ...**

□

69.2 Proj Constructions

69.2.1 Graded Ideal

Definition 69.2.1.1 Let S be a graded ring. A **homogeneous ideal** is simply an ideal $I \subseteq S$ which is also a graded submodule of S .

Equivalently, it is an ideal generated by homogeneous elements.

Equivalently, if $f \in I$, f has a unique decomposition

$$f = f_0 + \cdots + f_n; f_i \in S_i \cap I$$

Notation Take the following notation for **homogeneous graded ideals**: Suppose S be

$$Spec^h(S) := \{p \in Spec(S) | p \text{ is graded ideal}\}$$

69.2.2 Proj Construction

Definition 69.2.2.1 Let S be a graded ring. We define:

$$Proj(S) := \{p \in Spec^h(S) | S_+ \not\subseteq p\}$$

And $Proj(S)$ is equipped with the Zariski subspace topology, as a subcollection of $Spec(S)$. It is called the **homogeneous spectrum** of the graded ring S .

It is easy to verify: $Proj(S)$ is a topological subspace.
The space is equipped with a continuous map via inclusion

$$Proj(S) \rightarrow Spec(S_0) \quad S_0 \hookrightarrow S$$

Definition 69.2.2.2 Define the **graded localization**

- Suppose S be a multiplicative homogeneous subset. Then, define:

$$(S^{-1}R)_n := \left\{ \frac{r}{s} \mid r \in R_k, s \in S \cap R_{k-n} \right\}$$

- Especially, let $f \in S$, define:

$$S_{(f)} := \left\{ \frac{r}{f^n} \mid r \in S, \deg(r) = nd, d = \deg(f) \right\}$$

$$M_{(f)} := \left\{ \frac{r}{f^n} \mid \deg(r) = nd, r \in M, d = \deg(f) \right\}$$

◇ **Lemma 69.2.2.3** Let S be a $\mathbb{Z}$ -graded ring containing a homogeneous invertible element of positive degree. Then,

$$G = \{ \mathbb{Z}\text{-graded primes of } S \} \rightarrow Spec(S_0)$$

is a homeomorphism.

◦ **Proof**

□

Definition 69.2.2.4

- For $f \in S$ homogeneous of degree > 0 , define:

$$D_+(f) := \{ p \in Proj(S) \mid f \notin p \}$$

- For homogeneous ideal $I \subseteq S$, define

$$V_+(I) := \{ p \in Proj(S) \mid I \subset p \}$$

Especially: For set $E \subseteq S$, take notion $V_+(E) := V_+((E))$.

◇ **Lemma 69.2.2.5** (Topology of Proj) Let $S = \bigoplus_{d \geq 0} S_d$

- $D_+(f)$ is open in $Proj(S)$;
- $D_+(ff') = D_+(f) \cap D_+(f')$;
- Let $g = g_0 + \cdots + g_m$ be an element of S with $g_i \in S_i$,

$$D(g) \cap Proj(S) = (D(g_0) \cap Proj(S)) \cup \bigcup_{i \geq 1} D(g_i)$$

- Let $g_0 \in S_0$ be a homogeneous element of degree 0, then

$$D(g_0) \cap Proj(S) = \bigcup_{f \in S_d, d \geq 1} D_+(g_0 f)$$

- The open set $D_+(f)$ form a basis for the topology of $Proj(S)$;

- Let $f \in S$ be homogeneous of positive degree. The ring S_f is a natural $\mathbb{Z}$ -grading. The ring maps

$$S \rightarrow S_f \leftarrow S_{(f)}$$

induce homeomorphisms:

$$D_+(f) \leftarrow \{\mathbb{Z}\text{-graded primes of } S_f\} \rightarrow \text{Spec}(S_{(f)})$$

- There exists an S such that $\text{Proj}(S)$ is not quasi-compact;
- The sets $V_+(I)$ is closed;
- Any closed subset $T \subseteq \text{Proj}(S)$ is of the form $V_+(I)$ for some homogeneous ideal $I \subseteq S$;
- For any graded ideal $I \subseteq S$ we have $V_+(I) = \emptyset$ iff $S_+ \subseteq \sqrt{I}$;

◦ **Proof**

□

♣ **Example 69.2.2.6** Let R be a ring, $S = R[X]$ with $\deg(X) = 1$. Then, the map

$$\text{Proj}(S) \rightarrow \text{Spec}(R)$$

is a bijection in fact a homeomorphism. Namely, suppose $p \in \text{Proj}(S)$.

◇ **Lemma 69.2.2.7** Let S be a graded ring. M be a graded S -module and $p \in \text{Proj}(S)$.

Let $f \in S$ be a homogeneous element of positive degree such that $f \notin p$ i.e. $p \in D_+(f)$.

Let $p' \subseteq S_{(f)}$ be the element of $\text{Spec}(S_{(f)})$ corresponding to p as lemma above, then

$$S_{(p)} = (S_{(f)})_{p'} \quad M_{(p)} = (M_{(f)})_{p'}$$

◦ **Proof ...**

□

◇ **Lemma 69.2.2.8** ...

◦ **Proof ...**

□

Localization

◇ **Lemma 69.2.2.9** Let S be a graded ring, M be a graded S -module. Let p be an element of $\text{Proj}(S)$, let $f \in S$ be a homogeneous element of positive degree such that

$$f \notin p \text{ or } p \in D_+(f)$$

Let $p' \subseteq S_{(f)}$ be the element of $\text{Spec}(S_{(f)})$ corresponding to p

◦ **Proof ...**

□

Prime Avoidance

Finite Type Algebra

69.3 Noetherian Graded Rings

69.3.1 Noetherian Graded Ring Criterion

◇ **Lemma 69.3.1.1** Let S be a graded ring. A set of homogeneous elements $f_i \in S_+$ generates S as an algebra over S_0 iff they generate S_+ as an ideal of S .

◦ **Proof ...**

□

◇ **Lemma 69.3.1.2** A graded ring S is Noetherian iff S_0 is Noetherian and S_+ is finitely generated as an ideal of S .

◦ **Proof**

• $\Rightarrow$

• $\Leftarrow$

□

69.3.2 Numerical Polynomial

Definition 69.3.2.1 Let A be an abelian group. We say that a function $f : n \mapsto f(n) \in A$ defined for all sufficient large integers n is a **numerical polynomial** if there exists $r \geq 0$, elements $a_0, \dots, a_r \in A$ such that

$$f(n) = \sum_{i=0}^r \binom{n}{i} a_i, n \gg 0$$

◇ **Lemma 69.3.2.2** ...

◦ **Proof ...**

□

◇ **Lemma 69.3.2.3** Suppose $f : n \mapsto f(n) \in A$ defined for all n sufficiently large and suppose that

$$n \mapsto f(n) - f(n-1)$$

is a numerical polynomial, then f is a numerical polynomial.

◦ **Proof ...**

□

◇ **Lemma 69.3.2.4** If M is a finitely generated graded S -module, and if S is finitely generated over S_0 , then each M_n is finite S_0 -module.

◦ **Proof ...**

□

♠ **Proposition 69.3.2.5** Suppose S be a Noetherian graded ring and M a finite graded S -module. Consider the following function

$$\mathbb{Z} \rightarrow K'_0(S_0) \quad n \mapsto [M_n]$$

If S_+ is generated by elements of degree 1, then this function is a numerical polynomial.

◦ **Proof** Apply induction on minimal generators of S_1 :

□

Remark

♣ **Example 69.3.2.6** (Hilbert Polynomial for Syzygies)

◇ **Lemma 69.3.2.7** ...

◦ **Proof** ...

□

69.3.3 Noetherian Local Rings

Look here: 65.2

Chapter 70 Blow-Up Algebras/Rees Algebras

Reference: Vasconcelos Integral Closure: Rees Algebras, Multiplicities, Algorithms

Definition 70.0.0.1 Let R be a ring, $I \subseteq R$ be an ideal:

- The **blow up algebra**, or **Rees algebra** associated to the pair (R, I) is the graded R -algebra
- Let $a \in I$ be an element. Denote

Chapter 71 Ext Groups and Free Resolutions

Remark

Chapter 72 Polynomials

72.1 Noether's Normalization Theorem

► **Theorem** (Noether's Normalization Theorem) Let k be an infinite field, and

$$A = K[a_1, \dots, a_n]$$

is a finitely generated algebra. Then, there exists a finite $y_1, \dots, y_m, m \leq n$.

- $y_1, \dots, y_m$ are algebraically independent;
- A is a finite $k[y_1, \dots, y_m]$ -algebra.

◦ **Proof**

□

72.2 Hilbert's Basis Theorem

► **Theorem** (Hilbert) Suppose that R be a Noetherian ring, so is $R[X]$.

◦ **Proof** Let $I \subseteq R[X]$ be an arbitrary ideal. We need to show: I is finitely generated. Define the collection

$$J_i = \{a_i \in R \mid \exists a_0, \dots, a_{i-1} \in R, \sum_{k=0}^i a_k X^k \in I\}$$

Then, J_i is an R -ideal:

For $\alpha, \beta \in J_i, \gamma \in R$, then they have correspondent polynomials $f, g \in I$. And

$$\begin{aligned} f + g \in I &\Rightarrow \alpha + \beta \in J_i \\ \gamma g \in I &\Rightarrow \gamma \beta \in J_i \end{aligned}$$

J_i are an ascending chain ideals: $\alpha \in J_i$ with $f \in I, \text{Lead}(f) = \alpha$. Then $fX \in I \Rightarrow \alpha \in J_{i+1}$. Thus,

$$J_i = (a_{n,1}, \dots, a_{n,r})_R$$

Let r be the maximal number of generating elements of J_i from 0 to n . Suppose $a_{n,1}, \dots, a_{n,r}$ are correspondent to the polynomials $f_{n,1}, \dots, f_{n,r} \in R[X]$, then define

$$I' := (f_{i,j} \mid i = 0, \dots, n; j = 1, \dots, r) \subseteq I$$

We will prove: $I' = I$. Suppose $g = \sum b_i X^i \in I$, with $\deg(g)$. Consider the degree of g : Apply induction

- $\deg(g) < n$: $b_d \in J_d$, there exists representation

$$b_d = \sum_{i=1}^r r_i a_{d,i}$$

Then,

$$\hat{f} = f - \sum_{i=1}^r r_i f_{d,i} \in I \Rightarrow f \in I$$

- $g \geq n$: Similarly.

Thus, $I = I'$. □

★ **Corollary 72.2.0.1** Suppose R be a Noetherian ring, so is $R[X_1, \dots, X_n]$.

- **Proof** Notice,

$$R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$$

□

★ **Corollary 72.2.0.2** Suppose R be a Noetherian ring, then any of finitely algebra over R is Noetherian. In particular, every **affine algebra** is Noetherian.

- **Proof** Consider the quotient. □

◇ **Lemma 72.2.0.3** Let K be a field, then $K[X]$ is Noetherian.

- **Proof** K can be seen as ring with only finite chain of length 1. □

★ **Corollary 72.2.0.4** The polynomial ring $K[X_1, \dots, X_n]$ is Noetherian.

- **Proof** As above. □

Remark This is a fundamental result, that the space A_K^n is Noetherian.

72.3 Hilbert's Nullstellensatz

◇ **Lemma 72.3.0.1**

- If A is an integral domain and algebraic over K , then A is a field;
- Let k be a field and K a field extension which is finitely generated as a k -algebra, then K is algebraic over k . Equivalently, for a field A , which is contained in a k -affine domain, then A is algebraic.

- **Proof**

-
-

□

► **Theorem**

- **Proof**

□

► **Theorem**

- **Proof**

□

◇ **Lemma 72.3.0.2**

- **Proof**

□

72.3.1 Hilbert's Nullstellensatz, and Applications

► **Theorem** (Hilbert) Suppose that

- **Proof**

□

72.3.2 Textual Research: The Original Version by Hilbert

Chapter 73 Completion

73.1 Completion

Definition 73.1.0.1 Suppose R be a ring and I be an ideal. We define

- the **completion** of R with respect to I to be the limit
-

73.2 Completion for Noetherian Rings

73.3 Taking Limits of Modules

Chapter 74 Flatness

74.1 Flat Modules and Flat Ring Maps

Results

-
-

Definition 74.1.0.1

-
-
-
-

◇ Lemma 74.1.0.2

- Proof

□

74.2 Characterizing Flatness

74.3 Criteria for Flatness

Chapter 75 Differentials and Smoothness

75.1 Differentials

75.1.1 Derivative

Definition 75.1.1.1 Suppose $\varphi : R \rightarrow S$ be a ring map (thus, S is an R -module), and M be an S -module. An R -derivation is a map

$$D : S \rightarrow M$$

such that

- D is additive: $D(a + b) = D(a) + D(b)$;
- Annihilates elements of $\varphi(R)$:

$$D(\varphi(r)) = 0, r \in R$$

- Leibnitz's Rule:

$$D(ab) = aD(b) + bD(a)$$

Take the following notion of $Der_R(S, M)$ for such $D \in Hom_S(S, M)$.

And notice that,

$$D(ra) = D(\varphi(r)a) = \varphi(r) \cdot D(a)$$

Thus, an equivalent definition is:

- $D : S \rightarrow M$ is R -linear;
- D satisfies Leibnitz's rule.

◇ **Lemma 75.1.1.2** $Der_R(S, M)$ is an S -submodule of $Hom_S(S, M)$

Especially, for S -module map $\alpha : M \rightarrow N$, consider the composition $\alpha \circ D \in Der_R(S, N)$. Thus, $M \mapsto Der_R(S, M)$ is covariant.

◦ **Proof ...**

□

75.1.2 Kahler Differentials

Definition 75.1.2.1 Consider the following map

$$\begin{aligned} \bigoplus_{(a,b) \in S^2} S([a, b]) \oplus \bigoplus_{(f,g) \in S^2} S([f, g]) \oplus \bigoplus_{r \in R} S[r] &\rightarrow \bigoplus_{a \in S} S[a] \\ [(a, b)] &\mapsto [a + b] - [a] - [b] \\ [(f, g)] &\mapsto [fg] - g[f] - f[g] \\ [r] &\mapsto [\varphi(r)] \end{aligned}$$

And then, define $Coker(\bigoplus_{(a,b) \in S^2} S([a, b]) \oplus \bigoplus_{(f,g) \in S^2} S([f, g]) \oplus \bigoplus_{r \in R} S[r] \rightarrow \bigoplus_{a \in S} S[a])$, and a morphism

$$d : S \rightarrow Coker$$

Such that $d(a) = \overline{[a]} \in Coker$ which implies

$$d(a + b) = da + db \quad d(fg) = fdg + gdf \quad d(\varphi(r)) = 0; a, b, f, g \in S, r \in R$$

which is element of $Der_R(S, \Omega_{S/R} = coker)$

The pair $(\Omega_{R/S}, d)$ is called **the module of Kahler differentials** or the **module of differentials** of S over R .

◇ **Lemma 75.1.2.2** The module of differentials of S over R has the following universal property: The map

$$Hom_S(\Omega_{S/R}, M) \rightarrow Der_R(S, M)$$

is an isomorphism of functors.

◦ **Proof ...**

♠ **Proposition 75.1.2.3** Especially, if $R \rightarrow S$ is surjective, then $\Omega_{S/R} = 0$.

◦ **Proof ...**

★ **Corollary 75.1.2.4** ...

◦ **Proof ...**

Universal Property

♠ **Proposition 75.1.2.5** Given $f \in Der_R(S, M)$, there exists factorization

$$\begin{array}{ccc} \Omega_{B/A}^1 & \xrightarrow{\quad} & M \\ \uparrow \exists! & \nearrow & \\ S & & \end{array}$$

◦ **Proof ...**

Kahler Differentials-Zariski Construction

Definition 75.1.2.6 Equivalently: Consider the kernel of

$$I := Ker(S \otimes_R S \rightarrow S)$$

Then,

$$\Omega_{S/R} \cong I/I^2$$

With differential map

$$S \rightarrow \Omega_{S/R} \quad a \mapsto [1 \otimes a - a \otimes 1]$$

Now verify the equivalence between the two definitions:

- $I/I^2 \rightarrow \text{Coker}$: Define the following map

$$\beta : I \rightarrow \text{Coker} \quad \sum s_i \otimes t_i \mapsto \sum s_i dt_i$$

Then, for elements in I^2 , take

$$\begin{cases} \omega_1 = 1 \otimes a - a \otimes 1 \\ \omega_2 = 1 \otimes b - b \otimes 1 \end{cases}$$

We have

$$\begin{aligned} \beta(\omega_1 \omega_2) &= \beta((1 \otimes a - a \otimes 1)(1 \otimes b - b \otimes 1)) \\ &= \beta(1 \otimes (ab) - b \otimes a - a \otimes b + ab \otimes 1) \\ &= d(ab) - bda - adb + abd(1) \\ &= 0 \end{aligned}$$

- $\text{Coker} \rightarrow I/I^2$: $d : S \rightarrow \Omega_{S/R}$ is a derivative. With:

–
–

Limits

◇ **Lemma 75.1.2.7** Let I be a directed set, a system of $(R_i \rightarrow S_i, \varphi_{ii'})$ be a system of ring maps over I , then we have

$$\Omega_{S/R} = \text{colim}_i \Omega_{S_i/R_i}$$

where $R \rightarrow S = \text{colim}(R_i \rightarrow S_i)$.

◦ **Proof ...**

□

◇ **Lemma 75.1.2.8**

◦ **Proof ...**

□

Sequence

♠ **Proposition 75.1.2.9** Consider ring map $A \rightarrow B \rightarrow C$, there exists a canonical exact sequence:

...

of C -modules.

◦ **Proof ...**

□

Localization, Quotients

title

75.2 De Rham Complex

Recall:

$$\Omega_{B/A}^i := \wedge_B^i(\Omega_{B/A}); i \geq 0$$

Then:

Definition 75.2.0.1 The **de Rham complex** of B over A is the complex of $((\Omega_{B/A}^i), d)$ with the chain map $d : \Omega_{B/A}^i \rightarrow \Omega_{B/A}^{i+1}$ constructed as following:

$$d : \Omega^p \rightarrow \Omega^{p+1} \quad f_0 df_1 \wedge \cdots \wedge df_p \mapsto df_0 \wedge df_1 \wedge \cdots \wedge df_p$$

Verify the chain condition

◦ **Proof**

$$\Omega_{B/A}^0 \longrightarrow \Omega_{B/A}^1 \longrightarrow \Omega_{B/A}^2 \longrightarrow \cdots$$

$$\Omega^0 \longrightarrow \Omega^1 \longrightarrow \Omega^2 \longrightarrow \cdots$$

□

75.3 Finite Order Differential Operators

Definition 75.3.0.1 Let $R \rightarrow S$ be a ring map, M, N be S -modules. Let $k \geq 0$ be an integer. We inductively define a **differential operator**

$$D : M \rightarrow N$$

of order k to be an R -linear map such that $m \mapsto D(gm) - gD(m)$ is a differential operator of order $k - 1$ for all $g \in S$.

Especially: The differential operator of order 0 is an S -linear map.

Denote as $\text{Diff}^k(M, N)$.

Verify the properties:

- By induction, $g \in S, gD(fm) - fgD(m)$ is differential operator of order $k - 1$;
- Apply induction, $(D + D')(gm) - g(D + D')(m)$ is differential operator of order $k - 1$;

$$\text{Diff}^0(M, N) \subseteq \text{Diff}^1(M, N) \subseteq \cdots$$

Take the following notion

$$\text{Diff}_{S/R}^k(M, N)$$

♠ **Proposition 75.3.0.2** Let $R \rightarrow S$ a ring map, L, M, N be S -modules.

If $D : L \rightarrow M$ and $D' : M \rightarrow N$ are differential operators of order k and k' , then $D' \circ D$ is a differential operator of order $k + k'$.

◦ **Proof ...**

□

♠ **Proposition 75.3.0.3** Let $R \rightarrow S$ be a map. Let M be an S -module. Then, there exists an S -module $P_{S/R}^k(M)$ and a canonical isomorphism

$$\text{Diff}_{S/R}^k(M, N) = \text{Hom}_S(P_{S/R}^k(M), N)$$

Functorial in the S -module N .

Via Yoneda's lemma, this tells us: A 'universal' differential operator

$$D_{\text{univ}} : M \rightarrow P_{S/R}^k(M)$$

such that: For each $D : M \rightarrow N$, which is a , there exists a unique factorization $\alpha : P_{S/R}^k(M) \rightarrow N$ such that

$$D = \alpha \circ D_{\text{univ}}$$

◦ **Proof ...**

□

75.3.1 Modules of Principle Part

Definition 75.3.1.1 Such $P_{S/R}^k(M)$ is called the **sequence of principal parts** of order k of M , or sometimes **Jet module**.

♣ **Example 75.3.1.2** ...

Exact Sequence induced by $R \rightarrow S$

◇ **Lemma 75.3.1.3** Let $R \rightarrow S$ be a ring map, M be an S -module. There is a canonical short exact sequence

$$0 \rightarrow \Omega_{S/R} \otimes_S M \rightarrow P_{S/R}^1(M) \rightarrow 0$$

functorial in M .

◦ **Proof ...**

□

Tensor Product

♠ **Proposition 75.3.1.4** ...

◦ **Proof ...**

□

75.4 The Naive Cotangent Complex

75.5 Smoothness

75.5.1 Formally Smooth Maps

Definition 75.5.1.1 ...

75.5.2 Smoothness and Differentials

75.5.3 Smooth Algebra over Fields

75.5.4 Smooth Ring Maps in the Noetherian Case

75.5.5 Formal Smoothness of Fields

75.6 Etale Ring Maps

Chapter 76 Some of Rings

76.1 Local complete intersection Rings

Definition 76.1.0.1 Let k be a field, S be a finite type k -algebra:

- We say that S is a **global complete intersection** over k if there exists a presentation

$$S = k[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

such that: $\dim(S) = n - c$.

- We say that S is a **local complete intersection** over k if there exists a covering

$$\operatorname{Spec}(S) = \cup D(g_i)$$

such that each of the ring D_g is a global intersection over k .

◇ Lemma 76.1.0.2 ...

76.1.1 Properties

76.1.2 Topology and Homology

76.1.3 Normal Bundle

76.1.4 Cohen-Macaulay Property

76.2 Syntomic Morphisms

76.3 Gorenstein Rings

76.4 Regular Local Rings

76.5 Cohen-Macaulay Rings, Modules

76.6 Universal Catenary Rings

76.7 Henselian Local Rings

Chapter 77 Etale Map

Chapter 78 Japanese Rings, Nagata Rings

Chapter 79 Groebner Bases

Problems

- The ideal description problem: Does every ideal $I \subseteq k[X_1, \dots, X_n]$ has a finite generating set

$$\oplus k[X_1, \dots, X_n] f_i \rightarrow I \rightarrow 0 \text{ is exact} \quad \text{or} \quad I = \langle f_1, \dots, f_r \rangle$$

- The ideal membership problem. Given $f \in K[X_1, \dots, X_n]$ and an ideal $I = \langle f_1, \dots, f_s \rangle$. Determine if

$$f \in I$$

or weaker version, we should determine if $V(f_1, \dots, f_s) \subseteq V(f)$.

- Solving the polynomial equation

$$\begin{cases} f_1(X_1, \dots, X_n) = 0 \\ \dots \\ f_s(X_1, \dots, X_n) = 0 \end{cases}$$

Equivalently, asking the points in affine variety $V(f_1, \dots, f_s)$.

- The Implicitization Problem: Let V be a subset of k^n , parametrically is

$$\begin{aligned} x_1 &= g_1(t_1, \dots, t_m) \\ &\dots \\ x_n &= g_n(t_1, \dots, t_m) \end{aligned}$$

If: g_i is polynomial/rational function. Then, V is a variety/part of one variety. Find a family of polynomial equations defines the variety.

Consider some fundamental cases:

- ♣ **Example 79.0.0.1** $n = 1$: Then, $K[X]$ is PID. It has Euclidean algorithm, to show

$$I = (g)$$

Those problems has a simple description.

- ♣ **Example 79.0.0.2** Degree $\deg(f_i) = 1$, linear algebra problem.
Also the implicitization problem, $\deg(g_i) = 1$.

79.1 Ordering of Monomials

Take the following notation

$$x^\alpha = x^{\alpha_1} \dots x^{\alpha_n}; \alpha \in \mathbb{Z}_{\geq 0}^n$$

we want to define an ordering, such that

$$x_i^{m+1} > x_i^m \quad x_1 > \cdots > x_n$$

And the ordering is totally ordered. One and only one of

$$x^\alpha > x^\beta \quad x^\alpha = x^\beta \quad x^\alpha < x^\beta$$

should be true.

Operations: $x^\alpha > x^\beta \Rightarrow x^\alpha x^\gamma > x^\beta x^\gamma$.

Definition 79.1.0.1 A **monomial ordering** on $k[x_1, \dots, x_n]$ is any relation $>$ on $\mathbb{Z}_{\geq 0}^n$, or on x^α , which satisfies

1. $>$ is total order;
2. $\alpha > \beta \Rightarrow \alpha + \gamma > \beta + \gamma$;
3. $>$ is well-ordered.

◇ **Lemma 79.1.0.2** $>$ is well-ordering iff every strictly decreasing sequence in $\mathbb{Z}_{\geq 0}^n$

$$\alpha(1) > \alpha(2) > \dots$$

eventually terminates.

◦ **Proof**

- $\Rightarrow$
- $\Leftarrow$

□

Definition 79.1.0.3 (Lexicographic Order) Let $\alpha, \beta \in \mathbb{Z}_{\geq 0}^n$. We say:

$$\alpha >_{lex} \beta \Leftrightarrow \beta - \alpha \in \mathbb{Z}^n, \text{ the leftmost nonzero element is positive}$$

♠ **Proposition 79.1.0.4** The lex ordering on $\mathbb{Z}_{\geq 0}^n$ is a monomial ordering.

◦ **Proof**

□

Definition 79.1.0.5 (Graded Lex Order)

Definition 79.1.0.6 (Graded Reverse Lex Order)

Definition 79.1.0.7 Let $f = \sum_{\alpha} a_{\alpha} x^{\alpha}$ be a nonzero polynomial in $k[x_1, \dots, x_n]$. And let $>$ be a monomial order:

-
-
-

•

◇ **Lemma 79.1.0.8** Let $f, g \in k[X_1, \dots, X_n]$ be nonzero polynomials. Then:

- $\text{Multideg}(fg) = \text{multideg}(f) + \text{multideg}(g)$;
- If $f + g \neq 0$. Then, $\text{multideg}(f + g) \leq \max \text{multideg}(f), \text{multideg}(g)$

◦ **Proof**

□

79.2 A Division Algorithm for Polynomial Rings $k[X_1, \dots, X_n]$

79.3 Hilbert's Nullstellensatz, and Groebner's Base

79.4 Properties of Groebner's Bases

79.5 Buchberger's Algorithm

Chapter 80 Elimintation Theory

Part XXX

Valuation Theory

Chapter 81 Valuation Rings

Definition 81.0.0.1 Valuation rings:

- Let K be a field with $A, B \subseteq K$ local rings. We say that B **dominates** A if $A \subseteq B$ and $m_A = A \cap m_B$;
- Let A be a ring. We say A is a **valuation ring** if A is a local domain and if A is maximal for the relation of domination among local rings contained in the fraction field of A ;
- Let A be a valuation ring with fraction field K . If $R \subseteq K$ is a subring of K , then we say A is **centered** on R if $R \subseteq A$.

◇ **Lemma 81.0.0.2** (Existence of Valuation Ring) Let K be a field and $A \subseteq K$ be a local ring. Then there exists a valuation ring with fraction field K dominating A .

◦ **Proof** It is suffice to show: For collection of totally ordered collection of local subrings □

◇ **Lemma 81.0.0.3** Let A be a valuation ring, then A is a normal domain.

◦ **Proof** ... □

◇ **Lemma 81.0.0.4** Let A be a valuation ring with maximal ideal m and fraction field K , and $x \in K$. Then, either $x \in A$ or $x^{-1} \in A$ or both.

◦ **Proof** ... □

◇ **Lemma 81.0.0.5** ...

◦ **Proof** ... □

81.1 Valuation

◇ **Lemma 81.1.0.1** Let A be a valuation ring with field of fraction K . The set

...

◦ **Proof** ... □

Definition 81.1.0.2 Let A be a valuation ring:

- The totally ordered abelian group $(\Gamma, \geq)$ above is called **the value group** of valuation ring A ;
- Map $v : A - \{0\} \rightarrow \Gamma$ and $v : K^* \rightarrow \Gamma$ is called valuation associated to A ;

- The valuation ring A is called **discrete valuation ring** if $\Gamma \cong \mathbb{Z}$.

◇ **Lemma 81.1.0.3** ...

- **Proof** ...

□

Chapter 82 Valuation Theory

Chapter 83 Ultrametric Space

Part XXXI

Representation of Finite Groups

Part XXXII

Algebraic Representation Theory

Part XXXIII

Noncommutative Ring Theory

Preparations:

- Rings;
- Modules;
- (Associative) Algebras;
- Field Extension;

Reference

- T.Y.Lam
- Noncommutative Noetherian Ring, J.C.McConnell

Chapter 84 Free Modules, Projective and Injective Modules

84.1 Free Modules

84.1.1 IBN

Category of Left/Right Modules

Definition 84.1.1.1

-
-

Free Module

◇ Lemma 84.1.1.2 (Generation Lemma)

◦ Proof ... □

★ Corollary 84.1.1.3

◦ Proof ... □

IBN Properties

Definition 84.1.1.4 ...

84.1.2 Stable Finiteness

Definition 84.1.2.1 Recall: A ring is **Dedekind-finite** if

$$c, d \in S, cd = 1 \Rightarrow dc = 1$$

A ring R is **stably finite** if the matrix $M_n(R)$ are Dedekind-finite.

Remark Dedekind finiteness: In $M_n(F)$, we have

$$AB = I \Rightarrow BA = I$$

With:

...

But for $M_\omega(F)$, a counterexample is

...

♠ Proposition 84.1.2.2 The following properties of R are equivalent:

- R is stably finite;
- For any n , $R^n \cong R^n \oplus N \Rightarrow 0$ in $\mathfrak{m}_R$;
- For any n , any epimorphism $R^n \rightarrow R^n$ in $\mathfrak{m}_R$ is an isomorphism.

◦ **Proof ...**

□

♠ **Proposition 84.1.2.3** For any nonzero element R , stable finiteness implies IBN. But not conversely.

◦ **Proof ...**

□

♠ **Proposition 84.1.2.4** ...

◦ **Proof ...**

□

84.1.3 The Rank Condition

Definition 84.1.3.1

-
-

♠ **Proposition 84.1.3.2** ...

◦ **Proof ...**

□

84.1.4 The Strong Rank Condition

84.2 Projective Modules

84.2.1 Definitions, Examples

Definition 84.2.1.1 ...

84.3 Injective Modules

Chapter 85 Flat Modules and Homological Dimension

85.1 Flat and Faithfully Flat Modules

Definition 85.1.0.1 ...

Chapter 86 Jacobson Radical of Noncommutative Rings

Definition 86.0.0.1 (Jacobson Radical) Define: The Jacobson radical

$$\text{rad}(R) = \bigcap_{I \in L} I \quad L = \{I \subseteq R \mid I \text{ is the left ideal of } R\}$$

◇ **Lemma 86.0.0.2** For $y \in R$, the following statements are equivalent:

- $y \in \text{rad}(R)$;
- $1 - xy$ is left-invertible for any $x \in R$;
- $yM = 0$ for any simple left R -module M .

◦ **Proof**

-
-
-

□

Annihilators

Definition 86.0.0.3 ...

Jacobson Semisimple

Nilpotent

86.1 Jacobson Radical Under Change of Rings

86.2 Group Ring and the J-Semisimplicity

Chapter 87 Division Algebra

Reference

- A First Course in Noncommutative Rings, T.Y.Lam, Chapter 5;
-

87.1 Finite Dimensional Division Algebra

87.1.1 Discussion of Complex Numbers, Quaternion and Frobenius's Theorem

Introduction We have the extension of field

$$\mathbb{R} \rightsquigarrow \mathbb{C} \quad \dim_{\mathbb{R}}(\mathbb{C}) = 2$$

Moreover, we want to find some $\mathbb{R}$ -algebras, and be able to divide numbers.

Definition 87.1.1.1 A unital algebra D in which every nonzero element is invertible is called a division algebra.
i.e. for every pair $a \in D - \{0\}$, such x, y exists.

$$ax = 1 \quad ya = 1$$

A question is:

Is it possible to define multiplication on an n -dimensional real space so that it becomes a real division algebra?

Standard Results:

-
-
-

Assume: D is an n -dimensional $\mathbb{R}$ -algebra.

◇ **Lemma 87.1.1.2** For every $x \in D$, there exists a $\beta \in \mathbb{R}$ such that

$$x^2 + \beta x \in \mathbb{R}$$

◦ **Proof ...**

□

Define:

$$V := \{v \in D \mid v^2 \leq 0\}$$

◇ **Lemma 87.1.1.3** If $u, v \in V$ are linearly independent, then so are u, v and 1.

◦ **Proof ...**

□

Jordan Product

Definition 87.1.1.4 (Jordan Product) For $x, y \in D$, we write

$$x \circ y := xy + yx$$

Frobenius's Theorem

► **Theorem** (Frobenius's Theorem) A **finite dimensional division algebra** D is isomorphic to $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.

87.2 Structure of Finite Dimensional Algebras

Chapter 88 Wedderburn-Artin-Theory

88.1 Simple, Semisimple Modules

Definition 88.1.0.1

- A nonzero module M is **simple** or **irreducible** if it contains no proper non-zero submodule.
- A module M is called **semisimple** or **completely module** R -module if M is a direct sum of simple modules.

Definition 88.1.0.2

An R -module is **cyclic** with generator $m \in M$ such that

$$M = Rm$$

88.1.1 Simple Modules

Definition 88.1.1.1

An **irreducible representation** of an algebra A is defined as a representation

$$\varphi : A \rightarrow \text{End}_F(V)$$

which contains no proper nonzero subspace W of V that is invariant under $\varphi(a), a \in A$.

Equivalently, the associated module with structure $A \times V \rightarrow V, (a, v) \mapsto \rho(a)(v)$. The module V is irreducible.

♠ Proposition 88.1.1.2

The following are equivalent for R -module M :

- M is simple;
- M is cyclic and every nonzero element is a generator;
- $M \cong R/I$ for some maximal left ideal I ;

◦ Proof

- (1) $\Rightarrow$ (2) M is simple. Then for any $m \in M$, either $Rm = 0$ or $Rm = M$. Take

$$N := \{m \in M \mid Rm = 0\}$$

Then, N is a submodule, $N = 0$. This implies, that for every $m \neq 0$, $Rm = M$;

- (2) $\Rightarrow$ (3) Take

$$\phi : R \rightarrow Rm \quad r \mapsto rm$$

And $I := \text{Ker}(\phi)$. I is a submodule of R , and $R/I \cong Rm = M$.

I must be maximal. If not,

- (3) $\Rightarrow$ (1) R/I contains no submodule.

□

♣ **Example 88.1.1.3** Let D be a division ring and V is vector space over D . Then D is simple $\Leftrightarrow \dim_D(V) = 1$.

♣ **Example 88.1.1.4** Let V be a nonzero vector space over division ring D .

V can be regard as a module over $R = \text{End}_D(V)$, via $fv := f(v)$. We claim that V is a simple R -module: For each pair $v, w \in V$ with $v \neq 0$, there exist some f such that $f(v) = w$. This is true.

♣ **Example 88.1.1.5** Let R be a ring and L be an ideal. Then L is simple L -module iff L is minimal. Omitted.

Minimal Left Ideals

◇ **Lemma 88.1.1.6** Let R be a simple ring having a minimal left ideal L . Then every simple R -module is isomorphic to L .

◦ **Proof** ...

□

Maximal Left Ideal

Definition 88.1.1.7 A left ideal U of a ring R is called a **maximal left ideal** if $U \neq R$ and U is not properly contained in a proper left ideal.

♣ **Example 88.1.1.8** Let $R = M_n(D)$, with D a division ring. And let

$$U = RE_{11} \oplus \cdots \oplus RE_{n-1,n-1}$$

It is easy to check: U is a maximal left ideal of R .

And:

$$R = U \oplus L$$

where $L = RE_{nn}$ is the minimal left ideal. Especially we have $R/U \cong L$.

◇ **Lemma 88.1.1.9** Let R be a unital ring. Then every proper left ideal L of R is contained in a maximal left ideal.

◦ **Proof** Let $\mathcal{S}$ be the set of all proper left ideals of R contains L ,

$$S = \{I \mid I \text{ is left ideal and } L \subseteq I\}$$

Now, let $\{K_i \mid i \in I\}$ be a chain in S . Then

$$K = \cup K_i$$

is its upper bound. K is a left ideal, $1 \notin K \Rightarrow K \neq R$. By Zorn's lemma,

□

◇ **Lemma 88.1.1.10** Let R be a ring. Every simple R -module is isomorphic to R/U for some maximal left ideal U of R .

Conversely, if U is a maximal ideal of R such that $R^2 \not\subseteq U$. Then R/U is simple R -module.

◦ **Proof** ...

□

★ **Corollary 88.1.1.11** Every nonzero unital ring has a simple module.

◦ **Proof ...**

□

88.1.2 Semisimple Modules

♣ **Example 88.1.2.1** ...

◇ **Lemma 88.1.2.2** If M is the sum of a family of simple submodules $\{M_i | i \in I\}$. Then, there exists a subset $K \subseteq I$ such that

$$M = \bigoplus_{k \in K} M_k$$

◦ **Proof ...**

□

★ **Corollary 88.1.2.3** If a module is the sum of a family of simple submodules, then it is semisimple.

◦ **Proof Omitted.**

□

◇ **Lemma 88.1.2.4** The following statements are equivalent for a nonzero module M :

- M is semisimple;
- Every submodule of M is a direct summand and $Rm \neq 0$ for every nonzero $m \in M$;

◦ **Proof ...**

□

★ **Corollary 88.1.2.5** A **unital** module M is semisimple iff every submodule of M is a direct summand.

◦ **Proof Omitted.**

□

Endomorphisms Rings of Semisimple Modules

► **Theorem** (Schur's Lemma) If M is a simple R -module. Then the endomorphism ring $End_R(M)$ is a division ring.

◦ **Proof** Let $f \in End_R(M)$ and $f \neq 0$. Then, $ker(f)$ is a submodule of M , $Ker(f) = 0$. Then, $f \in Aut_R(M)$. f is invertible. There exists some $f^{-1} \in Aut_R(M)$.

□

A slightly more general version:

- A nonzero homomorphism from an arbitrary module to a simple module is surjective, and;
- A nonzero homomorphism from a simple module to an arbitrary module is injective.

◇ **Lemma 88.1.2.6** If M, N are simple modules, when they are not isomorphic, there are no nonzero homomorphisms from one to another.

◦ **Proof ...**

□

Semisimple Rings

88.1.3 Chain Conditions

88.2 Wedderburn-Artin Theorem

Part XXXIV

Serre's Problem on Projective Modules(Quillen,Suslin)

Part XXXV

Lie Algebra, and its Representation Theory

Part XXXVI

Algorithms

Part XXXVII

Matroid Theory

Part XXXVIII

Graph Theory

Chapter 89 Fundamental of Concepts in Graph Theory

89.1 Graphs

89.1.1

Definition 89.1.1.1

- (Directed Graph)
- (Undirected Grpah)

89.1.2 Vertexes

Definition 89.1.2.1 ...

89.1.3 Matrix Representation

89.2 Path, Circle and Connectedness

Definition 89.2.0.1

- A walk in a graph G is a sequence as following:

...

where $e_i = (x_i, x_{i+1})$ or $e_i = \{x_i, x_{i+1}\}$

- A path
- A cycle in a graph G is a

89.2.1 Connectedness

Undirected Graph

◇ **Lemma 89.2.1.1** Suppose G

◦ **Proof** ...

□

Definition 89.2.1.2 ...

♠ **Proposition 89.2.1.3** ...

◦ **Proof** Take the negation G is disconnected $\Leftrightarrow$ There exists some collection X such that

□

Then, the component for graphs is well-defined

Definition 89.2.1.4 Let G be a undirected graph, then,

$$V(G) = \cup V(G_i)$$

Directed Graph

♠ **Proposition 89.2.1.5** ...

◦ **Proof** ...

□

Definition 89.2.1.6 ...

Euler Tour

Definition 89.2.1.7

-
-

♠ **Proposition 89.2.1.8** A connected graph has one and only one Eulertour, when the graph is Euler.

◦ **Proof**

- $\Rightarrow$
- $\Leftarrow$

□

89.2.2 Leaf, Tree, Forest

Tree, Leaves

Definition 89.2.2.1 Suppose following graphs are undirected.

- (Tree) A
 - Connected
 - Undirected
 - with no cycle

graph is called **tree**.

- The vertice of grad 1 is called **leaf**.

◇ **Lemma 89.2.2.2** The following proposition holds: For undirected graph G , we have

- Every tree with at least two vertices contains at least 2 leaves;
- Suppose G be a tree with
 - n Vertices
 - m edges

- p connected components.

Then we have

$$n = m + p$$

◦ **Proof**

-
-

□

♠ **Proposition 89.2.2.3** The followings are equivalent:

-
-
-
-
-
-
-

◦ **Proof ...**

□

Arborezenence

Definition 89.2.2.4 ...

Minimal Spanned Tree

Definition 89.2.2.5 ...

89.2.3 Graph

Chapter 90 Optimal Tree and Path

90.1 Minimum Spanning Tree Problem

Kruskals Algorithms

Prim Algorithms

90.2 Graph Traversal

Definition 90.2.0.1

- A **connected and undirected graph**, which contains no circle, is called **tree**.
- A **undirected graph**, which contains no circle, is called **forest**;
- The vertex e , such that $|\delta(e)| = 1$, is called **leaf**, representing endpoints.

◇ **Lemma 90.2.0.2** The following propositions hold:

- Every tree with at least 2 vertexes contains 2 leaves;
- Suppose G be a forest with n vertexes, m edges and p components. Then:

$$n = m + p$$

◦ **Proof ...**

□

♠ **Proposition 90.2.0.3** Suppose G be a undirected graph with n vertexes. The following are equivalent

- G is a tree;
- G has $n - 1$ vertexes and contains no circle;
- G has $n - 1$ vertexes and is connected;
- G is connected and by deleting a vertex, we get a disconnected graph;
- G is a graph with $\delta(X) \neq \emptyset$, for every $\emptyset \subseteq X \subseteq V(G)$, and the property is destroyed by deleting one vertex;
- G is a forest.
- Every two path in G is connected by a unique path;

◦ **Proof ...**

□

Definition 90.2.0.4

- A directed graph G is called **arborescence**, if:
 - At every vertex $e \in E(G)$, the $\delta^+(e) \leq 1$;
 - The correspondent undirected graph is a tree.

- In an arborescence, there exists one and only one of vertex such that $\delta^-(r) = \emptyset$, which is called the **root**;
- As a dual of concept forest, such directed graph is called **Silvanescence**.

Graph Traversal Algorithm

♠ Proposition 90.2.0.5 ...

- Proof ...

□

DFS

BFS

Distance

Definition 90.2.0.6 (Unweighted) For two vertexes x, y in a graph G , the distance $dist(x, y)$ is the minimal number of $x - y$ path.

If there exists no $x - y$ path, then $dist(x, y) := \infty$.

90.3 Weighted Graph

Definition 90.3.0.1 A **weighted graph** is a graph G with a function

$$c : E(G) \rightarrow \mathbb{R}$$

which is called the **weight of edge**.

Let $H \subseteq G$ be a subgraph. Then,

$$c(E(H)) := \sum_{e \in E(H)} c(e)$$

is called the **weight** of subgraph H .

Distance

A main problem is: **The shortest path problem**.

Dijkstra's Algorithm

Definition 90.3.0.2

- Input: Weighted graph (G, c) with $s \in V(G)$.
- Output: Arboriculture $A = (R, T)$ in G , such that R contains all , and

$$disc_{(G,c)}(s, v) = disc_{(A,c)}(s, v); \forall v \in R$$

The Pseudocode is given here: 1

Algorithm 1 Dijkstra's Algorithm**Require:** Weighted Graph (G, c) , Knot $s \in V(G)$;**Ensure:** Arborescences $A = (R, T)$ in G , which contains all , and satisfy $dist_{G,c}(s, v) = dist_{(A,c)}(s, v)$ $R \leftarrow 0; Q \leftarrow \{s\}; l(s) \leftarrow 0 \quad \backslash \backslash R$: The ensured vertex with shortest path; Q :**while** $Q \neq \emptyset$: **do** Select $v \in Q$ with $l(v)$ smallest; $Q \leftarrow Q - \{v\}; R \leftarrow R \cup \{v\}$; $\backslash \backslash$ To find out the neighbors of v ; **for** $e = (v, w) \in \delta_G^+(v)$ with $w \notin R$ **do** **if** $w \notin Q$ or $l(v) + c(e) < l(w)$ **then** $l(w) \leftarrow l(v) + c(e); p(w) \leftarrow v; Q \leftarrow Q \cup \{w\}$; $\backslash \backslash$ If we find a shorter path from s to w via v , $\backslash \backslash$ then: $l(w) := l(v) + c(e); p(w) = v$, the previous edge; $Q := Q \cup \{w\}$, extending the **end if** **end for****end while** $T \leftarrow \{p(v) | v \in R - \{s\}\}$;♠ **Proposition 90.3.0.3** Dijkstras Algorithmus is correct. And has a

$$\mathcal{O}(n^2 + m)$$

where $n = |V(G)|, m = |E(G)|$.◦ **Proof ...**

□

Remark**Remark****Definition 90.3.0.4** ...

Chapter 91 Flow in Network

91.1 Network

Definition 91.1.0.1 ...

Part XXXIX

Combinatorics

Part XL

Set Theory

Part XLI

Category Theory

Chapter 92 Fundamentals

92.1 Definitions

92.1.1 Categories

92.1.2 Functors

92.1.3 Limits

92.2 Adjunctions

Chapter 93 Presheaves, Yoneda Lemma

93.1 Presheaves

Definition 93.1.0.1 Let C be a category:

- A **presheaf** is a functor

$$\mathfrak{F} : C^{op} \rightarrow Set$$

93.1.1 Yoneda Embedding

Definition 93.1.1.1 Define:

$$Y : C \rightarrow Psh(C)$$

With Hom -functor $Hom : C^{op} \times C \rightarrow Set$, there exists adjunction

◇ **Lemma 93.1.1.2** ...

◦ **Proof** ...

□

♠ **Proposition 93.1.1.3** Yoneda functor is:

- fully faithful;
- Preservation and Reflection the Isomorphisms;
- Dense(Density);
- Interaction With Density Comonads;
- Interaction With Codensity Monads;

◦ **Proof** ...

□

93.1.2 Yoneda's Lemma

► **Theorem** (Yoneda) Let $\mathcal{F} : C^{op} \rightarrow Set$ be a presheaf. We have following bijection

$$Nat(h_A, \mathcal{F}) \cong \mathcal{F}(A)$$

Natural in $A \in Obj(C)$. Especially, determining a natural isomorphism of functors

$$Nat(h_-, \mathcal{F}) \cong \mathcal{F}$$

◦ **Proof** ...

□

93.1.3 Properties of Categories of Presheaves

♠ Proposition 93.1.3.1 ...

◦ Proof ...

□

93.2 Representable Functors

93.2.1 Generalized Element

- <https://ncatlab.org/nlab/show/generalized+element>
- https://www.voutsadakis.com/TEACH/LECTURES/CATEGORY/Chapter4.pdf?utm_source=chatgpt.com P31

In category Set , the element can be seen as the maps

$$x : 1 \rightarrow A; A \in Ob(Set)$$

And in general

Definition 93.2.1.1 Let: C be an category and $A \in C$. Then, a morphism

$$x : S \rightarrow A$$

is a generalized element of A of shape S .

Especially, the generalized element is associated to the Yoneda embedding

$$Y : C \hookrightarrow [C^{op}, Set] \quad X \mapsto GenEl(X)$$

Under this embedding, $GenEl(X) : C^{op} \rightarrow Set, Ob(C) \ni U \mapsto (U \rightarrow X)$.

93.3 Copresheaf

93.4 Restsricted Yoneda Embedding, Yoneda Extension

Part XLII

Topos

Part XLIII

∞ -Categories(Kerodon)

Part XLIV

Logic

Part XLV

Model Theory

Part XLVI

Real Algebraic Geometry

Part XLVII

Recursion Theory

Part XLVIII

Sheaf Theory

Part XLIX

General Topology

Part L

Homological Algebra

Contents

Chapter 94 Projective and Injective Modules, Free Modules

94.1 Free Modules

Definition 94.1.0.1

Universal Properties

♠ **Proposition 94.1.0.2** Suppose R be a ring, M be $R\text{-Mod}$ and $I \in \text{Set}$ be index, then we have a correspondence

$$\text{Hom}_{\text{Mod-}R}(R^{\oplus I}, M) \simeq \text{Hom}_{\text{Set}}(I, M)$$

◦ **Proof ...**

□

94.2 Projective Modules

Definition 94.2.0.1 An object P of a category C is a **projective object** if it has the **left lifting property** against epimorphisms.

This means, that P is projective if for any morphism $f : P \rightarrow B$ and any epimorphism $q : A \rightarrow B$, f factors through

$$\begin{array}{ccc} & A & \\ \exists \nearrow & \downarrow \forall q \in \text{Epi} & \\ P & \xrightarrow{\forall f} & B \end{array}$$

An equivalent way to say to say this is that

Definition 94.2.0.2 An object P is projective precisely if the *hom-functor* $\text{Hom}(P, -)$ preserves epimorphisms.

With property of epimorphism, $P \rightarrow A$ is unique up to isomorphism.

♠ **Proposition 94.2.0.3** The followings are equivalent in Set :

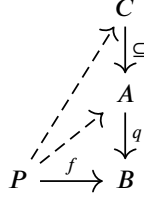
- Every object is projective;
- The axiom of choice holds.

◦ **Proof** $\Rightarrow$ In ZF, $AC \Leftrightarrow$ Every surjection $f : A \xrightarrow{B}$ has a section structure

$$g \circ f = id_A$$

We take $f = id_B, P = B$.

$\Leftarrow$ If AC holds, then we could give a function $P \rightarrow A$. Now we associate $f^{-1}(b)$ with $\bar{b} \in f^{-1}(b)$ by AC . Thus we could select $g : P \rightarrow \{\bar{b} | b \in B\}$ such that $f(x) = q(g(b))$.



□

Remark If we replace category of sets by another base topos will behave essentially precisely like set, but in general will not validate the axiom of choice.

Homological algebra in such a more general context is abelian sheaves, and the theory is the theory of cohomology of abelian sheaf.

A main task is the resolution of R -mod.

Remark Although mostly, we have a forgetful functor $\mathcal{F} \rightarrow \text{Sets}$, but the functor is not full, so not preserve the functions.

Definition 94.2.0.4 The **projective module** over R is the projective objects in $R - \text{Mod}$.

♠ **Proposition 94.2.0.5** The following properties are equivalent:

- P is projective module;
- If $g : B \rightarrow C$ is surjective, then

$$g_* : \text{Hom}(P, B) \rightarrow \text{Hom}(P, C)$$

is also surjective;

- For short exact sequence $0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$ Then,

$$0 \longrightarrow \text{Hom}_R(P, A) \longrightarrow \text{Hom}(P, B) \longrightarrow \text{Hom}(P, C) \longrightarrow 0$$

is exact;

- Short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is split exact;
- P is a direct item of free modules, i.e. there exist some V such that

$$F = P \oplus V$$

where F is free.

◦ **Proof**

-
-
-
-

□

Free Modules are Projective

♠ **Proposition 94.2.0.6** Let R be a ring. Every free R -module is projective.

◦ **Proof** Consider diagram

...

□

94.2.1 Sum

◇ **Lemma 94.2.1.1** ...

◦ **Proof** ...

□

♠ **Proposition 94.2.1.2** (Criterion of Projective Property for Sum) Suppose $P = \sum P_i$, then
 P is projective $\Leftrightarrow P_i$ is projective.

◦ **Proof** ...

□

★ **Corollary 94.2.1.3** The direct sum of projective modules is projective.

◦ **Proof** Omitted.

□

94.2.2 Tensor

♠ **Proposition 94.2.2.1**

- Suppose P be a projective right R -module, then: $P \otimes_R - : {}_R M \rightarrow Ab$ is an exact functor.
- Similarly for M_R .

◦ **Proof** ...

□

Flatness

Definition 94.2.2.2 ...

94.2.3 Projective Basis Theorem

94.2.4 Projective Resolution

Definition 94.2.4.1 Let M be R -module, then a **left resolution** is a complex $P_i = 0, i < 0$ with a map $\epsilon : P_0 \rightarrow M$ and the complex

$$\cdots \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

is exact. An resolution is **projective** iff P_i is projective.

♠ **Proposition 94.2.4.2** Every R -module M has a projective resolution.

More generally, if an abelian category $\mathcal{A}$ has enough projectives, then every object M in $\mathcal{A}$ has a projective resolution.

◦ **Proof** Consider the following constructions: By selecting $M_{i-1} := \ker(P_i \rightarrow P_{i-1})$ with new projective resolution $P_{i+1} \rightarrow M_i$. Now consider the compose

$$\begin{array}{ccccccc}
 & & P_{i+1} \rightarrow M_i \rightarrow P_i & & & & \\
 & & \nearrow & & \searrow & & \\
 P_1 & \xrightarrow{\quad} & P_0 & \longrightarrow & M & \longrightarrow & 0 \\
 & \searrow & \nearrow & & & & \\
 & & M_0 & & & & \\
 & \nearrow & \searrow & & & & \\
 0 & & & & 0 & &
 \end{array}$$

□

Comparison Theorem

► **Theorem** Let $P \xrightarrow{\epsilon} M$ be a projective resolution of M and $f' : M \rightarrow N$ be a map in $\mathcal{A}$. Then, for every resolution $Q \xrightarrow{\eta} N$ there is a chain map $f \rightarrow Q$ lifting f' in the sense of

$$\eta \circ f_0 = f' \circ \epsilon$$

$$\begin{array}{ccccccc}
 \dots & \longrightarrow & P_1 & \longrightarrow & P_0 & \xrightarrow{\epsilon} & M \longrightarrow 0 \\
 & & \downarrow & & \downarrow f_0 & & \downarrow f' \\
 \dots & \longrightarrow & Q_1 & \longrightarrow & Q_0 & \xrightarrow{\eta} & N \longrightarrow 0
 \end{array}$$

◦ **Proof** ...

□

Horseshow Lemma

► **Theorem** Suppose given a commutative diagram

$$\begin{array}{ccccccc}
 & & & & 0 & & \\
 & & & & \downarrow & & \\
 \dots & P'_2 & \xrightarrow{P_2} & P'_1 & P'_0 & A' & 0 \\
 & & & & & \downarrow & \\
 & & & & & A & \\
 \dots & P'_2 & & P'_1 & P'_0 & A' & 0 \\
 & & & & & \downarrow & \\
 & & & & & 0 &
 \end{array}$$

where the column is exact and the row are projective resolutions. Then, set

$$P_n := P'_n \oplus P''_n$$

The right hands column lift an exact sequence of complexes

$$0 \rightarrow P' \xrightarrow{i} P \xrightarrow{\pi} P'' \rightarrow 0$$

◦ **Proof ...**

□

94.3 Injective Modules

Definition 94.3.0.1 The **injective object** in $\mathcal{A}$ satisfies the universal lifting property: Given an injection $f : A \rightarrow B$ and a map $\alpha : A \rightarrow I$ there exists at least one map $\beta : B \rightarrow I$ such that $\alpha = \beta \circ f$

$$\begin{array}{ccccc} 0 & \longrightarrow & A & \xrightarrow{f} & B \\ & & \downarrow \alpha & \swarrow \beta & \\ & & I & & \end{array}$$

We say $\mathcal{A}$ has enough injectives if for every object A in $\mathcal{A}$, there is an injection $A \rightarrow I$ with I injective.

♠ **Proposition 94.3.0.2** In category *Sets*, every nonempty object is injective.

◦ **Proof ...**

□

Product

◇ **Lemma 94.3.0.3** If I_α is a family of injectives, so is $\prod I_\alpha$.

◦ **Proof ...**

□

94.3.1 Baer's Criterion

► **Theorem** (Baer) A right R -module $E \in \text{Mod} - R$ is injective iff for every right ideal J of R , every map $J \rightarrow E$ can be extended to a map

$$R \rightarrow E$$

★ **Corollary 94.3.1.1** ...

◦ **Proof ...**

□

Abelian Groups

★ **Corollary 94.3.1.2** $\mathcal{A}b$ has enough injectives.

◦ **Proof ...**

□

94.3.2 Embedding Theorem

94.3.3 Injective Resolution

Definition 94.3.3.1 ...

Comparison Theorem

Adjoint Functors

◇ **Lemma 94.3.3.2** For every right R -module M the natural map

$$\tau : \text{Hom}_{\text{Ab}}(M, A) \rightarrow \text{Hom}_{\text{mod-}R}(M, \text{Hom}_{\text{Ab}}(R, A))$$

is an isomorphism, where

◦ **Proof** ...

□

Definition 94.3.3.3 A pair of functions $L : \mathcal{A} \rightarrow \mathcal{B}$ and $R : \mathcal{B} \rightarrow \mathcal{A}$ are adjoint if there is a natural bijection for all $A \in \mathcal{A}$ and $B \in \mathcal{B}$ such that

$$\tau : \text{Hom}_{\mathcal{B}}(L(A), B) \xrightarrow{\cong} \text{Hom}_{\mathcal{A}}(A, R(B))$$

Natural: For $f : A \rightarrow A'$ and $g : B \rightarrow B'$,

$$\begin{array}{ccccc} \text{Hom}_{\mathcal{B}}(L(A'), B) & & \text{Hom}_{\mathcal{B}}(L(A), B) & & \text{Hom}_{\mathcal{B}}(L(A), B') \\ & & & & \\ \text{Hom}_{\mathcal{A}}(A', R(B)) & & \text{Hom}_{\mathcal{A}}(A, R(B)) & & \text{Hom}_{\mathcal{A}}(A, R(B')) \end{array}$$

★ **Corollary 94.3.3.4** ...

◦ **Proof** ...

□

♠ **Proposition 94.3.3.5**

◦ **Proof** ...

□

★ **Corollary 94.3.3.6** ...

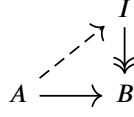
◦ **Proof** ...

□

94.4 Projective and Injective Objects in Abelian Category

94.4.1 Projective Object

Definition 94.4.1.1 Object $X \in \mathcal{C}$ is **projective** if it satisfies the left lifting property for **epimorphism**.



♠ **Proposition 94.4.1.2** Let $X \in \mathcal{A}$, $\mathcal{A}$ be an abelian category. The followings are equivalent:

- X is projective.
- $\text{Hom}(X, -)$ is exact.

◦ **Proof**

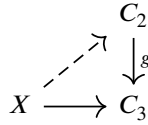
- $\Rightarrow$ Consider the exact sequence

$$0 \longrightarrow C_1 \xrightarrow{f} C_2 \xrightarrow{g} C_3 \longrightarrow 0$$

Then the morphisms induced by the morphisms above

$$0 \longrightarrow \text{Hom}(X, C_1) \xrightarrow{f^*} \text{Hom}(X, C_2) \xrightarrow{g^*} \text{Hom}(X, C_3) \longrightarrow 0$$

Injective easy to verify, and surjective given by the diagram



$\text{Im}(f) = \text{Ker}(g)$ from exactness of $\text{Hom}(X, -)$.

- $\Leftrightarrow$ If $\text{Hom}(X, -)$ is exact, now consider the following exact sequence

$$0 \rightarrow \text{Ker}(g) \rightarrow A \rightarrow B \rightarrow 0$$

And apply $\text{Hom}(X, -)$, surjectivity of $\text{Hom}(X, A) \rightarrow \text{Hom}(X, B)$.

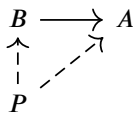
□

Note Without the projective property, $\text{Hom}(X, -)$ is just left-exact. See discussion here:

Projective Resolution

94.4.2 Injective Object

Definition 94.4.2.1 An object is **injective** if it satisfies the right lifting property for **monomorphism**



Preservation of Injective Objects

◇ **Lemma 94.4.2.2** Given a pair of additive adjoint functors

$$(L)$$

between abelian categories such that the left adjoint L is a **left exact functor**, then the right adjoint preserve injective objects.

◦ **Proof** ...

□

♠ **Proposition 94.4.2.3** ...

◦ **Proof** ...

□

Existence of Enough Injectives

Given some categories with enough injectives:

-
-

◇ **Lemma 94.4.2.4** ...

◦ **Proof** ...

□

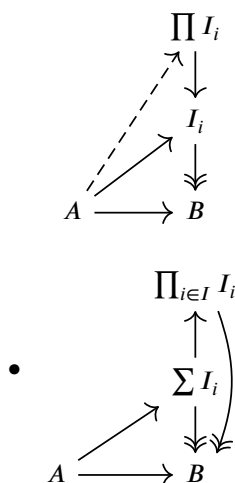
Constructions

♠ **Proposition 94.4.2.5**

- The (small) product of injective objects is injective;
- The direct sum of finite injective objects is injective;

◦ **Proof**

- Consider the following diagram



□

And the injectivity of arbitrary direct sum can be judged with Baer's criterion.

Injective Resolution

Definition 94.4.2.6 An **injective resolution** is a cochain complex $J^* \in Ch^*(\mathcal{A})$ equipped with a **quasi-isomorphism**

$$i : X \xrightarrow{\sim} J^*$$

such that $J^n \in \mathcal{A}$ is an injective object for all $n \in \mathbb{N}$. i.e. Quasi-isomorphism

$$\begin{array}{ccccccc} X & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 & \longrightarrow & \dots \\ \downarrow & & \downarrow & & & & \downarrow & & \\ J^0 & \longrightarrow & J^1 & \longrightarrow & \dots & \longrightarrow & J^n & \longrightarrow & \dots \end{array}$$

♠ **Proposition 94.4.2.7** If $\mathcal{A}$ has enough injectives, then every $X \in \mathcal{A}$ has an injective resolution.

◦ **Proof** We have

□

♠ **Proposition 94.4.2.8** ...

◦ **Proof** ...

□

Injective Resolution

Definition 94.4.2.9 For $X \in \mathcal{A}$ an object, an **injective resolution** is a cochain complex $J^* \in Ch^*(\mathcal{A})$ equipped with a quasi-isomorphism

$$i : X \xrightarrow{\sim} J^*$$

such that $J^n \in \mathcal{A}$ is an **injective object** for all $n \in \mathbb{N}$.

$$\begin{array}{ccccccc} X & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 & \longrightarrow & \dots \\ \downarrow i^0 & & \downarrow & & & & \downarrow & & \\ J^0 & \xrightarrow{d^0} & J^1 & \xrightarrow{d^1} & \dots & \longrightarrow & J^n & \xrightarrow{d^n} & \dots \end{array}$$

Projective Resolution

Definition 94.4.2.10 For $X \in \mathcal{A}$ is an object, a **projective resolution** of X is a chain complex $J_* \in Ch_*(\mathcal{A})$ equipped with a quasi-isomorphism

$$p : J_* \xrightarrow{\sim} X$$

such that $J_n \in \mathcal{A}$ is a projective object for all $n \in \mathbb{N}$.

$$\begin{array}{ccccccc} \dots & \longrightarrow & J_n & \longrightarrow & \dots & \longrightarrow & J_1 & \longrightarrow & J_0 \\ \downarrow & & \downarrow & & & & \downarrow & & \downarrow \\ \dots & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 & \longrightarrow & X \end{array}$$

♠ **Proposition 94.4.2.11** Let $\mathcal{A}$ be an abelian category with enough injectives, such as $R - \text{Mod}$ for some ring R , then every object $X \in \mathcal{A}$ has an injective resolution.

◦ **Proof ...**

□

♠ **Proposition 94.4.2.12** Let $\mathcal{A}$ be an abelian category with enough projectives (such as $R\text{-Mod}$ for some ring R). Then every object $X \in \mathcal{A}$ has a projective resolution.

◦ **Proof ...**

□

94.4.3 Resolutions, and Ext^1

Suppose $\mathcal{A}$ be an abelian category.

Left Resolution

Definition 94.4.3.1 A left resolution of $M \in \mathcal{A}$ consists of:

- Chain complex A_* ;
- Morphism $\epsilon : A_0 \rightarrow M$;
- Exactness

$$\cdots \rightarrow A_2 \rightarrow A_1 \rightarrow A_0 \xrightarrow{\epsilon} M \rightarrow 0$$

Free Resolution

Definition 94.4.3.2 Let $M \in \mathcal{A}$. A free resolution

Conclusions

♠ **Proposition 94.4.3.3** Every finitely generated abelian group admits a free resolution with length 2.

◦ **Proof** Take the free object generated by

□

Some of contents will be given in: Syzygies.

Chapter 95 Chain Complexes

95.1 Some of Category Theory

95.1.1 Ab-Category

Definition 95.1.1.1 A category $\mathcal{A}$ is called an $\mathcal{A}b$ -category if every hom -set $\text{Hom}_{\mathcal{A}}(A, B)$ in $\mathcal{A}$ is given the structure of an abelian group in such a way that composition distributes over addition.

In particular, given a diagram in $\mathcal{A}$ of the form

$$A \xrightarrow{f} B \begin{array}{c} \xrightarrow{g'} \\ \xrightarrow{g} \end{array} C \xrightarrow{h} D$$

We have

$$h(g + g')f = hgf + hg'f$$

95.1.2 Preadditive Category

(Stack Project, Homological Algebra, 3)

Definition 95.1.2.1 A category $\mathcal{A}$ is **preadditive** if each morphism set $\text{Mor}_{\mathcal{A}}(x, y)$ is endowed with a structure of an abelian group such that:

- Each composition

$$\text{Mor}(x, y) \times \text{Mor}(y, z) \rightarrow \text{Mor}(x, z)$$

$$\text{is bilinear}((f + g) \circ h = f \circ h + g \circ h, f \circ (g + h) = f \circ g + f \circ h)$$

Additive functor

Definition 95.1.2.2

- An additive functor $F : \mathcal{B} \rightarrow \mathcal{A}$ between $\mathcal{A}b$ -categories $\mathcal{B}$ and $\mathcal{A}$ is a functor such that each

$$\text{Hom}_{\mathcal{B}}(B', B) \rightarrow \text{Hom}_{\mathcal{A}}(F(B'), F(B))$$

is a group homomorphism;

◇ **Lemma 95.1.2.3** Let $\mathcal{A}$ be a preadditive. Let $x \in \text{Ob}(\mathcal{A})$, the following are equivalent:

- x is an initial object;
- x is a final object;
- $\text{id}_x = 0 \in \text{Mor}_{\mathcal{A}}(x, x)$

Furthermore, if such 0 exists, then: A morphism $\alpha : y \rightarrow z$ factors through 0 $\Leftrightarrow \alpha = 0$

◦ **Proof ...**

□

Such object is called **zero object** in preadditive category.

Remark Existence:

◇ **Lemma 95.1.2.4** (Isomorphism of Finite Product, Coproduct) In preadditive category



We have $p_A \circ f = id_A$, $f \circ p_A = id_{A \times 0}$ □

◇ **Lemma 95.1.3.3** (Product Map) If $f \in Mor_{\mathcal{A}}(B, C)$, $g \in Mor_{\mathcal{A}}(D, E)$, then we have

$$f \times g : B \times D \rightarrow C \times E$$

◦ **Proof** Consider following diagram

$$\begin{array}{ccc} B & \longrightarrow & C \\ \uparrow & & \uparrow \\ B \times D & \dashrightarrow & C \times E \\ \downarrow & & \downarrow \\ D & \longrightarrow & E \end{array}$$

□

♠ **Proposition 95.1.3.4** If $\mathcal{A}$ is an additive category, then the product exists, with

$$A \times B = A \sqcup B$$

◦ **Proof** Firstly, $A \times B$ has following mappings $\iota_A : A \cong A \times 0 \rightarrow A \times B$ (B resp.); And then for $p_A : P \rightarrow A$, $p_B : P \rightarrow B$ we have

$$\iota_A p_A + \iota_B p_B : P \rightarrow A \times B$$

Thus, $A \times B$ is the coproduct. □

Also take the following notion

$$A \times B \cong A \oplus B$$

95.1.4 Karoubian Categories

(Stack Project, HA, 4)

Definition 95.1.4.1 Let $\mathcal{C}$ be a preadditive category.

We say $\mathcal{C}$ is **Karoubian** if every idempotent endomorphism ($f^2 = f$) of an object of $\mathcal{C}$ has a kernel.

$$f^2 = f \Rightarrow \exists \ker(f), (0 \rightarrow \ker(f) \rightarrow x \rightarrow y)$$

♠ **Proposition 95.1.4.2** (Equivalent Conditions for Karoubian Categories) The followings are equivalent:

- $\mathcal{C}$ is Karoubian;
- Each idempotent endomorphism of an object C has a cokernel;
- Given an idempotent endomorphism $p : z \rightarrow z$, there exists a direct sum decomposition $z = x \oplus y$ such that: p corresponds to the projection onto y

◦ **Proof**

- (1) $\Rightarrow$ (2)
- (2) $\Rightarrow$ (3)
- (3) $\Rightarrow$ (1)

□

♠ **Proposition 95.1.4.3** (Criteria for Karoubian Category) Let $\mathcal{D}$

- If $\mathcal{D}$ has countable products and kernels of maps which have a right inverse, then Karoubian; $\mathcal{D}$
- If $\mathcal{D}$ has countable ciproduct and cokernels of maps which have a left inverse, then Karoubian; $\mathcal{D}$

◦ **Proof**

-
-

□

Warning Preadditive Category $\subset$ Karoubian Category $\subset$ Additive Category

♣ **Example 95.1.4.4** Category of free modules.

♣ **Example 95.1.4.5** Category *Set*.

95.1.5 Abelian Category

Definition 95.1.5.1 Define: For $f \in \text{Mor}_{\mathcal{A}}(B, C)$,

- (Kernel) The **kernel** is the universal object given by $i : \text{Ker}(f) \rightarrow B$ with $fi = 0$
- (Cokernel) The **cokernel** is the universal object given by $j : C \rightarrow \text{Coker}(f)$ with $jf = 0$.

And:

- f is mono iff $fg = fh \Rightarrow g = h$
- f is epi iff $gf = hf \Rightarrow g = h$

Then,

Definition 95.1.5.2 The **abelian category** satisfies:

- f is Mono/injective $\Leftrightarrow \text{Ker}(f) = 0$;
- f is Epi/surjective $\Leftrightarrow \text{Coker}(f) = 0$;

Define:

$$\begin{aligned} \text{Coim}(f) &:= \text{Coker}(\text{Ker}(f) \rightarrow B) \\ \text{Im}(f) &:= \text{Ker}(C \rightarrow \text{Coker}(f)) \end{aligned}$$

Remark Sometimes take the notion:

- $\text{inj.} : \text{Ker}(f) = 0$;

- $\text{sur.} : \text{Coker}(f) = 0$;

♠ **Proposition 95.1.5.3** Category $\mathcal{A}$ is abelian iff it is additive, all kernels and cokernels exist, and the natural map

$$\text{Coim}(f) \rightarrow \text{Im}(f)$$

is an isomorphism for all morphisms f of $\mathcal{A}$.

◦ **Proof** Notice: $\text{Coim}(f) = \text{Coker}(\text{Ker}(f) \rightarrow B) = \text{Ker}(C \rightarrow \text{Coker}(f)) = \text{Im}(f)$, now consider following property

$$\text{Ker}(f) \rightarrow B \rightarrow \text{Coim}(f) = \text{Im}(f) \rightarrow C \rightarrow \text{Coker}(f)$$

Then, equivalent by universal property. Omitted. □

We could define the exactness:

Definition 95.1.5.4 For abelian category $\mathcal{A}$, we could define the **exactness**

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

With $\text{Im}(f) = \text{Ker}(g)$.

◇ **Lemma 95.1.5.5** Let: $\mathcal{A}$ be a preadditive category.

The addition on sets of morphisms make $\mathcal{A}^{opp}$ into a preadditive category.

Furthermore, $\mathcal{A}$ is preadditive/additive/abelian $\Leftrightarrow \mathcal{A}^{op}$ is preadditive/additive/abelian ;

◦ **Proof** ... □

◇ **Lemma 95.1.5.6** (Finite Limimts, Colimits) The finite limits, colimits exist in the abelian category.

◦ **Proof**

- Limit
- Colimit

□

95.1.6 Pullback, Pushout Diagram

Definition 95.1.6.1 The **intersection/pullback** of objects $x, y \in \mathcal{A}$, which satisfies the following diagram

$$\begin{array}{ccc} x \cap y & \xrightarrow{p_1} & x \\ \downarrow p_2 & \lrcorner & \downarrow \\ y & \hookrightarrow & z \end{array}$$

And the **cointersection/pushout**

$$\begin{array}{ccc} z & \twoheadrightarrow & x \\ \downarrow & & \downarrow \\ y & \twoheadrightarrow & x \cup_z y \end{array}$$

95.1.7 Lemmas

3-Lemma

Snake Lemma

♠ Proposition 95.1.7.1 ...

◦ Proof ...

□

4-Lemma

5-Lemma

95.1.8 Grothendieck Category

<https://ncatlab.org/nlab/show/additive+and+abelian+categories>

<https://ncatlab.org/nlab/show/Grothendieck+category>

Definition 95.1.8.1 (Grothendieck)

- (AB1) Existence of kernel,cokernel;
- (AB2) Isomorphism $Coim(f) \rightarrow Im(f)$, where $Coim(f) = Coker(Ker(f) \rightarrow A)$, $Im(f) = Ker(A \rightarrow Coker(f))$;
(AB1)+(AB2)=Abelian
- (AB3) An abelian category with all coproducts; Hence all colimits;
- (AB4) AB3 Satisfied, coproduct of monomorphism is monic;
- (AB5) AB3 Satisfied, filtered limits of exact sequences exist and are exact;
- (AB6) AB3 Satisfied, such that for increasing directed set J , $(B^j)_{j \in J}$ and families $B^j = (B_i^j)_{i \in I_j}$ (Thus, $B = ((B_i^j)_{i \in I_j})_{j \in J}$) then, we have

$$\cap_{j \in J} (\sum_{i \in I_j} B_i^j) = \sum_{(i_j) \in \prod I_j} (\cap_{j \in J} B_{i_j}^j)$$

- (AB3*) DF
- (AB4*) DF
- (AB5*) DF
- (AB6*) DF

95.1.9 Extension

Definition 95.1.9.1

- An **extension** of A by B in abelian category $\mathcal{A}$ is an (short) exact sequence

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

- The morphism between extensions: For short exact sequences $0 \rightarrow A \rightarrow E \rightarrow B \rightarrow 0; 0 \rightarrow A \rightarrow F \rightarrow B \rightarrow 0$. And the following diagram commutes:

$$\begin{array}{ccccccccc} 0 & & A & & E & & B & & 0 \\ & \searrow & & \searrow & & \searrow & & \searrow & \\ 0 & & A & & F & & B & & 0 \end{array}$$

- The isomorphic class: Let $A, B \in \text{Ob}(\mathcal{A})$, the set of isomorphic classes of extension of B by A denoted

$$\text{Ext}_{\mathcal{A}}(B, A)$$

is called the **Ext-group**.

Actually, we turn $\text{Ext}(-, -)$ into functor

- Pullback
- Pushout

◇ **Lemma 95.1.9.2** The construction

$$(E_1, E_2) \mapsto E_1 + E_2$$

defines a commutative group law on $\text{Ext}_{\mathcal{A}}(B, A)$, which is functorial on each variable.

◦ **Proof ...**

□

♠ **Proposition 95.1.9.3** (Six-Item Exact Sequence) There exists long exact sequence

$$\begin{array}{ccccccc} 0 & & \text{Hom}(C, X) & & \text{Hom}(B, X) & & \text{Hom}(A, X) \\ & & & & & & \\ & & \text{Ext}(C, X) & & \text{Ext}(B, X) & & \text{Ext}(A, X) & & 0 \end{array}$$

◦ **Proof ...**

□

♠ **Proposition 95.1.9.4** (Six-Item Exact Sequence)

$$\begin{array}{ccccccc} 0 & & \text{Hom}(X, A) & & \text{Hom}(X, B) & & \text{Hom}(X, C) \\ & & & & & & \\ & & \text{Ext}(X, A) & & \text{Ext}(X, B) & & \text{Ext}(X, C) & & 0 \end{array}$$

◦ **Proof ...**

□

95.1.10 Additive Functors

◇ **Lemma 95.1.10.1** Let $\mathcal{A}, \mathcal{B}$ abelian categories, $F : \mathcal{A} \rightarrow \mathcal{B}$ is functor. Then, the followings are equivalent

-
-

•

◦ **Proof** ...

□

◇ **Lemma 95.1.10.2** ...

◦ **Proof** ...

□

95.1.11 Localization

95.1.12 Jordan-Hölder

95.1.13 Serre Subcategories

Definition 95.1.13.1 Let $\mathcal{A}$ be an abelian category:

- A **Serre subcategory** of $\mathcal{A}$ is a nonempty full subcategory $\mathcal{C}$ such that for every exact sequence

$$A \longrightarrow B \longrightarrow C$$

with $A, C \in \text{Ob}(\mathcal{A})$ such that

$$A, C \in \text{Ob}(\mathcal{C}) \Rightarrow B \in \text{Ob}(\mathcal{C})$$

- A **weak Serre subcategory** $\mathcal{A}$ is a nonempty full subcategory $\mathcal{C}$ of $\mathcal{A}$ such that: Given any exact sequence

$$A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow A_4$$

with $A_0, A_1, A_3, A_4 \in \text{Ob}(\mathcal{C}) \Rightarrow A_2 \in \text{Ob}(\mathcal{C})$.

◇ **Lemma 95.1.13.2** (Equivalent Condition of Serre Subcategory) Let $\mathcal{A}$ be an abelian category, $\mathcal{C}$ be a subcategory of $\mathcal{A}$. Then, the following conditions are equivalent: $\mathcal{C}$ is Serre subcategory $\Leftrightarrow$

- $0 \in \text{Ob}(\mathcal{C})$;
- $\mathcal{C}$ is a strictly full subcategory of $\mathcal{A}$;
- Any subobject or quotient of $\mathcal{C}$ is an object of $\mathcal{C}$;
- If $A \in \text{Ob}(\mathcal{A})$ is an extension of objects of $\mathcal{C}$, then $A \in \text{Ob}(\mathcal{C})$.

◦ **Proof** $\Leftarrow$

$\Rightarrow$ Omitted.

□

◇ **Lemma 95.1.13.3** (Equivalent Condition of Weak Serre Subcategories)

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◦ **Proof ...** □

Serre Subcategory by Morphism

◇ **Lemma 95.1.13.4** Let $\mathcal{A}, \mathcal{B}$ be abelian categories, and $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor. Then, the **full subcategory** specified by $F(C) = 0$ forms a Serre subcategory of $\mathcal{A}$.

◦ **Proof** Notice that the functor is exact. □

Definition 95.1.13.5 Let $\mathcal{A}, \mathcal{B}$ be an exact functors, then the full subcategory of objects C such that $F(C) = 0$ is called the **kernel** of the functor F . Sometimes denoted by

$$\ker(F)$$

Quotient

◇ **Lemma 95.1.13.6** (Categorical Quotient) Let $\mathcal{A}$ be an abelian category, $C \subseteq \mathcal{A}$ be a Serre subcategory, there exists an abelian category and an exact functor

$$F : \mathcal{A} \rightarrow \mathcal{A}/C$$

which is **essentially surjective** and whose kernel is C .

The category $\mathcal{A}/C$ is specified by the following universal property: For any exact functor

$$G : \mathcal{A} \rightarrow \mathcal{B}$$

Such that $C \subseteq \ker(G)$, there exists a factorization $G = H \circ F$ for a unique exact functor $H : \mathcal{A}/C \rightarrow \mathcal{B}$.

◦ **Proof ...** □

◇ **Lemma 95.1.13.7** Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor, let $C \subseteq \mathcal{A}$ be Serre subcategory contained in the kernel of F . Then,

$$C = \ker(F) \Leftrightarrow \bar{F} : \mathcal{A}/C \rightarrow \mathcal{B} \text{ is faithful}$$

◦ **Proof ...** □

95.1.14 K-Groups of Abelian Category

0th Abelian Category

Definition 95.1.14.1 Let $\mathcal{A}$ be an abelian category, then, denote $K_0(\mathcal{A})$ the **zeroth K-group** of $\mathcal{A}$ and for exact sequence

Remark Another way, we to say this is that: There exists a presentation

$$\bigoplus_{(A \rightarrow B \rightarrow C) \in \text{Seq}} Z[A \rightarrow B \rightarrow C] \rightarrow \bigoplus_{A \in \text{Ob}(\mathcal{A})} Z[A] \rightarrow K_0(\mathcal{A}) \rightarrow 0$$

◇ Lemma 95.1.14.2

◦ Proof ...

□

◇ Lemma 95.1.14.3 ...

◦ Proof ...

□

◇ Lemma 95.1.14.4 ...

◦ Proof ...

□

95.1.15 Cohomological δ -Functor

Definition 95.1.15.1 Let $\mathcal{A}, \mathcal{B}$ be abelian categories, a **cohomological δ -functor** or simply **δ -functor** from $\mathcal{A} \rightarrow \mathcal{B}$ is given by the following datas:

- A collection

$$F^n : \mathcal{A} \rightarrow \mathcal{B}, n \geq 0$$

be additive functors, and

- For any short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ of $\mathcal{A}$, a collection

$$\delta_{A \rightarrow B \rightarrow C} : F^n(C) \rightarrow F^{n+1}(A), n \geq 0$$

of morphisms of $\mathcal{B}$.

These data are assumed to satisfy the following axioms:

- For any short exact sequence as above, we have the **long exact sequence**

$$\begin{array}{ccccccc} 0 & \longrightarrow & F^0(A) & \longrightarrow & F^0(B) & \longrightarrow & F^0(C) \\ & & & & & \searrow & \\ & & & & & F^1(A) & \longrightarrow & F^1(B) & \longrightarrow & F^1(C) \\ & & & & & \nwarrow & \\ & & & & & \dots & \end{array}$$

- For every morphism of short exact sequence $(0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0) \rightarrow (0 \rightarrow A' \rightarrow B' \rightarrow C' \rightarrow 0)$ of short exact sequences, the diagram

$$\begin{array}{ccc} F^n(C) & \xrightarrow{\delta_{A \rightarrow B \rightarrow C}} & F^{n+1}(A) \\ \downarrow & & \downarrow \\ F^n(C') & \xrightarrow{\delta_{A' \rightarrow B' \rightarrow C'}} & F^{n+1}(A') \end{array}$$

This implies, that $F^0(-)$ is left-exact.

Morphism of δ -Functors

Definition 95.1.15.2 ...

Universal δ -Functor

Definition 95.1.15.3 ...

◇ Lemma 95.1.15.4 ...

◦ Proof ...

□

◇ Lemma 95.1.15.5 ...

◦ Proof ...

□

95.2 Chain Complex

Definition 95.2.0.1

- A **$\mathbb{Z}$ -graded chain complex** in $R - Mod$ is:

- A collection of objects $\{C_n\}_{n \in \mathbb{Z}}$;
- Morphisms $\partial_n : C_n \rightarrow C_{n-1}$ such that

$$\partial_n \circ \partial_{n+1} = 0, n \in \mathbb{N}$$

Then:

- * ∂_n are called the **differentials** or **boundary maps**;
- * The elements of C_n are called n -chains;
- * For $n \geq 1$ the elements in the kernel

$$Z_n := Z_n(C_*) = \text{Ker}(\partial_n)$$

are called **n-cycles**;

- * The image

$$B_n := B_n(C_*) = \text{im}(\partial_{n+1})$$

of $\partial_n : C_n \rightarrow C_{n-1}$ are called the **n-boundaries**;

There exists an inclusion $\partial^2 = 0 \Rightarrow 0 \hookrightarrow B_n \hookrightarrow Z_n \hookrightarrow C_n$; Define $H_n := Z_n/B_n$, then the **chain homology** of C_*

$$0 \rightarrow B_n \rightarrow Z_n \rightarrow H_n \rightarrow 0$$

- A **chain map** $f : V_* \rightarrow W_*$ is a collection of morphism $(f_n : V_n \rightarrow W_n)_{n \in \mathbb{Z}}$ in $\mathcal{A}$ such that all the diagram

$$\begin{array}{ccc} V_{n+1} & \longrightarrow & V_n \\ \downarrow & & \downarrow \\ W_{n+1} & \longrightarrow & W_n \end{array}$$

- Take the notation of $Ch(R - Mod)$ for the category. Furthermore, for the category $\mathcal{A}$, take the notion of $Ch(\mathcal{A})$.

Especially, when $R = \mathbb{Z}$, is the chain complex of abelian groups.

♠ **Proposition 95.2.0.2** $Ch(\mathcal{A})$ is an abelian category, iff $\mathcal{A}$ is an abelian category.

- **Proof** $\Rightarrow$ Omitted. There exists a faithful functor $\mathcal{A} \rightarrow Ch(\mathcal{A})$;
 $\Leftarrow$ $\mathcal{A}$ is additive, with structure induced by $\mathcal{A}$. And the biproduct is well-defined. The kernel commutes with $\square$

★ **Corollary 95.2.0.3** $Ch(R - Mod)$ is an abelian category.

- **Proof** Omitted. $\square$

Exactness

◇ **Lemma 95.2.0.4** $Ker, Coker$ commutes with component operation $(-)_n : Ch(\mathcal{A}) \rightarrow \mathcal{A}$.

- **Proof**

- Ker :
- $Coker$:

$\square$

♠ **Proposition 95.2.0.5** The followings are equivalent:

- $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is exact in $Ch(\mathcal{A})$;
- $0 \rightarrow A_n \rightarrow B_n \rightarrow C_n \rightarrow 0$ is exact for each $n \in \mathbb{Z}$ in $\mathcal{A}$;

- **Proof** By definition, omitted. $\square$

♠ **Proposition 95.2.0.6** Suppose $A, B, C \in \mathcal{A}, \mathcal{A}$ abelian, $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is exact, then

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0 \text{ splits} \Leftrightarrow B = A \oplus C$$

- **Proof** Consider the following diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & A & \xrightarrow{f} & B & \xleftarrow{g} & C \longrightarrow 0 \\
 & & \swarrow & & \downarrow & \uparrow & \searrow \\
 & & & & A \oplus C & &
 \end{array}$$

Take $f\pi_A + h\pi_B : A \oplus C \rightarrow B$

π_A, π_B satisfied by ??.

$\square$

♠ **Proposition 95.2.0.7** (Left-Exactness of $Hom(-, X)$) Suppose $X \in \mathcal{A}$, then $Hom(-, X)$ is left-exact: For exact

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

We get the left-exact sequence

$$0 \rightarrow \text{Hom}(C, X) \rightarrow \text{Hom}(B, X) \rightarrow \text{Hom}(A, X)$$

◦ **Proof** The morphisms induced by the morphisms above.

- $\text{Hom}(C, X) \rightarrow \text{Hom}(B, X)$ is injective: We have

$$h \circ g = 0, g \text{ is surjective} \Rightarrow h = 0$$

- $\text{Im}(f^*) = \text{Ker}(g^*)$:

- $\text{Im}(g^*) \subseteq \text{Ker}(f^*)$, easy to verify, that for $h \in \text{Hom}(C, X)$,

$$f^* g^*(h) = h \circ g \circ f = 0$$

- $\text{Ker}(f^*) \subseteq \text{Im}(g^*)$, we have for $h \in \text{Hom}(B, X)$,

$$f^*(h) = h \circ f = 0$$

Now, we can get the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0 \\ & & & & \downarrow h & \swarrow & \\ & & & & X & & \end{array}$$

Notice that $\text{Coker}(f) = C$. Thus the morphism $C \rightarrow X$ is induced.

□

Similarly

♠ **Proposition 95.2.0.8** (Left Exactness of $\text{Hom}(X, -)$) Suppose $X \in \mathcal{A}$, then $\text{Hom}(-, X)$ is right-exact: For exact

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

We can get a right-exact sequence

$$0 \rightarrow \text{Hom}(X, A) \rightarrow \text{Hom}(X, B) \rightarrow \text{Hom}(X, C)$$

◦ **Proof**

- $f^* : \text{Hom}(X, A) \rightarrow \text{Hom}(X, B)$ is injective: We have

$$f^*(h) = 0 \Rightarrow f \circ h = 0 \Rightarrow h = 0$$

- Similarly $\text{Im}(f^*) = \text{Ker}(g^*)$.

□

Furthermore, we have a longer exact sequence: Here

95.2.1 Operations on Chain Complexes

Product, Direct Sum

Definition 95.2.1.1 For a family of chain complexes $\{C_\alpha\}$ in $Ch(\mathcal{A})$, where $\mathcal{A}$ is abelian category, define:

- Product $(\prod A_\alpha, \prod d_\alpha)$, with

$$\prod d_\alpha : \prod A_{\alpha,n} \rightarrow \prod A_{\alpha,n-1}$$

- Sum $(\oplus A_\alpha, \oplus d_\alpha)$, with

$$\oplus d_\alpha : \oplus A_{\alpha,n} \rightarrow \oplus A_{\alpha,n-1}$$

♠ **Proposition 95.2.1.2** The two operations commute with homology, i.e

$$\oplus H_n(A_\alpha) \cong H_n(\oplus A_\alpha) \quad \prod H_n(A_\alpha) = H_n(\prod A_\alpha)$$

◦ **Proof ...**

□

Subcomplex, Quotient Complex

Definition 95.2.1.3 A subcomplex

Especially: In Ab the quotient structure is defined as:

Definition 95.2.1.4 Given $A \hookrightarrow B$, then

$$A/B = \text{coker}(A \rightarrow B)$$

Thus, define the quotient complex

Definition 95.2.1.5 Define: For subcomplex $(B_n) \hookrightarrow (A_n)$ define

$$((A_n/B_n), (\tilde{d}_n))$$

Induced by:

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & B_{n-1} & \longrightarrow & B_n & \longrightarrow & B_{n+1} \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & A_{n-1} & \longrightarrow & A_n & \longrightarrow & A_{n+1} \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & A_{n-1}/B_{n-1} & \longrightarrow & A_n/B_n & \longrightarrow & A_{n+1}/B_{n+1} \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

The exactness is given by 9-lemma.

Tensor Product

Definition 95.2.1.6 For $X, Y \in Ch_*(A)$ write $X \otimes Y \in Ch_*(A)$ for chain complex whose component is given by:

$$(X \otimes Y)_n := \bigoplus_{i+j=n} X_i \otimes_R Y_j$$

with $\partial^{X \otimes Y}(x, y) = (\partial^X x, y) + (-1)^{\deg(X)}(x, \partial^Y y)$.

♣ **Example 95.2.1.7** Let R be some ring and $I_* \in C_*(R\text{-Mod})$ be the chain complex given by

$$I_* = [\cdots \rightarrow 0 \rightarrow 0 \rightarrow R \xrightarrow{\partial_0^I} R \oplus R]$$

where $\partial_0^I = (-id, id)$. We may think of

$$I_0 = R \oplus R \cong R[\{(0), (1)\}] \quad I_1 = R \cong R[(0 \rightarrow 1)]$$

Thus,

$$\partial_0^I : (0 \rightarrow 1) \mapsto (1) - (0)$$

Write out the tensor product $S_* := I_* \otimes I_*$, by definition we have

$$\begin{aligned} S_0 &= \\ S_1 &= \\ S_2 &= \end{aligned}$$

♠ **Proposition 95.2.1.8** For category $\mathcal{A}$, we have

$$Ch(\mathcal{A}) \text{ is monoidal} \Leftrightarrow \mathcal{A} \text{ is monoidal}$$

◦ **Proof ...** □

Chain $[X, Y]$

Definition 95.2.1.9

- Define the category $Ch_*^{ub}(\mathcal{A}) := \{C \in Ch_*(\mathcal{A}) | C \text{ is unbounded}\}$
- Define:

$$\begin{aligned} [X, Y]_n &:= \prod_{i \in \mathbb{Z}} Hom_{RMod}(X_i, Y_{i+n}) \\ df &:= d_Y \circ f - (-1)^n f \circ d_X \quad f \in [X, Y]_n \end{aligned}$$

Thus,

$$[-, -] : Ch_*(\mathcal{A})^{op} \times Ch_*(\mathcal{A}) \rightarrow Ch_*^{ub}(\mathcal{A})$$

is the **inner hom** of the category of chain complexes.

♠ **Proposition 95.2.1.10** The collection of cycles of the internal $Hom[X, Y]_*$ in degree 0 coincides with the external hom functor

$$Z_0([X, Y]) = Hom_{Ch_*^{ub}(X, Y)}$$

◦ **Proof ...**

□

♠ **Proposition 95.2.1.11** The monoidal category $(Ch., \otimes)$ is a closed monoidal category, the internal hom is the standard internal hom of chain complexes.

◦ **Proof ...**

□

95.2.2 Shifting and Translation

Definition 95.2.2.1 Define the chain/cochain complex

$$C[p]_n := C_{n+p} \quad C[p]^n = C^{n-p}$$

Then, define the differential map

$$d :=$$

95.2.3 Bicomplex

Definition 95.2.3.1 A **double complex** or **bicomplex** in $\mathcal{A}$ is a family $\{C_{p,q}\}$ of object $\mathcal{A}$ together with maps

$$d^h : C_{p,q} \rightarrow C_{p-1,q} \quad d^v : C_{p,q} \rightarrow C_{p,q-1}$$

Such that: $d^h \circ d^h = d^v \circ d^v = d^v d^h + d^h d^v = 0$. Lattice

$$\begin{array}{ccccccc} & & \cdots & & \cdots & & \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & C_{p-1,q+1} & \longrightarrow & C_{p-1,q} & \longrightarrow & C_{p-1,q-1} \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & C_{p,q+1} & \longrightarrow & C_{p,q} & \longrightarrow & C_{p,q-1} \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & C_{p+1,q+1} & \longrightarrow & C_{p+1,q} & \longrightarrow & C_{p+1,q-1} \longrightarrow \cdots \\ & & \cdots & & \cdots & & \cdots \end{array}$$

Tot Complexes

Definition 95.2.3.2 Define:

$$Tot_n^\Pi := \prod_{p+q=n} C_{p,q} \quad Tot_n^\oplus = \bigoplus_{p+q=n} C_{p,q}$$

With the differential

$$\begin{aligned} d : Tot_n^\Pi &\rightarrow Tot_{n-1}^\Pi \\ d : Tot_n^\oplus &\rightarrow Tot_{n-1}^\oplus \end{aligned}$$

such that $d \circ d = 0$.

♠ **Proposition 95.2.3.3** C is a bounded double complex with exact rows/columns $\Rightarrow Tot(C)$ is acyclic.

◦ **Proof ...** □

95.2.4 Truncations

Definition 95.2.4.1 ...

95.2.5 Mapping Cone

95.3 Long Exact Sequence

95.4 Chain Homotopies

Definition 95.4.0.1 A category

- has **enough projective** if for every object X if there is a projective object Q equipped with an epimorphism $Q \rightarrow X$;
- has **enough injectives** iff for every object X there is an injective Q equipped with monomorphism $X \rightarrow P$.

♠ **Proposition 95.4.0.2** Assuming the axiom of choice, the category $R - Mod$ has enough projective module.

◦ **Proof ...** □

95.4.1 Introduction

Consider the following vector space decomposition over field:

◇ **Lemma 95.4.1.1** Every exact sequence in $Vect_k$ splits.

$$C_n = Z_n \oplus B'_n; Z_n = Im(\partial_{n+1}), B'_n := C_n / Z_n = Coker(Z_n \rightarrow C_n) \cong B_{n-1}$$

$$Z_n = B_n \oplus H'_n, H'_n := Z_n / B_n = H_n(C)$$

$$\begin{array}{ccccccc}
 & \downarrow & & 0 & & & \\
 & C_{n+1} & & \downarrow & & & \\
 & \uparrow \partial_{n+1} & & \downarrow & & & \\
 0 \longrightarrow & Im(\partial_{n+1}) = B_n & \xrightarrow{\partial_n} & Ker(\partial_n) = Z_n & \longrightarrow & H_n(C_\bullet) & \longrightarrow 0 \\
 & \downarrow \partial_n & & \downarrow & & & \\
 & C_n & & \downarrow & & & \\
 & \downarrow \partial_n & & \downarrow & & & \\
 0 \longrightarrow & Im(\partial_n) = B_{n-1} & \longrightarrow & Ker(\partial_{n-1}) = Z_{n-1} & \longrightarrow & H_{n-1}(C_\bullet) & \longrightarrow 0 \\
 & \downarrow & & \downarrow & & & \\
 & 0 & & C_{n-1} & & & \\
 & & & \downarrow & & & \\
 & & & \dots & & &
 \end{array}$$

Therefore, we can form the compositions

$$C_n \xrightarrow{\pi} Z_n \xrightarrow{\pi} B_n \cong B'_{n+1} \subseteq C_{n+1}$$

By the subspace $C_{n+1}/Z_{n+1} \hookrightarrow C_{n+1}$. Define the **split map** $s_n : C_n \rightarrow C_{n+1}$. We have

$$\partial_{n+1} s_n \partial_{n+1} = \partial_{n+1}$$

Verify:

...

And:

- ∂s :
- $s\partial$:

Split

Definition 95.4.1.2 A complex C is called split if there exists some $s_n : C_n \rightarrow C_{n+1}$ such that

$$d = dsd$$

such s is called **splitting map**.

Especially: If C is exact, then we say that C is **split exact**.

Chain Contraction

Definition 95.4.1.3

- We say that a chain map $f : C \rightarrow D$ is **null homotopic** if there are maps $s_n : C_n \rightarrow D_{n+1}$ such that

$$f = ds + sd$$

- The maps $\{s_n\}$ are called a **chain contraction** of f .

Chain Homotopies

Definition 95.4.1.4

◇ **Lemma 95.4.1.5** (Chain Homotopy Invariance for H_*)

-
-

◦ **Proof ...**

□

♠ **Proposition 95.4.1.6**

◦ **Proof ...**

□

Quotient Category

♠ **Proposition 95.4.1.7** ...

◦ **Proof ...**

□

95.4.2 Resolution

95.5 Homology Exact Sequence

Definition 95.5.0.1 An **exact squence** in $\mathcal{A}$ is a chain complex $C_\bullet$ in $\mathcal{A}$ with vanishing chain homology in each degree

$$\forall n \in \mathbb{N}, H_n(C) = 0$$

Especially, a **short exact sequence** is an exact sequence of the form

$$\cdots \rightarrow 0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0 \rightarrow \cdots$$

Mostly written as

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

95.5.1 Some of Skills in Chain

3 × 3-Lemma

♠ **Proposition 95.5.1.1** ...

◦ **Proof** ...

□

Snake Lemma

5-Lemma

Addendum

95.5.2 Existence of ∂ in Exact Chain Complexes

So, the following diagram commutes:

$$\begin{array}{ccc} H_*(A) & & H_*(B) \\ & \searrow & \downarrow \\ & & H_*(C) \end{array} \quad (\text{The exact triangle})$$

♠ **Proposition 95.5.2.1**

◦ **Proof** ...

□

95.5.3 Resolution

♠ **Proposition 95.5.3.1** Suppose A_* be a resolution of M , then

$$H_n(A_*) = \begin{cases} M & n = 0 \\ 0 & n \in \mathbb{Z} - \{0\} \end{cases}$$

◦ **Proof** Omitted.

□

95.5.4 Functor Ext^1

Definition 95.5.4.1 Select an injective resolution of X :

$$0 \rightarrow X \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

Apply $H(A, -)$ and deleting $Hom(A, X)$ we get

$$0 \rightarrow Hom(A, I^0) \xrightarrow{d^0} Hom(A, I^1) \xrightarrow{d^1} Hom(A, I^2) \rightarrow \dots$$

Define:

$$Ext^1(A, X) := ker(d^1)/im(d^0)$$

We have the following isomophic construction via projective resolution:

♠ **Proposition 95.5.4.2** In in the category of $\mathbb{Z}$ -module/abelian groups, we have the exact sequence as followings:

$$\begin{array}{ccccccc} 0 & \longrightarrow & Hom(C, X) & \longrightarrow & Hom(B, X) & \longrightarrow & Hom(A, X) \longrightarrow \\ & & & & & \searrow & \\ & & & & & Ext^1(C, X) & \longrightarrow Ext^1(B, X) \longrightarrow Ext^1(A, X) \longrightarrow 0 \end{array}$$

◦ **Proof ...**

□

In General: In $\mathcal{A}b$ -category we have the long exact sequence (here)

Extension and Ext

Ref: Weibel 3.4

Definition 95.5.4.3

- An **extension** of A by B is an exact sequence

$$0 \rightarrow A \rightarrow X \rightarrow B \rightarrow 0$$

- Two extensions of A by B are **equivalent** if

$$\begin{array}{ccccccc} 0 & \longrightarrow & B & \longrightarrow & X & \longrightarrow & A \longrightarrow 0 \\ & & \parallel & & \downarrow & & \parallel \\ 0 & \longrightarrow & B & \longrightarrow & X & \longrightarrow & A \longrightarrow 0 \end{array}$$

♣ **Example 95.5.4.4** (Extension of $\mathbb{Z}/p$) Consider the following equivalent class of split extensions:

$$0 \longrightarrow \mathbb{Z}/p \xrightarrow{p} \mathbb{Z}/p^2 \xrightarrow{i} \mathbb{Z}/p \longrightarrow 0, i=1, \dots, p-1$$

◇ **Lemma 95.5.4.5** $Ext^1(A, B) = 0 \Leftrightarrow$ Every extension of A by B splits.

◦ **Proof ...**

□

► **Theorem ...**

◦ **Proof ...**

Definition 95.5.4.6 (Baer Sum)

95.6 Mapping Cones and Cylinders

95.6.1 Mapping Cones

Definition 95.6.1.1 Let $f : B \rightarrow C$ be a map of chain complexes. The **mapping cone** of f is the chain complex $\text{cone}(f)$ given by:

$$[\text{cone}(f)]_n := B_{n-1} \oplus C_n$$

With differentials

$$d(b, c) = (-d(b), d(c) - f(b)) \quad \begin{pmatrix} -d_B & 0 \\ -f & +d_C \end{pmatrix} :$$

$$\begin{array}{ccc} B_{n-1} & \xrightarrow{-} & B_{n-2} \\ \oplus & \searrow - & \oplus \\ C_n & \xrightarrow{+} & C_{n-1} \end{array}$$

Especially, denote $\text{cone}(C) := \text{cone}(id_C)$

♠ **Proposition 95.6.1.2** $f : C \rightarrow D$ is **null homotopic** iff if f extends $(-s, f) : \text{cone}(C) \rightarrow D$.

◦ **Proof ...**

95.6.2 Mapping Cylinder

Chapter 96 Derived Functors

96.1 δ -Functors

Definition 96.1.0.1 A (covariant) homological/cohomological δ -functor between $\mathcal{A}$ and $\mathcal{B}$ is a collection of additive functors $T_n : \mathcal{A} \rightarrow \mathcal{B}$ and

$$\delta_n : T_n(C) \rightarrow T_{n-1}(A)$$

$$\delta^n : T^n(C) \rightarrow T^{n+1}(A)$$

defined for each short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ in $\mathcal{A}$. Here we make the convention that $T^n = T_n = 0, n < 0$.

The two conditions are imposed:

- Long exact sequence

$$\begin{aligned} \cdots \rightarrow T_{n+1}(C) \xrightarrow{\delta} T_n(A) \rightarrow T_n(B) \rightarrow T_n(C) \xrightarrow{\delta} T_{n-1}(A) \rightarrow \cdots \\ \cdots \rightarrow T^{n-1}(C) \xrightarrow{\delta} T^n(A) \rightarrow T^n(B) \rightarrow T^n(C) \xrightarrow{\delta} T^{n+1}(A) \rightarrow \cdots \end{aligned}$$

- For two exact $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ and $0 \rightarrow A' \rightarrow B' \rightarrow C' \rightarrow 0$, we have commutes squares

$$\begin{array}{ccc} T_n(C') & \xrightarrow{\delta} & T_{n-1}(A') \\ \downarrow & & \downarrow \\ T_n(C) & \xrightarrow{\delta} & T_{n-1}(A) \end{array}$$

♣ **Example 96.1.0.2** Consider the functors

$$H_* : Ch_{\geq 0}(\mathcal{A}) \rightarrow \mathcal{A}$$

$$H^* : Ch^{\geq 0}(\mathcal{A}) \rightarrow \mathcal{A}$$

H_* gives a homological δ -functor, and H^* gives a cohomological δ -functor. See here:

♠ **Proposition 96.1.0.3** Let S be a category of short exact sequences

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0 \quad (*)$$

in $\mathcal{A}$. Then, δ_i is a natural transformation from the functor sending from the functor sending $(*)$ to $T_i(C)$, to the functor sending $(*)$ to $T_{i-1}(A)$.

- **Proof** Consider the two functors on on the category of short exact sequences S to $\mathcal{B}$:

$$F : (0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0) \mapsto T_i(C)$$

$$G : (0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0) \mapsto T_{i-1}(A)$$

Thus, the natural transformation

$$\begin{array}{ccc} S & \xrightarrow{F} & \mathcal{B} \\ & \downarrow & \\ S & \xrightarrow{G} & \mathcal{B} \end{array}$$

We could consider the following example:

$$\begin{array}{ccc}
(0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0) & T_i(C) \longrightarrow T_{i-1}(A) \\
\downarrow (f,g,h) & \downarrow T_i(h) \quad \downarrow T_{i-1}(f) \\
(0 \rightarrow A' \rightarrow B' \rightarrow C' \rightarrow 0) & T_i(C') \longrightarrow T_{i-1}(A')
\end{array}$$

□

♣ **Example 96.1.0.4** (p-torsion) If p is an integer, the functors $T_0(A) := A/pA$ and

$$T_1(A) = {}_pA = \{a \in A \mid pa = 0\}$$

fit together to form a homological δ -functor, or a cohomological δ -functor from $\mathcal{A}b$ to $\mathcal{A}b$: Apply the snake lemma to

$$\begin{array}{ccccccc}
0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\
& & \downarrow p & & \downarrow p & & \downarrow \\
0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0
\end{array}$$

We get the exact sequence:

$$0 \rightarrow {}_pA \rightarrow {}_pB \rightarrow {}_pC \xrightarrow{\delta} A/pA \rightarrow B/pB \rightarrow C/pC \rightarrow 0$$

♣ **Example 96.1.0.5** (Generalization) Similarly, for a ring R with element $r \in R$, then

$$T_0(M) := M/rM \quad T_1 := {}_rM$$

And then, we will build a homological/cohomological functor $\delta : R\text{-Mod} \rightarrow \mathcal{A}b$,

$$T_n(M) = \text{Tor}_n^R(R/r, M)$$

Especially, if r is nonzerodivisor, equivalently to say is

$${}_rR := \{s \in R \mid rs = 0\} = 0$$

Then, we have

$$T_1^R(M) = \text{Tor}_1^R(R/r, M) = {}_rM; \text{Tor}_n(R/r, M) = 0, n \geq 2$$

And:

$$\text{Tor}_1^R(R/r, M) = {}_rM / \{sm \mid rs = 0, r \in R, m \in M\}$$

In general, ${}_rR \neq 0$.

96.1.1 Morphism of δ -Functors

Definition 96.1.1.1 Suppose S, T be two collection of δ -functors. A **morphism of δ -functors** is a family of natural transformations between these functors, which commute with δ . i.e. For $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ we have

$$\begin{array}{ccc}
S_n(C) & \xrightarrow{\delta_n} & S_{n-1}(A) \\
\downarrow f_n(C) & & \downarrow f_{n-1}(A) \\
T_n(C) & \xrightarrow{\delta_n} & T_{n-1}(A)
\end{array}$$

And

Definition 96.1.1.2

- A homological functor T is **universal**, if: Given
 - Another δ -Functor S ;
 - A morphism $f_0 : S_0 \rightarrow T_0$

There exists a unique morphism $\{f_n : S_n \rightarrow T_n\}$ of δ functor extends f_0 ;

- Dual Definition **couniversal** for extension $f_0 : T_0 \rightarrow S_0$ to $\{f_n : T_n \rightarrow S_n\}$.

♠ Proposition 96.1.1.3 Functor

- $H_*(-) : Ch_{\geq 0}(\mathcal{A}) \rightarrow \mathcal{A}$ is universal δ -functor;
- $H^*(-) : Ch^{\geq 0}(\mathcal{A}) \rightarrow \mathcal{A}$ is universal δ -functor;

◦ **Proof ...** □

♠ Proposition 96.1.1.4 Consider the following functor: Suppose $F : \mathcal{A} \rightarrow \mathcal{B}$ is exact, then the following functor:

$$T_0 = F \quad T_n = 0, n \neq 0$$

Then, $T = \{T_n\}$ is a universal δ -functor.

◦ **Proof ...** □

96.2 Left Derived Functors

Definition 96.2.0.1 Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a right exact functor between two abelian categories. If $\mathcal{A}$ has enough projectives, we can construct the **left derived functors** $L_i F, i \geq 0$ of F as follows: If $A \in \mathcal{A}$, choose a projective resolution $P \rightarrow A$ and define

$$L_i F(A) := H_i(F(P))$$

$$\begin{array}{ccccccc} \dots & \longrightarrow & P_2 & \longrightarrow & P_1 & \longrightarrow & P_0 \\ & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & A \end{array} \Rightarrow \text{Exact } P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow A \Rightarrow \text{Exact } F(P_2) \rightarrow \dots F(P_1) \rightarrow F(P_0) \rightarrow 0$$

Consider the exact sequence $F(P_1) \rightarrow F(P_0) \rightarrow F(A) \rightarrow 0$, we have

$$L_0 F(A) = H_0(F(P)) = F(A)$$

96.2.1 Universal δ -Functority for $L_* F(-)$

The aim is to show $L_* F$ form a universal homological δ -functor.

◇ **Lemma 96.2.1.1** The objects $L_i F(A)$ of $\mathcal{B}$ are well defined up to isomorphism. That is: If $Q \rightarrow A$ is a second projective resolution, there exists a canonical isomorphism

$$L_i F(A) = H_i(F(P)) \xrightarrow{\cong} H_i(F(Q))$$

In particular, a different choice of projective resolution would yield new functor $\hat{L}_i F$, which are naturally isomorphic to the functors $L_i F$. This implies, that the value of $L_i F(A)$

◦ **Proof** Apply comparison theorem. □

★ **Corollary 96.2.1.2** If A is projective, then $L_i F(A) = 0, i \neq 0$.

◦ **Proof** Consider the projective resolution of $0 \rightarrow P \xrightarrow{id} P \rightarrow 0$, then $L_i F(A) = H_i(F(P)) = 0$ □

96.2.2 F -Acyclic Functor

Definition 96.2.2.1 An object Q is called F -Acyclic if

$$L_i F(Q) = 0, i \neq 0$$

96.2.3 $L_i F$ is Universal δ -Functor

◇ **Lemma 96.2.3.1** If $f : A' \rightarrow A$ is any map in $\mathcal{A}$, there exists a natural map $L_i F(f) : L_i F(A') \rightarrow L_i F(A)$.

◦ **Proof** ... □

♠ **Proposition 96.2.3.2** ...

◦ **Proof** ... □

► **Theorem** Each $L_i F$ is an additive functor from $\mathcal{A}$ to $\mathcal{B}$

◦ **Proof** ... □

♠ **Proposition 96.2.3.3** (Preversing Derived Functor)

◦ **Proof** ... □

► **Theorem** The derived functors $L_* F$ form a homological δ -functor.

◦ **Proof** ... □

Dimension Shifting

♠ **Proposition 96.2.3.4** ...

◦ **Proof** ... □

Homology, Cohomology

♠ **Proposition 96.2.3.5** The functors $H_* : Ch_{\geq 0}(\mathcal{A}) \rightarrow \mathcal{A}$ and $H^* : Ch^{\geq 0}(\mathcal{A}) \rightarrow \mathcal{A}$ are universal δ -functors.

◦ **Proof ...**

□

Effaceable Functors

Definition 96.2.3.6

- An additive functor $F : \mathcal{A} \rightarrow \mathcal{B}$ is **effaceable** if for each object A of $\mathcal{A}$ there is a monomorphism $u : A \rightarrow I$ such that $F(u) = 0$;
- Conversely, **coeffaceable** iff there exists surjection $u : P \rightarrow A$ such that $F(u) = 0$.

♠ **Proposition 96.2.3.7**

- If T_* is a homological δ -functor such that T_n is coeffaceable except T_0 , then T_* is universal;
- If T^* is a cohomological δ -functor such that T_n is effaceable, then T^* is universal.

◦ **Proof ...**

□

96.3 Right Derived Functors

96.3.1 Right Derived Functors

Definition 96.3.1.1

$$R^i F(A) := H^i(F(I))$$

♠ **Proposition 96.3.1.2** (Right-Derived Functor is Dual of Left-Derived Functor)

◦ **Proof ...**

□

96.3.2 Ext Functor

Ext Functor on R -modules

Definition 96.3.2.1 Define:

$$Ext_R^i(A, B) := R^i Hom_R(A, -)(B)$$

♠ **Proposition 96.3.2.2** The followings are equivalent:

-

-
-
-

◦ Proof ...

□

♠ **Proposition 96.3.2.3** The followings are equivalent:

-
-
-
-

◦ Proof ...

□

Ext Functor in Abelian Categories

Definition 96.3.2.4

-
-

◇ **Lemma 96.3.2.5** The two concepts above are equivalent in abelian category.

◦ Proof ...

□

♣ **Example 96.3.2.6** ...

Ext Functor in Derived Category

Definition 96.3.2.7 Consider the derived category $D(\mathcal{A})$. Define

$$Ext^p(X, A) := Hom_{D(\mathcal{A})}(X, A[p])$$

96.4 Adjoint Functors and Left/Right Exactness

► **Theorem** Suppose $L : \mathcal{A} \rightarrow \mathcal{B}$ and $R : \mathcal{B} \rightarrow \mathcal{A}$ be adjoint pair. Then, there exists a natural isomorphism

$$Hom(L(A), B) \xrightarrow{\cong} Hom(A, R(B))$$

◦ Proof ...

□

★ **Corollary 96.4.0.1**

- $\otimes_R B : \text{mod} - R \rightarrow \text{Ab}$ is left adjoint to $\text{Hom}_{\text{Ab}}(B, -)$, so $- \otimes_R B$ is right exact;
- More generally, if B is an $R - S$ bimodule, then $\otimes_R B$ takes $\text{mod} - R$ to $\text{mod} - S$ and is a left adjoint, so it is right exact.

◦ **Proof ...** □

♠ **Proposition 96.4.0.2** ...

◦ **Proof ...** □

96.4.1 Tor Functor

Definition 96.4.1.1 Define: For functor $T(A) = A \otimes_R B$, define the abelian group

$$\text{Tor}_n^R(A, B) := (L_n T)(A)$$

In particular, $\text{Tor}_0^R(A, B) = A \otimes_R B$.
And when A

96.4.2 Adjoint Limit and Colimits

96.5 Balancing Tor and Ext

More discussions in next chapter.

Chapter 97 Tor and Ext

97.1 Tor for Abelian Groups

97.2 Tor and Flatness

Definition 97.2.0.1

- Consider $B \in R - Mod$. Then, B is **flat** iff $- \otimes B$ is exact.
- Dual to define $A \in Mod - R$. Then, A is **flat** iff $A \otimes -$ is exact.

Pontrjagin Dual

Definition 97.2.0.2 The Pontrjagin dual B^* of a left-module B is the right R -module

$$B^* \in Hom_{Ab}(B, \mathbb{Q}/\mathbb{Z})$$

and element $r \in R$ acts via $(fr)(b) = f(rb)$. So B^* is a right R -module.

97.3 Ext for Nice Rings

97.4 Ext and Extensions

97.5 Derived Functor and Inverse Limits

Notion In this section, we take

- $\mathcal{A} = Ab$;
- I is the poset $\cdots \rightarrow 2 \rightarrow 1 \rightarrow 0$;

Definition 97.5.0.1 (Tower of Abelian Groups) Define

$$\{A_i\} : \cdots \rightarrow A_2 \rightarrow A_1 \rightarrow A_0$$

97.5.1 Mittag-Leffers System

Definition 97.5.1.1 (Mittag-Leffers System) Let I be a directed set, (A_i, φ_{ij}) be an inverse system in $Sets$.

Limit

◇ **Lemma 97.5.1.2** (Existence)

◦ **Proof ...**

□

◇ **Lemma 97.5.1.3** (Exactness) Let

...

97.6 Universal Coefficient Theorem

Chapter 98 Homological Dimension

Chapter 99 Spectral Sequence

99.1 Introductions

- (Leray) Sheaf, Spectral Sequence;
- (Koszul, 1945) Algebraic;

99.1.1 Spectral Sequence

Definition 99.1.1.1 Let $\mathcal{A}$ be an abelian category

- A **spectral sequence** in $\mathcal{A}$ is given by a system

$$(E_r, d_r)_{r \geq 1}$$

where: Each E_r is an object of $\mathcal{A}$, each $d_r : E_r \rightarrow E_r$ is a morphism such that

$$d_r \circ d_r = 0 \quad E_{r+1} = \ker(d_r) / \text{Im}(d_r)$$

- A morphism of spectral sequences

$$f : (E_r, d_r)_{r \geq 1} \rightarrow (E'_r, d'_r)_{r \geq 1}$$

is given by a family of morphisms $(f_r)_{r \geq 1}$ such that

$$f_r \circ d_r = d'_r \circ f_r$$

And $f_{r+1} : E_{r+1} \rightarrow E'_{r+1}$ is identified by the morphism induced by $f_r : E_r \rightarrow E'_r$

$$\begin{array}{ccccc} \text{Im}(d_r) & \xrightarrow{\subseteq} & \ker(d_r) & \longrightarrow & E_{r+1} \\ \downarrow f_r|_{\text{Im}(d_r)} & & \downarrow f|_{\ker(d_r)} & & \downarrow \\ \text{Im}(d'_r) & \xrightarrow{\subseteq} & \ker(d'_r) & \longrightarrow & E'_{r+1} \end{array}$$

Remark Sometimes denoted as $(E_r, d_r)_{r \geq 0}$. We loose the definition: There exists isomorphism $E_r \rightarrow \ker(d_r) / \text{Im}(d_r)$.

Definition 99.1.1.2 ...

Spectral Sequence

Definition 99.1.1.3 Given a double complex $(E_r, d_r)_{r \geq 1}$, define

$$0 = B_1 \subseteq B_2 \subseteq \dots \subseteq Z_r \subseteq \dots \subseteq Z_2 \subseteq Z_1 = E_1$$

By following procedure:

$$B_{r+1} = \text{Im}(d_r) \quad Z_{r+1} = \text{Ker}(d_r)$$

Then, it is clear that $d_2 : Z_2 / B_2 \rightarrow Z_2 / B_2$.

Limit of Spectral Sequence

Definition 99.1.1.4 Let $\mathcal{A}$ be an abelian category, $(E_r, d_r)_{r \geq 1}$ be a spectral sequence.

- If the subobjects $Z_\infty = \cap Z_r$ and $B_\infty = \cup B_r$ of E_1 exist. Then, we define the **limit of spectral sequence** to be the object

$$Z_\infty = \cap Z_r \quad B_\infty = \cup B_r$$

- We say that the spectral sequence degenerates at E_r if the differentials $d_r, d_{r+1}, \dots$ are all zero.

If the spectral sequence degenerates at E_r , then we have

$$E_r = E_{r+1} = \dots = E_\infty$$

Also, almost any abelian category we will encounter has countable sums and intersections.

99.1.2 Exact Couple

Definition 99.1.2.1 Let $\mathcal{A}$ be an exact category:

- An **exact couple** is a datum (A, E, α, f, g) where A, E are objects of $\mathcal{A}$ and α, f, g are morphisms as in following diagram

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & A \\ & \searrow g & \\ & E & \end{array}$$

•

◇ **Lemma 99.1.2.2** ...

◦ **Proof** ...

□

99.1.3 Differential Object

Definition 99.1.3.1 Let $\mathcal{A}$ be an abelian category.

A **differential object** of $\mathcal{A}$ is a pair (A, d) consisting of

99.1.4 Filtered Differential Object

99.1.5 Filtered Complexes

99.1.6 Double Complexes

99.1.7 Abelian Groups

Chapter 100 Simplicial Methods in Homological Algebra

100.1 Simplicial Objects

100.2 Operations on Simplicial Objects

Chapter 101 Hochschild and Cyclic Homology

Chapter 102 Derived Category

102.1 Triangulated Category

Definition 102.1.0.1 Define: Let $\mathcal{D}$ be an additive category, let

$$[1] : \mathcal{D} \rightarrow \mathcal{D} \quad E \mapsto E[1]$$

be an additive functor which is an auto-equivalence of $\mathcal{D}$:

- A **triangle** is sextuple

$$(X, Y, Z, f, g, h)$$

where

- $X, Y, Z \in \text{Ob}(\mathcal{D})$;
- $f : X \rightarrow Y; g : Y \rightarrow Z; h : Z \rightarrow X[1]$ are morphisms of $\mathcal{D}$;

- A morphism of triangles

$$(X, Y, Z, f, g, h) \rightarrow (X', Y', Z', f', g', h')$$

with morphisms

$$\begin{cases} a : X \rightarrow X' \\ b : Y \rightarrow Y' \\ c : Z \rightarrow Z' \end{cases}$$

of $\mathcal{D}$ such that

$$\begin{cases} b \circ f = f' \circ a \\ c \circ g = g' \circ b \\ a[1] \circ h = h' \circ c \end{cases}$$

With visualization

$$\begin{array}{cccc} X & Y & Z & X[1] \\ & \searrow & \searrow & \searrow \\ X' & Y' & Z' & X'[1] \end{array}$$

102.1.1 Triangulated Category

Definition 102.1.1.1 Axioms:

- TR1: Any triangle category isomorphic to a distinguished triangle is a distinguished triangle.
Any triangle of the form $(X, X, 0, id, 0, 0)$ is distinguished.
For any morphism $f : X \rightarrow Y$ in $\mathcal{D}$, there exists a **distinguished triangle** of the form

$$(X, Y, Z, f, g, h)$$

- TR2: The triangle (X, Y, Z, f, g, h) is distinguished iff $(Y, Z, X[1], g, h, -f[1])$ is
- TR3: Given the diagram

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

$$X' \longrightarrow Y' \longrightarrow Z' \longrightarrow X[1]$$

- TR4: Given objects $X, Y, Z \in \text{Ob}(\mathcal{D})$, and morphisms $f : X \rightarrow Y, g : Y \rightarrow Z$ and distinguished triple

$$(X, Y, Q_1, f, p_1, d_1) \quad (X, Z, g \circ f, p_2, d_2) \quad (Y, Z, Q_2, g, p_3, d_3)$$

There exists morphisms

$$a : Q_1 \rightarrow Q_2 \quad b : Q_2 \rightarrow Q_3$$

such that:

$$- (Q_1, Q_2, Q_3,)$$

-

-

We will call:

-
-

Remark

Homological Functor

Definition 102.1.1.2

- Let $\mathcal{D}$ be a pre-triangulated category, $\mathcal{A}$ be an abelian category.

An additive category $H : \mathcal{D} \rightarrow \mathcal{A}$ is **homological** iff for every distinguished triangle (X, Y, Z, f, g, h) the sequence

$$H(X) \rightarrow H(Y) \rightarrow H(Z)$$

is exact in the abelian category $\mathcal{A}$

- Conversely, $H : \mathcal{D}^{op} \rightarrow \mathcal{A}$ if the corresponding functor

$$H^{op} : \mathcal{D} \rightarrow \mathcal{A}^{op}$$

is homological.

.

Remark If $H : \mathcal{D} \rightarrow \mathcal{A}$, we write

$$H^n(X) := H(X(n))$$

Definition 102.1.1.3 By TR2, a distinguished triangle (X, Y, Z, f, g, h) determines a long exact sequence

$$H^{-1}(Z) \xrightarrow{\text{above}} H^0(X) \xrightarrow{\text{above}} H^0(Y) \xrightarrow{\text{above}} H^0(Z) \xrightarrow{H(h)} H^1(X)$$

δ -Functor

Definition 102.1.1.4 Let $\mathcal{A}$ be an abelian category. Let $\mathcal{D}$ be a triangulated category. A δ -functor from $\mathcal{A}$ to $\mathcal{D}$ is given by a functor

$$G : \mathcal{A} \rightarrow \mathcal{D}$$

102.1.2 Elementray Results

◇ **Lemma 102.1.2.1** Let $\mathcal{D}$ be a pre-triangulated category, let (X, Y, Z, f, g, h) be distinguished triangle, then

$$g \circ f = 0 \quad h \circ g = 0 \quad f[1] \circ h = 0$$

◦ **Proof ...**

□

◇ **Lemma 102.1.2.2** Let $\mathcal{D}$ be a pre-triangulated category. For any object W of $\mathcal{D}$, the functor $\text{Hom}_{\mathcal{D}}(W, -)$ is homological, and the functor $\text{Hom}_{\mathcal{D}}(-, W)$ is cohomological.

◦ **Proof ...**

□

◇ **Lemma 102.1.2.3** Let $\mathcal{D}$ be a pretriangulated category, and

$$(a, b, c) : (X, Y, Z, f, g, h) \rightarrow$$

◦ **Proof ...**

□

♠ **Proposition 102.1.2.4** Let $\mathcal{D}$ be trianglatedd category. Any commutative category

$$X \quad Y$$

$$X' \quad Y'$$

can be extended to the diagram:

$$X \quad Y \quad Z \quad X[1]$$

$$X' \quad Y' \quad Z' \quad X'[1]$$

$$X'' \quad Y'' \quad Z \quad X''[1]$$

$$X[1] \quad Y[1] \quad Z[1] \quad X[2]$$

where: All square commutes, except for the lower right square which is anticommutative.

Moreover, each of column and rows are distinguished triangles.

Finally, the morphisms on the bottom row/right column are obtained from the morphisms of the top row/left column applying $[1]$.

◦ **Proof ...** □

102.1.3 Constructions

Triangulated Functor

Localization of Triangulated Functor

Definition 102.1.3.1 Let: $\mathcal{D}$ be a pre-triangulated category, we say a multiplicative system S is **compatible with triangulated structure** if the following two conditions hold:

- MS5: For a morphism f of $\mathcal{D}$ we have

$$f \in S \Leftrightarrow f[1] \in S$$

- Given a solid commutative square

$$\begin{array}{ccccccc} X & \longrightarrow & Y & \longrightarrow & Z & \longrightarrow & X[1] \\ \downarrow s & & \downarrow s' & & \downarrow \exists s'' & & \downarrow s[1] \\ X' & \longrightarrow & Y' & \longrightarrow & Z' & \longrightarrow & X'[1] \end{array}$$

whose rows are distinguished triangles with $s, s' \in S$, there exists a morphism

$$s'' : Z \rightarrow Z'$$

in S such that (s, s', s'') is a morphism of triangles.

◇ **Lemma 102.1.3.2** ...

Quotient of Triangulated Category

◇ **Lemma 102.1.3.3**

102.2 The Homotopy Category

Definition 102.2.0.1 Let $\mathcal{A}$ be additive category:

- We set $Comp(\mathcal{A}) = CoCh(\mathcal{A})$ be the **category of (cochain) complexes**.
- A complex $K^\bullet$ is **bounded below** if:

...

-
-
-

-
-

102.3 Derived Category

102.3.1 Filtered Derived Category

Part LI

Syzygies

Chapter 103 Introduction: What is Syzygy?

Notions:

103.1 Hilbert's Theorem on Syzygies

† **Problem 103.1.1** Let: $S = \mathbb{C}[z_0, \dots, z_r]$ in $r + 1$ variables over the complex numbers. Let E be a finitely generated graded S -module. Choose homogeneous generators:

$$m_1, \dots, m_b \in E$$

with $\deg(m_i) = a_{0,i}$. These determine a surjective mapping

$$\oplus S(-a_{0,j}) \xrightarrow{\epsilon} E \rightarrow 0 \quad (f_1, \dots, f_b) \mapsto \sum f_i \cdot m_i$$

Then, choose $\ker(\epsilon)$ and continue apply the method above, we build a sequence

$$\dots \longrightarrow P_2 = \oplus S(-a_2, j) \longrightarrow P_1 = \oplus S(-a_{1,j}) \longrightarrow E \longrightarrow 0$$

If at each stage we choose the generators minimally in a suitable sense, we can arrange that the matrices defining the δ_i not have non-zero constant entries. In this case: It is elementary that $P_\bullet$ is unique up to isomorphism.

► **Theorem** (Hilbert) The process just described terminates after at most $r + 1$ steps.

In other words, any finitely generated S -module E has a minimal graded free resolution of length at most $r + 1$, up to isomorphism.

Remark Syzygies and Varieties:

Notice that, the base ring S is Noetherian. Any variety can be represented as an ideal generated by finite numbers of generators,

$$X = V(f_1, \dots, f_r)$$

Then, consider the syzygy of

$$\oplus S(-d) \rightarrow A(X) \rightarrow 0$$

♣ **Example 103.1.0.1** (Resolution of the Twisted Quartic Curve) Consider the following curve

$$\mathbb{P}^1 \hookrightarrow \mathbb{P}^3 \quad [s, t] \mapsto [s^4, s^3t, st^3, t^4]$$

Take the ideal $I_C \subseteq S = \mathbb{C}[x, y, z, w]$ of C , generated by a quadric and three cubics:

$$Q = yz - xw, \begin{cases} F_1 = z^3 - yw^2 \\ F_2 = xz^2 - y^2w \\ F_3 = y^3 - x^2z \end{cases}$$

Then, the module of Syzygies among these generators is spanned by the four relations:

$$\begin{cases} z^2Q - yF_1 + wF_2 = 0 \\ ywQ - xF_1 + zF_2 = 0 \\ xzQ - yF_2 - wF_3 = 0 \\ y^2Q - xF_2 - zF_3 = 0 \end{cases}$$

† **Problem 103.1.2** *Is there any structure to the totality of all resolutions of some large class of S -modules.*

There are very few modules E for which one can write down the resolution $P_\bullet$ explicitly. Instead, much of our attention will be focused on the question relating to the **grading** of $P_\bullet$. This involves two sorts of invariants.

Definition 103.1.0.2 The (graded) Betti numbers $b_{i,j} = b_{i,j}(E)$ are defined by writing

$$P_i = \bigoplus_j S(-j)^{b_{i,j}}$$

with $b_{i,j}$ numbers in degree of j .

Another definition of Betti numbers is given in algebraic topology. Those definitions coincide in Homological algebra.

† **Problem 103.1.3** *Which Betti numbers $b_{ij}(E)$ are nonzero?*

Definition 103.1.0.3 Define:

$$\text{reg}(E) := \max\{j - i \mid b_{i,j}(E) \neq 0\}$$

Then, $\text{reg}(E) \leq m \Leftrightarrow E$ is generated in degrees $\leq m$ and for every i the i th module of syzygies of E is generated in degrees $\leq i + m$.

Chapter 104 Hilbert's Syzygies Theorem, Free Resolution and Hilbert Functions

104.1 The Study of Syzygies

104.1.1 Hilbert's Functions

Definition 104.1.1.1 Let:

- Space $\mathbb{P}^r := \mathbb{P}_K^r, S = K[X_0, \dots, X_r]$;
- Let $M = \bigoplus M_d$ be graded S -module, which is finitely generated;
Then, M_d is a finite-dimensional vector space over K . (Each S_d is finite-dimensional and M_d is finitely generated)
- Define:

$$H_M(d) := \dim_K(M_d)$$

104.2 Examples of Free Resolutions

► **Theorem** Let $v_1, \dots, v_{ds}$

◦ **Proof**

□

104.3 The Local Setting

Fix a Noetherian local ring $(A, \mathfrak{m})$ of dimension n .

104.4 Some Local Invariants

104.5 Graded Syzygies

Chapter 105 Introduction to Boij-Söderberg Theory

Part LII

Algebraic Topology

Part LIII

Differential Topology

Part LIV

Topological K-Theory and Characteristic Classes

Chapter 106 Vector Bundles

106.1 Quasi-Vector Bundles

Let $k = \mathbb{R}, \mathbb{C}$

Definition 106.1.0.1 A **quasi-vector bundle** with base X is given by:

- A finite dimensional k -vector space E_x for every point $x \in X$;
- A topology $E = \coprod E_x$;

A quasi-vector bundle

Remark This also is called family of vectors.

♣ **Example 106.1.0.2** ...

♣ **Example 106.1.0.3** ...

106.2 Vector Bundles

Definition 106.2.0.1 An n -dimensional vector bundle is a map $p : E \rightarrow B$ together with

- a real vector space structure on $p^{-1}(b)$ (**fibre**) for each $b \in B$;
- There exists a covering of B by U_α such that: Equipped with homeomorphism

$$h_\alpha : p^{-1}(U_\alpha) \xrightarrow{\cong} U_\alpha \times \mathbb{R}^n$$

which is a homomorphism. And $h_\alpha|_{p^{-1}(b)} : p^{-1}(b) \rightarrow \{b\} \times \mathbb{R}^n$ is a vector isomorphism.
such h_α is called **trivialization** of the vector bundle.

In principle, $K = \mathbb{C}, \mathbb{R}$.

♣ **Example 106.2.0.2** ...

♣ **Example 106.2.0.3** ...

♣ **Example 106.2.0.4** (Tangent Bundle)

♣ **Example 106.2.0.5** (Normal Bundle)

♣ **Example 106.2.0.6** ...

106.2.1 title

106.3 Classifying Vector Bundles

106.3.1 Pullback Bundles

106.3.2 Clutching Functions

106.3.3 Cell Structures on Grassmannians

106.4 Homotopy Theory of Vector Bundles

► **Theorem** Let X be a compact space and E be a vector bundle over $X \times I$, $I = [0, 1]$. Let

$$\begin{array}{ccc} \alpha_t : X \rightarrow X \times I & \prod : X \times I \rightarrow X \\ x \mapsto (x, t) & (x, u) \mapsto x \end{array}$$

Then the vector bundles $E_0 = \alpha_0^*(E)$ is isomorphic to $E_1 = \alpha_1^*(E)$.

◦ **Proof ...**

□

► **Theorem ...**

◦ **Proof ...**

□

► **Theorem ...**

◦ **Proof ...**

□

106.5 Metrics and Forms on Vector Bundles

Chapter 107 K-Theory

107.1 The Functor $K(X)$

Chapter 108 Operations

Chapter 109 Bott Periodicity

Chapter 110 Computation of Some K-Groups

Chapter 111 Applications of K-Theory

Chapter 112 Stiefel-Whitney Classes

Chapter 113 Cell-Structure for Grassman- nian Manifolds

Chapter 114 The Cohomology Ring

Chapter 115 Chern-Weil Theory

Part LV

Classical Algebraic Geometry

Contents

Chapter 116 Fundamentals

116.1 Affine and Projective Spaces

116.1.1 Affine Spaces, Affine Varieties, Quasi-Affine Varieties

Definition 116.1.1.1

- By **affine space** over field K , we mean simply the vector space K^n . This is usually denoted as

$$A_K^n \quad \text{or} \quad A^n$$

(And the origin point plays no role in the affine space)

- An **affine variety** is the common zero locus of a collection of polynomials $f_\alpha \in K[X_1, \dots, X_n]$.
- An open subset of an affine variety is a **quasi-affine variety**.

116.1.2 Projective Space, Projective Varieties

Projective Space, Projective Varieties

Definition 116.1.2.1

- By **projective space** over field K , we mean the quotient of $K^{n+1} - \{0\}$ specified by

$$(x_1, \dots, x_n) \sim (\lambda x_1, \dots, \lambda x_n), \lambda \in K^\times$$

- For vector space V ,

$$\mathbb{P}(V) := (V - \{0\}) / \sim$$

- Polynomials: Homogeneous with degree d

$$F(\lambda X_0, \dots, \lambda X_n) = \lambda^d F(X_0, \dots, X_n)$$

- A **projective variety** is define to be zero locus of a family numbers of homogeneous polynomials F_α .

Action of $PGL_{n+1}K$

Definition 116.1.2.2

- Firstly, define the matrix algebra $PGL_{n+1}K$

$$PGL_{n+1}K := GL_{n+1}K / \sim$$

- Secondly, define the action on $\mathbf{P}^n$ via action of $GL_{n+1}K$ on $\mathbf{R}^{n+1}$

$$\begin{aligned} PGL_{n+1}K \times \mathbb{P}^n &\rightarrow \mathbb{P}^n \\ ([A], [v]) &\mapsto [Av] \end{aligned}$$

- Two varieties $X, Y \subseteq \mathbf{P}^n$ are **projectively equivalent** iff they are congruent modulo the map.

$$X \sim Y \Leftrightarrow \exists \varphi \in PGL_{n+1}K, \varphi(X) = Y$$

Remark Actually, in Harris's book, the projective space is defined as: The one-dimensional subspaces of the vector space V .

$$\mathbb{P}V := \{L \subseteq V \mid V \in Vect_K, \dim(V) = 1\}$$

Homogeneous Coordinates, Dual Space

♣ **Example 116.1.2.3** When we consider the coordinate of $\mathbb{P}V$, we may think of, to write those coordinate in the form of

$$Z_0, \dots, Z_n$$

However, this is a misleading. Z_i are not function on $\mathbb{P}^n$: By definition we have

$$[Z_0, \dots, Z_n] = [\lambda Z_0, \dots, \lambda Z_n], \lambda \in K$$

On their otherwise, (Z_i/Z_j) are functions, the denominator is nonzero.

Definition 116.1.2.4

- Define the dual of projective space via

$$(\mathbb{P}V)^* := \mathbb{P}V^*$$

where V^* has elements of linear functions $Z_i : V \rightarrow K$.

- Similarly: The homogeneous functions of degree d can be naturally identified with

$$Sym^d(V^*) \cong F[X_0, \dots, X_n]^d$$

Remark The **coordinate** of points in $\mathbb{P}V$ is given by

$$\begin{aligned} \mathbb{P}V &\rightarrow \mathbb{P}^n \\ v &\mapsto [Z_0(v) : \dots : Z_n(v)] \end{aligned}$$

Affine Coverings

Definition 116.1.2.5

$$U_i \subseteq \mathbb{P}^n, U_i := \{[Z_0, \dots, Z_n] | Z_i \neq 0\}$$

Then, the ratios $z_j := Z_j/Z_i$ are well defined. Give a bijection

$$U_i \cong \mathbb{A}^n \quad [Z_0, \dots, Z_n] \mapsto (z_1, \dots, z_n)$$

Geometrically, we can think of the map as associating to a line $L \subseteq K^{n+1}$ not contained in the hyperplane $Z_i = 0$. Its point p with the affine plane intersects with the affine plane $Z_i = 1 \subseteq K^{n+1}$.

Remark In fact, this is a projection. See here: 115.5.0.2

Definition 116.1.2.6 We may thus think of projective space as a compactification of affine space. The function z_j of U_i are called **affine/Euclidean coordinates** on the projective space or on the open set U_i .

◇ **Lemma 116.1.2.7** Let $X \subseteq \mathbb{P}^n$ be a variety. Then $X_i = X \cap U_i$ is an affine variety.

◦ **Proof** Suppose that X is specified by polynomials. And one of polynomials is $F \in K[Z_0, \dots, Z_n]$ which is homogeneous. Then, for U_i

$$f = F(Z_0, \dots, Z_n)/Z_i^d = F(Z_0, \dots, 1, \dots, Z_n); d = \deg(F)$$

□

Thus: Any projective space is the union of affine spaces, any projective variety is the union of affine varieties.

◇ **Lemma 116.1.2.8** Conversely: Every affine variety is the intersection of U_0 with a projective variety X . Given by

$$F_\alpha(Z_0, \dots, Z_n) := Z_0^{d_\alpha} \cdot f_\alpha(Z_1/Z_0, \dots, Z_n/Z_0)$$

◦ **Proof** Omitted. It is the inverse process of the lemma above.

□

Remark If $\mathbb{R}, \mathbb{C}$ are given usual topology, then

$$\mathbb{P}^n(\mathbb{R}) \quad \mathbb{P}^n(\mathbb{C})$$

are compact. The two can be seen as a compactification of $\mathbb{R}^n, \mathbb{C}^n$, which can be imbedded in $\mathbb{R}^{n+1}, \mathbb{C}^{n+1}$.

Dual Projective Space

Definition 116.1.2.9 (Hyperplane) The hyperplane can be specified by the linear equation

$$H(l) := \{[v] \in \mathbb{P}V^* | l(v) = 0\}, l \in V^*$$

It is equivalent to the definition later.

Definition 116.1.2.10 The hyperplanes in a projective space $\mathbb{P}^n$ is again a projective space, denoted by $\mathbb{P}^{n*}$.

$$\mathbb{P}^{n*} := \{H(l) | l \in V^*\} \cong \mathbb{P}(V^*)$$

More generally, if $\Lambda \cong \mathbb{P}^{n-k-1}$, then

$$\mathbb{P}^{n-k-1*} = \{H(l) | \Lambda \subseteq H(l)\}$$

Especially: $W \subseteq V, (V/W)^* = \text{Ann}(W)$. We have $\mathbb{P}(\text{Ann}(W)) \subseteq \mathbb{P}(V^*)$.

116.1.3 Examples

Linear Spaces

Definition 116.1.3.1 (Linear Subspace in Projective Space) An inclusion of vector spaces is $W \cong K^{k+1} \hookrightarrow V \cong K^{n+1}$. This induces

$$\mathbb{P}K^{k+1} \hookrightarrow \mathbb{P}K^{n+1}$$

The image Λ is called the **linear subspace** of $\mathbb{P}^n$.

In the case:

- $n = k + 1$ Hyperplane;
- $k = 1$ Line;

Operations on Linear Spaces:

Definition 116.1.3.2 (Span of Linear Subspaces) Suppose $\begin{cases} \Lambda = \mathbb{P}W \\ \Lambda' = \mathbb{P}W' \end{cases}$, which are the linear subspaces of $\mathbb{P}^n$. Then, define

$$\overline{\Lambda, \Lambda'} := \text{The smallest linear subspace containing } \Lambda, \Lambda'$$

In general, for any subset $\Phi, \Gamma \subseteq \mathbb{P}^n$, the span $\overline{\Gamma \cup \Phi}$.

Remark It is easy to show, that

$$\overline{\Gamma \cup \Phi} = \mathbb{P}(W + W')$$

Remark We will define the dimension later.

♠ **Proposition 116.1.3.3**

$$\dim(\overline{\Lambda, \Lambda'}) = \dim(\Lambda) + \dim(\Lambda') - \dim(\Lambda \cap \Lambda')$$

◦ **Proof** Omitted. □

Remark Especially, define

$$\dim \mathbb{P}^0 := -1$$

Then: We can get the following inequality

$$\dim(\Lambda \cap \Lambda') = \dim(\Lambda) + \dim(\Lambda') - \dim(\overline{\Lambda + \Lambda'}) \geq \dim(\Lambda) + \dim(\Lambda') - \dim(\mathbb{P}V)$$

Finite Sets

♠ **Proposition 116.1.3.4** Any finite subset Γ of $\mathbb{P}^n$ is a variety.

◦ **Proof** Given any $q \notin \Gamma$. We can find a polynomial on $\mathcal{P}^n$ vanishing on the point $p_i \in \Gamma$ but not at q (Especially, it is a homogeneous linear form L_i) Take its product

$$F := \prod L_i$$

And F_q is associated to $q \notin \Gamma$. Then, $\Gamma = \cap_{q \notin \Gamma} V(F_q)$. □

† **Problem 116.1.1** The smallest degree polynomial that suffice to describe a given variety $\Gamma \subseteq \mathbb{P}^n$?

◇ **Lemma 116.1.3.5** Suppose $\text{card}(\Gamma) = d$. And Γ is not contained in a line. Then Γ may be described as the zero locus of polynomials of degree $d - 1$ and less.

◦ **Proof** Notice that: When $d = 3, n \geq 2$.

The space of polynomials of degree $m = 2$, the space of polynomials

$$\dim V_m = \binom{n+m}{n} = \binom{n+2}{n} \geq 6$$

And the valuation map

$$V_m \rightarrow k^3 \quad F \mapsto (F(P_1), F(P_2), F(P_3))$$

is a linear map. The dimension of kernel ≥ 3 .

In general case, for other $d - 3$ points, We can add the degree of polynomial by multiplying the linear homogeneous form. □

Especially, to describe the lower bound of n for $\mathbb{P}^n$ and m :

Remark 'And less' is unnecessary: If a variety $X \subseteq \mathbb{P}^n$ is the zero locus of the polynomials F_α of degree $d_\alpha \leq m$, then it may also be represented as the zero locus of

$$X^l \cdot F_\alpha$$

where for each α , the monomial X^l range over all monomials of degree $m - d_\alpha$.

Dependent and Independent

Another direction in which we can go is to focus our attention on sets of points that satisfy no more linear relation than they have to. To be precise, we say

Definition 116.1.3.6 We say that l points $p_i = [v_i] \in \mathbf{P}^n$ are **independent** if the corresponding vector v_i are independent. Or equivalently,

$$\dim[\text{Span}(v_1, \dots, v_l)] = l - 1$$

◇ **Lemma 116.1.3.7**

- Any $n + 2$ points in $\mathbf{P}^n$ are dependent.
- $n + 1$ points are dependent $\Leftrightarrow$ They lie in a hyperplane;

◦ **Proof**

- Notice that,

$$\text{Span}(v_1, \dots, v_{n+2}) = \mathbb{R}^{n+1}$$

There must be two vectors which are not linear independent.

$$\lambda w_1 + \mu w_2 = 0$$

Then, $[w_1], [w_2]$ are on the same projective line

$$L = \{[\lambda p + \mu q] \mid [p], [q] \in \mathbb{P}^n, (\lambda : \mu) \in \mathbb{P}^1\}$$

- $n + 1$ points $[v_i]$ are dependent $\Leftrightarrow \dim \text{span}(v_1, \dots, v_{n+1}) \leq n \Leftrightarrow$ There exists a vector $w \notin \text{span}(v_1, \dots, v_{n+1})$. So

$$\text{span}(v_1, \dots, v_{n+1}) \hookrightarrow \mathbb{A}_K^n \hookrightarrow \mathbb{A}_K^{n+1}$$

Take projective space.

□

General Position

Definition 116.1.3.8 We say a finite set $\Gamma \subseteq \mathbf{P}^n$ is in **general position** if no $n + 1$ or fewer of them are dependent.

We can ask: What is the smallest degree needed to cut out a set of points in general position in $\mathbf{P}^n$?

► **Theorem** If $\Gamma \subseteq \mathbb{P}^n$ is any collection of $d \leq 2n$ points in general position. Then Γ may be described as the zero locus of quadratic polynomials.

- **Proof** Let $\Gamma = \{p_1, \dots, p_{2n}\}$ consisting of exactly $2n$ points. The general case:

Suppose now that $q \in \mathbb{P}^n$ is any point such that: Every quadratic polynomial vanishing on Γ vanishes at q
 $\Rightarrow q \in \Gamma$

Suppose $\Gamma = \Gamma_1 \cup \Gamma_2$ is a decomposition of Γ into sets of cardinality n , then each Γ_i spans a hyperplane

$$\Lambda_i \subset \mathbb{P}^n$$

Then, union of hypersurfaces $\Lambda_1 \cup \Lambda_2$ is the zero locus of polynomials vanishing on Γ . (Detail: The two hypersurfaces can be described by linear equation. Then consider their product). So $p \in \Lambda_1 \cup \Lambda_2$.

Now, let $p_1, \dots, p_k$ be any minimal subset (in the sense of cardinal) of Γ such that: $q \in \text{Span}(p_1, \dots, p_k)$. By preceding, we can take $k \leq n$. (General Position)

Remark

Suppose that $\Sigma \subseteq \Gamma - \{p_1, \dots, p_k\} = \{p_{k+1}, \dots, p_{2n}\}$; $\text{card}(\Sigma) = n - k + 1$

$$p_1 \notin \Lambda = \text{span}(p_2, \dots, p_k, \Sigma)$$

If not, then Λ is a hypersurface, contradict to the general position. Also hold $q \notin \Lambda$ because

$$q \in \Lambda, p_1 \notin \Lambda \Rightarrow q \in \Lambda \cap \text{span}\{p_1, \dots, p_k\}$$

Contraidict to minimal subset.

Finally $q \in \text{span}(\Gamma - \{p_2, \dots, p_k\} - \Sigma)$. (One of Λ_1, Λ_2). And we take arbitrary Σ . So

$$q \in \cap \text{span}(\Gamma - \{p_2, \dots, p_k\} - \Sigma) = \{p_1\}$$

□

► **Theorem** In general for $k \geq 2$, any collection Γ of $d \leq kn$ points in general position may be described by polynomials of degree k and less.

◦ **Proof** Similarly.

□

An associated proposition is: 115.1.3.18

◇ **Lemma 116.1.3.9** Consider $n + 2$ points in general points in $\mathbb{P}^n$. For $n + 1$ points of them, there exists projective transformation

$$\varphi(P_i) = [e_i]$$

And $\varphi(P_{n+1}) = [1 : \dots : 1]$.

◦ **Proof** Suppose that $P_i = [v_i]$. Then, notice that there exists a transformation

$$\varphi : \text{Span}(v_0, \dots, v_n) \rightarrow \text{Span}(e_0, \dots, e_n)$$

such that $\varphi(v_i) = e_i$. Then take $[\varphi]$.

□

Remark $\dim PGL_{n+1} = (n + 1)^2 - 1 = n(n + 2)$, and the constriant for $n + 2$ points $(n + 2) \cdot n$.

► **Theorem** Any two ordered subsets of $n + 2$ points in general position in $\mathbf{P}^n$ are projectively equivalent.

◦ **Proof** Apply the transformation above. □

Cross Ratio

† **Problem 116.1.2** When could two ordered subsets of $d \geq n + 3$ points in general position in $\mathbf{P}^n$ be projectively equivalent? The question is answered in the case of $n = 1$:

Definition 116.1.3.10 (Cross-Ratio) Define:

$$\lambda(z_1, \dots, z_4) := \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_3)(z_2 - z_4)}$$

Especially, the function $\lambda(z_1, z_2; z_3, -) : \mathbf{P}^1 \rightarrow \mathbf{P}^1$ (Linear Homogeneous Form) and $z_1 \mapsto 1, z_2 \mapsto \infty, z_3 \mapsto 0$.

♠ Proposition 116.1.3.11

- (n=1) $z_1, \dots, z_4; z'_1, \dots, z'_4 \in \mathbf{P}^1$ are projectively equivalent $\Leftrightarrow \lambda(z_1, \dots, z_4) = \lambda(z'_1, \dots, z'_4)$;
- In general, for points $p_i, p'_i \in \mathbf{P}^n$:
 $(p_1, \dots, p_{n+3})$ and $(p'_1, \dots, p'_{n+3})$ are equivalent $\Leftrightarrow \lambda(p_1, p_2; p_3, p_i) = \lambda(p'_1, p'_2; p'_3, p'_i); i = 4, \dots, n + 3$.

◦ **Proof**

-
-

□

Hypersurfaces

Definition 116.1.3.12 A hypersurface X is a subvariety of $\mathbf{P}^n$ described as the zero locus of one single homogeneous polynomial

$$F(Z_0, \dots, Z_n)$$

And with the definition later, we have $\dim \Gamma = V(F) = n - 1$.

◇ **Lemma 116.1.3.13** Every hypersurface X can be described as a polynomial F with no repeated prime factor.

◦ **Proof** Via Hilbert's Nullstellensatz, for prime f ,

$$I(V(f^e)) = (f) \Rightarrow V(f) = V(f^e)$$

Then, consider each component of $\Gamma = V(F) = V(\prod F_i^{e_i}) = \coprod V(F_i)$ □

Remark Hilbert's Nullstellensatz is the tool, which could make the definition in classical algebraic geometry and the definition in algebraic geometry coincided.

We must consider the condition, that for example, $f^n = 0$.

Analytic Subvarieties and Submanifolds

♣ **Example 116.1.3.14** Observe the polynomial

$$f \in \mathbb{C}[z_1, \dots, z_n]$$

with complex coefficient are holomorphic function (Complex Analysis in Multivariable)

An algebraic variety X in $\mathbb{A}_{\mathbb{C}}^n$ or $\mathbb{P}_{\mathbb{C}}^n$ will be particular a complex analytic subvariety of these complex manifolds.

Notably, this give us an a priorinotion of dimension of an algebraic variety $X \subseteq \mathbb{P}_{\mathbb{C}}^n$, and likewise a smooth and singular points of X .

Remark But the definition of dimension, smoothness and singularity is not algebraic...

► **Theorem** (Chow's Theorem) If $X \subseteq \mathbb{P}_{\mathbb{C}}^n$ is any **complex analytic subvariety** then X is an algebraic subvariety.

◦ **Proof** ... □

This is an inverse version of statement above.

The Twisted Cubic

Definition 116.1.3.15 The **twisted cubic** given in terms of affine coordinates on both spaces by

$$v : x \mapsto (x, x^2, x^3)$$

And its image is the curve. Or in terms of homogeneous coordinates on both

$$v : [X_0, X_1] \mapsto [X_0^3, X_0^2 X_1, X_0 X_1^2, X_1^3] = [Z_0, Z_1, Z_2, Z_3]$$

C lies on 3 quadric surfaces Q_0, Q_1 and Q_2 . Given by the zero locus of following homogeneous polynomials

$$\begin{cases} Q_1 : F_0(Z) = Z_0 Z_2 - Z_1^2 \\ Q_2 : F_1(Z) = Z_0 Z_3 - Z_1 Z_2 \\ Q_3 : F_2(Z) = Z_1 Z_3 - Z_2^2 \end{cases}$$

◇ **Lemma 116.1.3.16**

- Show that for any $0 \leq i \leq j \leq 2$, the surface Q_i and Q_j intersect in the union of C and a line L ;
- Generally, for any $\lambda = [\lambda_0, \dots, \lambda_2]$, let

$$F_\lambda = \lambda_0 F_0 + \lambda_1 F_1 + \lambda_2 F_2$$

And Q_λ is the surface defined by F_λ . Then,

$$Q_\lambda \cap Q_\mu = L_{\lambda, \mu} \cup C$$

◦ **Proof**

$$\bullet Q_1 \cap Q_2 = \{(Z_0, Z_1, Z_2, Z_3) | Z_0 Z_2 - Z_1^2 = 0, Z_0 Z_3 - Z_1 Z_2 = 0\}$$

$$Q_1 \cap Q_3 = \{(Z_1, Z_2, Z_3, Z_4) | Z_0 Z_2 - Z_1^2 = 0, Z_1 Z_3 - Z_2^2 = 0\}$$

We have

$$Z_3 F_0(Z) = Z_1 F_1(Z) \Rightarrow Z_1(Z_1 Z_3 - Z_2^2) = 0$$

$$Z_2 F_0(Z) = Z_0 F_2(Z) \Rightarrow Z_2(Z_1^2 - Z_0 Z_2) = 0$$

When

$$Z_1 = 0 \Rightarrow Z_0 Z_2 = 0, Z_0 Z_3 = 0 \Rightarrow Z_0 = 0$$

$$Z_2 = 0 \Rightarrow Z_1^2 = 0, Z_1 Z_3 = 0 \Rightarrow Z_1 = 0$$

- Given in the part of determinant variety.

□

Family:

Rational Normal Curve

Definition 116.1.3.17 This is the generalization of the twisted cubic:

$$v_d : P^1 \rightarrow P^d \quad [X_0 : X_1] \mapsto [X_0^d, X_0^{d-1} X_1, \dots, X_1^d] = [Z_0, \dots, Z_d]$$

The image $C \subseteq \mathbb{P}^d$ is readily seen to be common zero locus of polynomials

$$F_{i,j}(Z) = Z_i Z_j - Z_{i-1} Z_{j+1}; 1 \leq i \leq j \leq d-1$$

♠ **Proposition 116.1.3.18** If $p_1, \dots, p_{kd+1}$ are any points on a rational normal curve of $\mathbb{P}^d$, then any polynomial F of degree k on $\mathbb{P}^d$ vanishing on the points p_i vanish on C .

◦ **Proof ...**

□

Determinantal Representation of Rational Normal Curve

Definition 116.1.3.19 The rational normal curve can be represented as the zero locus of $\det \begin{pmatrix} z_i & z_{i+1} \\ z_{i+1} & z_{i+2} \end{pmatrix}$, in general, can be represented as the matrix

$$\begin{pmatrix} z_0 & z_1 & \dots & z_k \\ z_1 & z_2 & \dots & z_{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ z_{d-k} & \dots & \dots & z_d \end{pmatrix}$$

with rank 1.

Another Parametrization of the Rational Normal Curve

Definition 116.1.3.20 ...

The Family of Plane Conics

Other Rational Curves

- 1.
- 2.
- 3.

116.1.4 Quasi-Projective Varieties

Definition 116.1.4.1 An open subset of projective variety is **quasi-projective** variety.

116.2 Regular Functions, Regular Maps and Irreducibility

116.2.1 Zariski Topology

Definition 116.2.1.1 Define:

- Open and closed subsets in $\mathbb{A}^n$

$$\begin{cases} D(f) = \{x \in \mathbb{A}^n | f(x) \neq 0\} \\ V(f) = \{x \in \mathbb{A}^n | f(x) = 0\} \end{cases}, f \in K[X_1, \dots, X_n]$$

- Open and closed subsets in $\mathbb{P}^n$

$$\begin{cases} D(F) = \{x \in \mathbb{P}^n | F(x) \neq 0\} \\ V(F) = \{x \in \mathbb{P}^n | F(x) = 0\} \end{cases}, F \in K[X_0, \dots, X_n]^h$$

◇ **Lemma 116.2.1.2** We have:

- $\cap V(F_\alpha) = V(\{F_\alpha\}) = V(\prod F_\alpha)$;
- $\cup V(F_\alpha) = V(\prod F_\alpha)$

◦ **Proof** ...

□

♠ **Proposition 116.2.1.3** The Zariski topology is Noetherian.

◦ **Proof** According to Hilbert's basis theorem.

□

Quasi-Affine/Projective Variety

Definition 116.2.1.4 Especially: Let X be a variety. An open subset $U \subseteq X$ is called a **quasi-projective variety**. Equivalently, it is a locally closed subset of $\mathbb{P}^n$: For each

We have the following statement

The quasi varieties is much larger than varieties.

Remark Notice that: The topology is generated by those **principle elements**. But, what we must note, is that:

Not every open/closed subset in Zariski topology is principle.

An obvious logic proof: The polynomial ring in multivariables is not PID.

116.2.2 Regular Functions on Affine Varieties

Definition 116.2.2.1

- Let $X \subseteq \mathbb{A}^n$ be a variety. We define the **ideal** of X to be the ideal

$$I(X) := \{f \in K[z_1, \dots, z_n] \mid f \equiv 0 \text{ on } X\}$$

- Define the **coordinate ring**

$$A(X) := K[z_1, \dots, z_n]/I(X)$$

A special case is the regular function on open subsets:

Definition 116.2.2.2 (Affine) Let $U \subseteq X$ be an open subset. We say that a function f on U is **regular** at p if in some neighborhood V of p , the function can be expressed as a quotient

$$f|_{D(w) \cap X} = \frac{g}{h}; g, h \in K[z_1, \dots, z_n], h(p) \neq 0$$

f is **regular on** U if it is regular at every point of U .

◇ **Lemma 116.2.2.3** The ring of functions regular at every point of X is the **coordinate ring** $A(X)$.
More generally, for $U = D(f)$, we have

$$A(U) = A(X)_f$$

which is the localization.

◦ **Proof** Important tool: A step of Hilbert's Nullstellensatz.

$$h(z) \neq 0, z \in D(w) \Leftrightarrow w(z) \neq 0 \Rightarrow h(z) \neq 0 \Rightarrow D(w) \subseteq D(h)$$

$$f \text{ is regular on } U \Leftrightarrow \forall p \in D(w), f|_{D(h)} = \frac{g}{h}.$$

Now, we will prove that

$$A(X) = \{\text{The ring of functions regular at every point of } X\}$$

Equivalently, for $h \in K[X_1, \dots, X_n]$,

$$\text{image of } h \text{ is a unit in } A(X) \Leftrightarrow h \text{ is nowhere zero on } X$$

$\Rightarrow$ Obviously;

$\Leftarrow$ Firstly, Suppose h is nowhere zero on X . $V(h) \cap X = \emptyset$. According to Hilbert's Nullstellensatz (version for quotient correspondence), we have: In algebra $A(X)$

$$V(\bar{h}) = \emptyset \Rightarrow \sqrt{\bar{h}} = A(X) \Rightarrow 1 = g \cdot \bar{h}^e$$

This implies, that h is invertible in $A(X)$.

□

Continuation of Regular Functions on Irreducible Sets

♠ **Proposition 116.2.2.4** Suppose X be an irreducible variety. Then, any pair (U, f) has a unique continuation F on X such that

$$F|_U = f$$

◦ Proof

- Existence: Firstly, (U, f) has a local representation

$$f = \frac{g}{h^n}$$

Then, take the finite covering

$$X = \cup_{i=1}^n D(h_i)$$

Because of U is dense, so on every $D(h_i) \cap D(h_j)$, we have

$$\frac{g_i}{h_i^{n_i}} = \frac{g_j}{h_j^{n_j}}$$

Construct the global morphism

...

- Uniqueness: Omitted.

□

This construction is also applied for the proof on the Zariski topology of $\text{Spec}(A)$.

116.2.3 Regular Functions on Projective Varieties

116.2.4 Regular Maps

116.2.5 The Veronese Map

Definition 116.2.5.1 For any n, d , define the **Veronese map**

$$v_d(\mathbb{P}^n) : \mathbb{P}^n \rightarrow \mathbb{P}^N \quad [X_0, \dots, X_n] \mapsto [\dots, X^l, \dots]$$

where X^l is the polynomial of degree d in $X_0, \dots, X_n$.

◇ **Lemma 116.2.5.2** We have the following relation

$$N = \binom{n+d}{d} - 1$$

◦ **Proof** This is an enumeration problem. We should apply stars-and-bars method, for $n+d$ positions, n stars and d bars. □

♣ **Example 116.2.5.3** Consider the quadratic Veronese map

$$v_2 : \mathbb{P}^2 \rightarrow \mathbb{P}^5 \quad [X_0, X_1, X_2] \mapsto [X_0^2, X_1^2, X_2^2, X_0X_1, X_0X_2, X_1X_2]$$

And $Im(v_2)$ is called the **Veronese surface**.

It can be represented as:

$$\begin{cases} Z_0Z_1 = Z_3^2 \\ Z_0Z_2 = Z_4^2 \\ Z_1Z_2 = Z_5^2 \\ Z_3Z_5 = Z_1Z_4 \\ Z_4Z_5 = Z_2Z_3 \\ Z_3Z_4 = Z_0Z_5 \end{cases}$$

and has a determinantal representation of Veronese varieties

$$\begin{pmatrix} Z_0 & Z_3 & Z_4 \\ Z_3 & Z_1 & Z_5 \\ Z_4 & Z_5 & Z_2 \end{pmatrix}$$

has rank 1.

Properties of Ideals of Veronese Variety

With this example, we can find its generalization: Let I, J, K, L be the coordinate $(\alpha_0, \dots, \alpha_n)$, $\sum \alpha_i = d$.

$$X^I X^J = X^K X^L$$

Then, we can associate index to coordinates in $\mathbb{P}^N$. So,

♠ **Proposition 116.2.5.4** The Veronese variety is the intersection of quadric hypersurfaces

$$Im(v_d(\mathbb{P}^n)) = \cap V(Z_I Z_J - Z_K Z_L)$$

◦ **Proof**

- $\subseteq$ Omitted.
- $\supseteq$ Conversely, the index and pair (I, J) are bijective.

□

Subvarieties

In particular, we will consider the restriction of v_d on linear subspace $\Lambda \cong \mathbb{P}^k \subseteq \mathbb{P}^n$.

♣ **Example 116.2.5.5** Consider a projective line in $\mathbb{P}^2$. Then, the image is plane curve on Veronese surface S .
More generally:

◇ **Lemma 116.2.5.6** Suppose $Y \subseteq \mathbb{P}^n$ be a variety. Then, its image under Veronese map is a subvariety of $\mathbb{P}^N$.

- **Proof** Consider the induced mapping

$$k[y_0, \dots, y_N] \rightarrow k[x_0, \dots, x_n] \quad y_\alpha \mapsto x_0^{a_0} \dots x_n^{a_n}; \alpha = (a_0, \dots, a_n)$$

via this map, we could define the **Veronese ideals**, which are the preimage of ideals in $k[X]$. Thus the image, which is the intersection of polynomials in the ideal of preimage, is also a subvariety. □

Coordinate-Free Description

116.2.6 The Segre Map

Definition 116.2.6.1 Consider the map of Segre maps

$$\sigma : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^{(n+1)(m+1)-1} \quad ([X_0, \dots, X_n], [Y_0, \dots, Y_m]) \mapsto [\dots, X_i Y_j, \dots]$$

where the coordinates in the target space range over all pairwise product of coordinates X_i, Y_j .
The image is called **Segre variety**. Take the notation of

$$\Sigma_{m,n} = \cap_{i,j} V(Z_{i,j} \cdot Z_{k,l} - Z_{i,l} \cdot Z_{k,j})$$

It is also a determinantal variety, via zero locus of 2×2 minor.

♣ **Example 116.2.6.2** The first example is $\Sigma_{1,1} \subseteq \mathbb{P}^3$. It is the zero locus of

...

whose image is simply a quadric surface. Note that: The fibers of two projections maps

$$\mathbb{P}^{n+m} \rightarrow \mathbb{P}^n, \mathbb{P}^m$$

Consider a fiber of $\mathbb{P}^1 \times \mathbb{P}^1$,

♣ **Example 116.2.6.3** (Segre Threefold)

Subvarieties of Segre Varieties

116.2.7 Product, Fibre Product, Tensor Product

Product

Fibre Product

Tensor Product

116.2.8 Graphs

♠ **Proposition 116.2.8.1** Let $X \subseteq \mathbb{P}^n$ be any projective subvariety, and $\varphi : X \rightarrow \mathbb{P}^m$ is a regular map. Then,

$$\Gamma_\varphi \subseteq X \times \mathbb{P}^n = \mathbb{P}^n \times \mathbb{P}^m$$

is a variety.

◦ **Proof** ...

□

116.2.9 Category of Varieties

Definition 116.2.9.1 ...

116.3 Smoothness and Dimension

116.3.1 Smooth and Singular Points

(Zariski Affine) Tangent Space of Hyperplane

Definition 116.3.1.1 To begin, we will consider the situation: Assume that

$$V = V(f)$$

Let $P = (a_1, \dots, a_n) \in V$. The **tangent space** is the hypersurface V , in the form of

$$T_P V := \{(x_1, \dots, x_n) \in \mathbb{A}_k^n \mid \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x_i - a_i) = 0\}$$

where $\frac{\partial f}{\partial x_i}$ is the formal derivative of f with respect to x_i .

◇ **Lemma 116.3.1.2** Let $L \subseteq \mathbb{A}_k^n$ be an affine line through P . Then: P is a multiple root of $f|L \Leftrightarrow L \subseteq T_P V$

◦ **Proof** With parametrization

$$L : x_i = a_i + b_i t, i = 1, \dots, n$$

Then, let $g = f|L$,

$$g(t) = f(a_1 + b_1 t, \dots, a_n + b_n t); t \in \mathbb{A}^1$$

$P = (a_1, \dots, a_n) \in V$. We have

$$g(0) = f|L(P) = 0$$

Then, g has multiroot at 0 $\Leftrightarrow \frac{\partial g}{\partial t} = 0 \Leftrightarrow \sum_{i=1}^n b_i \frac{\partial f}{\partial t_i} = 0 \Leftrightarrow L \subseteq T_P V$ □

Definition 116.3.1.3

- P is called **smooth point** of hypersurface V if there is some i such that

$$\frac{\partial f}{\partial x_i}(P) \neq 0$$

or equivalently, $(\frac{\partial f}{\partial x_i})(P) \neq 0$. Take the notation of collection of smooth points in $V \subseteq \mathbb{A}_K^n$ by V_{smooth} .

- Elsewise is **singular**.

Under these definitions, we could get the conclusion immediately: For hypersurface V

- P is smooth $\Leftrightarrow T_P V$ is affine hyperplane;
- P is singular $\Leftrightarrow T_P V = \mathbb{A}_K^n$.

♠ **Proposition 116.3.1.4** For an irreducible affine hyperplane, V_{smooth} is open and dense in V .

◦ **Proof ...** □

Tangent Space of Irreducible Variety

Definition 116.3.1.5 Let $f \in K[X_1, \dots, X_n]$. Then, the **linear part** of f at $P = (a_1, \dots, a_n)$ is defined as

$$f_P^{(1)} = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(P)(x_i - a_i)$$

Definition 116.3.1.6

- Let V be an irreducible affine variety. Then,

$$T_P V := \cap_{f \in I(V)} f_P^{(1)} \subseteq \mathbb{A}_1^n$$

- For an irreducible variety,

$$\dim V := \min \{ \dim T_P V \mid P \in V \}$$

immediately follows, $\dim V \geq \dim T_P V$.

♠ **Proposition 116.3.1.7** The function

$$X \rightarrow p \mapsto \dim T_p X$$

is upper semi-continuous.

◦ **Proof ...**

□

♠ **Proposition 116.3.1.8** ...

◦ **Proof ...**

□

Definition 116.3.1.9 (Dimension)

-
-
-

Jacobian Criterion

Zariski Cotangent Space

Definition 116.3.1.10 Let X be a variety. Then, define the **Zariski cotangent space**

$$T_p^*(X) = \mathfrak{m}/\mathfrak{m}^2 \quad \mathfrak{m} \subseteq \mathcal{O}_{X,p}$$

Then, the Zariski tangent space can be defined as the dual of Zariski cotangent space

$$T_p(X) = (T_p^*(X))^*$$

Example: Real and Smooth Manifolds

116.3.2 Algebraic Characterizations

116.4 Ideals of Varieties, Irreducible Decomposition and Nullstellensatz

116.4.1 Generating Ideals

Correspondence:

$$\{\text{Subvarieties of } \mathbb{A}^n\} \begin{array}{c} \xrightarrow{I} \\ \xleftarrow{V} \end{array} \{\text{Ideals } I \subseteq K[z_1, \dots, z_n]\}$$

► **Theorem** For any ideal $I \subseteq k[z_1, \dots, z_n]$, the ideal of functions vanishing on the common zero locus of I is the radical of I . i.e.

$$I(V(I)) = \sqrt{I}$$

◦ **Proof ...**

□

116.4.2 Ideals of Projective Varieties

Definition 116.4.2.1 The projective v

♠ **Proposition 116.4.2.2**

116.4.3 Irreducible Varieties and Irreducible Decomposition

Definition 116.4.3.1 We say: A variety is **irreducible** if for any pair of closed subvarieties $Y, Z \subseteq X$,

$$Y \cup Z = X \Rightarrow X = Y \text{ or } X = Z$$

◇ **Lemma 116.4.3.2** $X \subseteq \mathbb{A}^n$ is irreducible $\Leftrightarrow$ Ideal $I(X) \subseteq k[x_1, \dots, x_n]$ is prime.

◦ **Proof ...**

□

♠ **Proposition 116.4.3.3** A variety X is irreducible iff every Zariski open subset of X is dense. i.e. Every two Zariski open subsets of X meet.

◦ **Proof** Suppose X be irreducible, $U \subseteq X$ be Zariski open, then for $V \subseteq X$
 $U \cap V = \emptyset \Leftrightarrow (X - U) \cup (X - V) = X$. Contradiction.

□

♠ **Proposition 116.4.3.4**

- Any variety X may be uniquely expressed as a finite union of irreducible subvarieties X_i with $X_i \not\subseteq X_j, i \neq j$;
-

◦ **Proof ...**

□

► **Theorem ...**

◦ **Proof ...**

□

116.4.4 General Object

116.4.5 Algebras

116.5 Cones, Projection and More about Product

♣ **Example 116.5.0.1** (Cones) Here we start with the hypersurface $\mathbb{P}^{n-1} \subseteq \mathbb{P}^n$, as a hypersurface given by

$$Z_n = 0$$

Take point $p = [0 : \dots : 1]$. Let $X \subseteq \mathbb{P}^{n-1}$ be any variety. Then, we then define the **cone**

$$\overline{X, p} = \cup_{q \in X} \overline{qp}$$

of the lines joining p to points of X .

$\overline{X, p}$ is easily to seen to be a variety: If we choose coordinates as earlier and $X \subseteq \mathbb{P}^{n-1}$ is the locus of polynomials

$$F_\alpha = F_\alpha(Z_0, \dots, Z_{n-1})$$

And the line $\overline{qp}$ is specified by: Suppose $q = [\alpha_0, \dots, \alpha_{n-1}, 0] \in X$

$$\begin{aligned} \overline{qp} &= \overline{Span((\alpha_0, \dots, \alpha_{n-1}, 0), (0, \dots, 0, 1))} \\ &= Span[\{(\lambda\alpha_0, \dots, \lambda\alpha_{n-1}, \mu) | \lambda, \mu \in K\}] \\ &= \{[(\alpha_0, \dots, \alpha_{n-1}), [\mu]] | \mu \in K\} \end{aligned}$$

Then, the cone $\overline{X, p}$ will be the locus of

$$F_\alpha \in K[Z_0, \dots, Z_n]$$

Definition 116.5.0.2 (Generation of Cones) Let: $\Lambda \cong \mathbb{P}^k \subseteq \mathbb{P}^n$ and $\Psi \cong \mathbb{P}^{n-k-1}$ be its complementary linear subspaces. i.e. $\Lambda \oplus \Psi = \mathbb{P}^n$.

Let $X \subseteq \Psi$ be any variety. We can define the **cone** $\overline{X, \Lambda}$ over X with,

$$\overline{X, \Lambda} = \cup_{p \in \Lambda} \overline{X, p}$$

or equivalently,

$$\overline{X, \Lambda} = \cup_{q \in X} \overline{q, \Lambda}$$

♠ **Proposition 116.5.0.3** Let Ψ and $\Lambda \subseteq \mathbb{P}^n$ be complementary linear subspaces as earlier, and $X \subseteq \Psi$ and $Y \subseteq \Lambda$ subvarieties.

The union of all lines joining points of X to points of Y is a variety.

◦ **Proof** Suppose

□

The Joint of Two Varieties

Definition 116.5.0.4 (H&F, P88)

116.5.1 Quadrics

♣ **Example 116.5.1.1** (Quadrics) Suppose $\text{char}(K) \neq 2$.

Definition 116.5.1.2 A quadratic hypersurface $Q \subseteq \mathbb{P}V = \mathbb{P}^n$ is given as the zero locus of a homogeneous quadratic polynomial

$$Q : V \rightarrow K$$

Then, the associated bilinear form can be denoted as Q_0 ,

$$Q_0(v, w) = \frac{Q(v+w) - Q(v) - Q(w)}{2}$$

Then,

$$\tilde{Q} : V \rightarrow V^* \quad v \mapsto (w \mapsto Q_0(v, w))$$

Classifying Problem of Quadratics

♠ **Proposition 116.5.1.3** (Gram-Schmidt) Every quadratic form can be written in the form of

$$\sum x_i^2$$

via choosing Basis.

◦ **Proof** In linear algebra. □

Rank of Quadrics:

More information about quadrics should is given here: 120.

116.5.2 Projection

Resultant

This is a short introduction, aim to provide the tool of proof for the theorem later.

♣ **Example 116.5.2.1**

Definition 116.5.2.2 Let the hyperplane $\mathbb{P}^{n-1} \subset \mathbb{P}^n$ and the point $p \in \mathbb{P}^n - \mathbb{P}^{n-1}$ (as what we did for cone). Then, we can define a map

$$\begin{aligned} \pi_p : \mathbb{P}^n - \{p\} &\rightarrow \mathbb{P}^{n-1} \\ q &\mapsto \overline{qp} \cap \mathbb{P}^{n-1} \end{aligned}$$

► **Theorem** The projection $\tilde{X}$ of X from p to $\mathbb{P}^{n-1}$ is a projective variety.

◦ **Proof ...** □

♣ **Example 116.5.2.3** (Definition of Cones via Projection)

Constructible Sets

116.6 Rational Function and Rational Maps

A useful lemma:

◇ **Lemma 116.6.0.1** (Hartshorne, P24) Suppose X, Y be varieties. And φ, ψ are two morphism $X \rightarrow Y$. Suppose there is a nonempty open subset $U \subseteq X$ such that

$$\varphi|_U = \psi|_U$$

Then, $\varphi = \psi$.

◦ **Proof** Suppose $Y \subseteq \mathbb{P}^n$. Then, by composing

$$Y \hookrightarrow \mathbb{P}^n$$

We can reduce this problem to the case $Y = \mathbb{P}^n$. Then, take the product of maps

$$\varphi \times \psi : X \rightarrow \mathbb{P}^n \times \mathbb{P}^n$$

Notice: The space $\mathbb{P}^n \times \mathbb{P}^n$ is a projective variety via segre map. And consider the following closed subset

$$\Delta = \{P \times P | P \in \mathbb{P}^n\}$$

which is closed in $\mathbb{P}^n \times \mathbb{P}^n$, specified by the equations (from the conditions of projective coordinate)

$$x_i y_j - x_j y_i = 0$$

So, $\Delta = \cap V(x_i y_j - x_j y_i) \cap (\mathbb{P}^n \times \mathbb{P}^n)$ is closed. And U is dense in X , Δ is closed, so $(\varphi \times \psi)(X) \subseteq \Delta$. Thus $\varphi = \psi$. (Apply the conclusion here 115.4.3.3) □

Definition 116.6.0.2 We loosely define, that an variety is **affine** if its isomorphic to an affine variety.

◇ **Lemma 116.6.0.3** Let $Y = V(f) \subseteq \mathbb{A}^n$ be a hypersurface.

Then, $\mathbb{A}^n - Y$ is isomorphic to the hypersurface in $\mathbb{A}^{n+1}$, given by $x_{n+1}f = 1$.

$$\mathbb{A}^n \supseteq V(f) \cong V(x_{n+1}f - 1) \subseteq \mathbb{A}^{n+1}$$

In particular, $\mathbb{A}^n - Y$ is affine, the coordinate ring is $k[x_1, \dots, x_n]_f$.

◦ **Proof** Define:

$$\begin{aligned} \varphi : H &\rightarrow \mathbb{A}^n - Y \\ (a_1, \dots, a_{n+1}) &\mapsto (a_1, \dots, a_n) \end{aligned}$$

which is correspondent to $A \rightarrow A_f$.

To show φ is an isomorphism, inverse $(a_1, \dots, a_n) \mapsto (a_1, \dots, a_n, \frac{1}{f(a_1, \dots, a_n)})$. $\square$

★ **Corollary 116.6.0.4** On any variety Y , there is a base for the topology consisting of open affine subsets.

◦ **Proof** Especially, we will finish the proof for projective/quasi-projective varieties. $\square$

116.6.1 Affine Rational Functions

Definition 116.6.1.1 Let $X \subseteq \mathbb{A}^n$ be an irreducible affine variety. $A(X)$ is an integral domain. We can form a quotient field

$$K(X) := \text{Quot}(A(X))$$

Elements $h \in K(X)$ can be written as $h = f/g$; $f, g \in A(X)$, which are regular.

Remark It is allowed, that at some point $p \in X$, we have the value at some points

$$\frac{\infty}{\infty} \quad \frac{0}{0}$$

So, the rational function is not a real function $X \rightarrow K$.

♠ **Proposition 116.6.1.2** Such $K(X)$ is a field.

◦ **Proof** $A(X)$ is integral domain. $\square$

♠ **Proposition 116.6.1.3** Suppose X be irreducible, and U be an open subset. Then, we have

$$K(U) = K(X)$$

◦ **Proof** Firstly, consider the open covering

$$U = \cup D(f_i)$$

Then, we could consider the pasting

$$\mathcal{O}_X(U) = \{(s_1, \dots, s_r) \in \prod A_{f_i} \mid s_i \mid D(f_i) \cap D(f_j) = s_j \mid D(f_i) \cap D(f_j)\}$$

or directly, define the affine ring of U be the subalgebra generated by $A_{f_1} \cup \dots \cup A_{f_r}$. Finally we have

$$\mathcal{O}_X(U) = A(U) = A[A_{f_1}, \dots, A_{f_r}]$$

via the embedding $A_{f_i} \hookrightarrow K(X)$

X is irreducible. So, there there exists a unique embedding

$$\mathcal{O}_X(U) \hookrightarrow K(X)$$

□

♠ **Proposition 116.6.1.4** Suppose X be an irreducible subset.
The definition via quotient field and the definition via regular function on open subset are equivalent.

◦ **Proof**

Take the following notation: $K(X)$ for the equivalence class via intersection, and $Quot(A(X))$ for the formal function field.

- $\subseteq$ Given two local regular functions $(U, \frac{f_U}{g_U}); (V, \frac{f_V}{g_V})$ can be seen as the element in $K(U), K(V)$. Then, $\frac{f_U}{g_U}|_{U \cap V} = \frac{f_V}{g_V}|_{U \cap V}$, correspondent to the element in $K(U \cap V) = K(X)$.
- $\supseteq$ With the lemma above.

□

116.6.2 Projective Rational Functions

Definition 116.6.2.1 Two formal definitions are given:

- Via affine opens: By $U = X \cap \mathbb{A}^n$ of any affine open;
- Via formal quotient of homogenous polynomials:

$$h(Z) = \frac{f(Z)}{g(Z)}; \deg(f) = \deg(g); f, g \in S(X) = K[X_1, \dots, X_n]^h / I(X)^h$$

♠ **Proposition 116.6.2.2** The two formal definitions are equivalent.

- **Proof** With the affinization map $\mathbb{A}^n \hookrightarrow \mathbb{P}^n$.

□

Remark Furthermore, we can develop the theory on the reducible variety. Any variety can be written as the union of

$$X = \cup_{i=1}^n X_i$$

Then, the quotient $K(X)$ is a ring. We have

$$\begin{aligned} I(X) &= \cap_i I(X_i) \\ A(X) &= K[X_1, \dots, X_n] / I(X), A(X_i) = K[X_1, \dots, X_n] / I(X_i) \\ K(X) &\hookrightarrow \prod K(X_i) \end{aligned}$$

116.6.3 Rational Maps

Rational Map in $\mathbb{A}^n$

Definition 116.6.3.1

- Let X be an irreducible variety. A **rational map** φ from X to $\mathbb{A}^n$ given by an n -tuple of rational functions. i.e.

$$\varphi(x) = (h_1(x), \dots, h_n(x)); h_i(x) \in K(X)$$

- Likewise, define a **rational map** φ from X to $\mathbb{P}^n$

$$\varphi(x) = [h_0(x), \dots, h_n(x)]$$

Remark: Another form of rational map in projective space.

Rational Maps

Definition 116.6.3.2 Let X be an irreducible variety and Y any variety. A **rational map**

$$\varphi : X \rightarrow Y$$

is defined to be an equivalence class of pairs (U, γ) with:

- $U \subseteq X$ a dense Zariski open subset; and
- $\gamma : U \rightarrow Y$ a regular map

Two pairs are equivalent if $\gamma|_{U \cap V} = \eta|_{U \cap V}$. Take the notation of the collection

$$\text{Rat}(X, Y)$$

With the discussion above, we have

$$\begin{aligned} \text{Rat}(X, Y) &\cong \text{Rat}(U, Y); U \text{ open} \\ \text{Rat}(X, W) &\hookrightarrow \text{Rat}(X, Y); W \subseteq Y \text{ open} \end{aligned}$$

And $\text{Rat}(X, W)$ is defined via $[(U, \gamma)], \gamma(U) \not\subseteq Y - W$.

Definition 116.6.3.3

- (Composition of Rational Maps) Consider the representative element $\varphi = [(U, f)] : X \rightarrow Y, \eta = [(V, g)] : Y \rightarrow Z$. Then, the composition

$$\eta \circ \varphi = [(f^{-1}V, g \circ f)]$$

- (Inclusion, Restriction) Inclusion $\eta : Z \rightarrow X$,

$$\eta = [(U, \iota)]$$

♣ **Example 116.6.3.4** The simplest example of a rational map that is not regular is the map

$$\begin{aligned} \varphi : \mathbb{A}^2 &\rightarrow \mathbb{P}^1 \\ (x, y) &\mapsto [x, y] \end{aligned}$$

Note that:

- This map is defined exactly on $\mathbb{A}^2 - \{(0, 0)\}$: Consider the intersection of

Completing, we may think of φ as a rational map from $\mathbb{P}^2$ to $\mathbb{P}^1$

♣ **Example 116.6.3.5** Generalization: For any plane $\wedge \cong \mathbb{P}^k$, the projection map

$$\pi_A : \mathbb{P}$$

116.6.4 Graphs

Definition 116.6.4.1 Let X be an irreducible projective variety and $\varphi : X \rightarrow \mathbb{P}^n$ be a rational map. Let $U \subseteq X$ be an open subset where φ is defined. The graph

$$\varphi|U$$

is as we have seen a closed subvariety $U \times \mathbb{P}^n$. The **graph** of the rational map φ will mean closure Γ_φ in $X \times \mathbb{P}^n$ of graph of $\varphi|U$.

$$\Gamma_\varphi = \overline{\varphi|U}$$

♠ **Proposition 116.6.4.2** Let X be irreducible.

The closure, denoted as Γ_φ , is independent of the choice of open subset U .

- **Proof** $\overline{\varphi|U} = \overline{\varphi|V}$. Notice that $\bar{U} = \bar{V}$. We just do the comparison for images. Notice that

$$\overline{\varphi(U)} = \overline{\varphi(V)}$$

Notice that $\varphi(U) \subseteq \varphi(X) \subseteq \overline{\varphi(V)}$, we have $\overline{\varphi(U)} = \overline{\varphi(V)}$. □

- Its closure is a closed subvariety of $X \times \mathbb{P}^n$;
- The projection $\pi_2(\Gamma_\varphi)$ of the graph Γ_φ to $\mathbb{P}^n$ is a gain a projective variety.

Definition 116.6.4.3 (Inverse Image) Let $Z \subseteq \mathbb{P}^n$. Then, the **inverse image** $\varphi^{-1}(Z)$ of Z in X to be the image

$$\varphi^{-1}(Z) := \pi_1(\pi_2^{-1}(Z))$$

where $\pi_1 : \Gamma_\varphi \rightarrow X, \pi_2 : \Gamma_\varphi \rightarrow \mathbb{P}^n$ are projections.

Let $Y \subseteq X$, we define the **image** or **total transform** of Y under φ to be the image

$$\varphi(Y) := \pi_2(\pi_2^{-1}(Y))$$

Remark Let $Z \subseteq Y$ be a subvariety such that the restriction of φ to Z is defined. The graph

$$\Gamma_\eta \subseteq Z \times \mathbb{P}^n \Rightarrow \Gamma_\eta \subseteq (\pi_1)^{-1}(Z) = \Gamma_\varphi \cap (Z \times \mathbb{P}^n) \subseteq X \times \mathbb{P}^n$$

But they may not be equal.

Rational Normal Curve

♠ Proposition 116.6.4.4 ...

◦ Proof ...

□

General Fiber

♠ Proposition 116.6.4.5 ...

◦ Proof ...

□

Domain of Regularity

♠ Proposition 116.6.4.6 Let $\varphi : X \dashrightarrow \mathbb{P}^n$ be a rational map. Γ_φ be a graph.

There is a unique maximal open subset $U \subseteq X$ such that the projection π_1 on the first factor induces an isomorphism

$$\Gamma_\varphi \supseteq \pi_1^{-1}(U) \Leftrightarrow U$$

There is a pair (U, f) in the equivalence class φ such that for all $(V, g) \in \varphi, V \subseteq U$.

◦ Proof ...

□

116.6.5 Birational Isomorphisms

Dominant

Consider the rational map $\varphi : X \rightarrow Y, \psi : Y \rightarrow Z$ and their composition. Then, we want to calculate the value, for $\varphi = [(U, f)], \psi = [(V, g)]$, we have

$$g(f(x)); x \in f^{-1}V \cap U$$

But then, a question is: If $f(U) \subseteq Y - V$, the composition is meaningless.

Definition 116.6.5.1 Let $\varphi : X \rightarrow Y$ be a rational map. Then, X is **dominant** if

$$\overline{f(X)} = Y$$

Now, $g \circ f$ is well-defined on $f^{-1}(V) \cap U$.

Remark This is not surjective.

Pullback

Definition 116.6.5.2 Let $\varphi : X \rightarrow Y$ be a rational map. In particular, if $f \in K(Y)$ is a rational function, $\varphi^* f$ is the **pullback**

$$\varphi^*(f) := f \circ \varphi$$

And induces the inclusion

$$K(Y) \hookrightarrow K(X)$$

Birationality

Definition 116.6.5.3 A rational map $\varphi : X \rightarrow Y$ is **birational** iff there exists a rational map $\psi : Y \rightarrow X$ such that

$$\gamma \circ \psi = id_Y \quad \psi \circ \gamma = id_X$$

► **Theorem** The following conditions are equivalent:

- $K(X) \cong K(Y)$;
- There exists nonempty open subsets $U \subseteq X$ and $V \subseteq Y$ isomorphic to one another.

◦ **Proof**

-
-

□

Definition 116.6.5.4 (Rational Variety) We say a variety is **rational** if:

- X is birational to $\mathbb{P}^n$;
- $K(X) \cong K(x_1, \dots, x_n)$;
- X possesses an open subset U isomorphic to an open subset of $\mathbb{A}^n$;

Elsewise irrational.

♣ **Example 116.6.5.5** (The Quadric Surface) Consider the quadric surface. It has a standard form of

$$Z_0 Z_3 - Z_1 Z_2 = 0$$

Take $\varphi : [Z_0, \dots, Z_3] \rightarrow [Z_0, Z_1, Z_2]$. Defined on points except $[0, 0, 0, 1]$.

Conversely

In general:

♠ **Proposition 116.6.5.6** In general: Let $Q \subseteq \mathbb{P}^n$ be a quadric hypersurface and $p \in Q$ be any point not lying on the vertex of Q . The map

$$\pi_p : Q \rightarrow \mathbb{P}^{n-1}$$

is a birational isomorphism.

◦ **Proof** Similarly,

□

♣ **Example 116.6.5.7** Consider the cubic curve $C \subseteq \mathbb{P}^2$ given by equation

$$F = ZY^2 - X^3 + X^2Z, C = V(F)$$

$\pi_p : C - \{[0, 0, 1]\} \rightarrow \mathbb{P}^1, [X, Y, Z] \mapsto [X, Y]$. It is the intersection of the line which goes through $[0, 0, 1]$ and $[X, Y, Z]$, and the plane $Z = 0$.

Inverse: $\mathbb{P}^1 \rightarrow C, [X, Y] \mapsto [X(Y^2 + X^2), Y(Y^2 + X^2), X^3]$.

$C - \{[0, 0, 1]\}$ is dense open: $\{[0, 0, 1]\} = V(X = 0, Y = 0)$ is closed.

♣ **Example 116.6.5.8** For m, n , the product $\mathbb{P}^m \times \mathbb{P}^n$ is birational with $\mathbb{P}^{m+n}$.

We could take affine opens $\mathbb{A}^{m+n} \cong \mathbb{A}^m \times \mathbb{A}^n$. With the irreducibility, $\mathbb{P}^{m+n}$ is birational to $\mathbb{P}^m \times \mathbb{P}^n$.

Or: Another way to construct the map is

$$F : \mathbb{P}^{m+n} - \{[Z_0 : \dots : Z_{m+n}] | Z_0 = 0\} \rightarrow \mathbb{P}^m \times \mathbb{P}^n \quad [Z_0 : \dots : Z_{m+n}] \mapsto ([Z_0, \dots, Z_m], [Z_0 : Z_{m+1} : \dots, Z_n])$$

$$F^{-1} : \mathbb{P}^m \times \mathbb{P}^n - \{([X_0, \dots, X_m], [Y_0, \dots, Y_n]) | X_0 = 0, Y_0 = 0\} \rightarrow \mathbb{P}^{m+n} \quad ([X_0, \dots, X_m], [Y_0, \dots, Y_n]) \mapsto [X_0 Y_0 : \dots : Y_m Y_0 : \dots]$$

Example: The Brationality from X of dimension r to a hypersurface Y of $\mathbb{P}^{r+1}$ (Hartshorne)

► **Theorem** (Theorem of Primitive Element) Let L be a finite separable extension field of a field K . Then there is an element $\alpha \in L$, which generates L as an extension field of K .

Furthermore, let $\beta_1, \dots, \beta_n$ be the set of generators of L over K , and if K is infinite. Then, α can be taken to be a linear combination

...

◦ **Proof** ...

□

Definition 116.6.5.9 ...

► **Theorem** ...

◦ **Proof** ...

□

► **Theorem** ...

◦ **Proof** ...

□

We have the following conclusion:

♠ **Proposition 116.6.5.10** Any variety X of dimension r is birational to a hypersurface Y of $\mathbb{P}^{r+1}$.

◦ **Proof** ...

□

116.6.6 Degree of Rational Map

♠ **Proposition 116.6.6.1** The general fiber of map $f : X \rightarrow Y$ is finite iff the inclusion f^* expresses the field $K(Y)$ as a finite extension of field $K(X)$.

In this case, $\text{char}(K) = 0$, and

$$\text{card}(f^{-1}\{x\}) = [K(Y) : K(X)]$$

◦ **Proof ...**

□

116.6.7 Blowing Up

Definition 116.6.7.1 In general: Let $X \subseteq \mathbb{A}^m$ be any affine variety and $Y \subseteq X$ any subvariety. Choose a set of generators

$$f_0, \dots, f_n \in A(X)$$

for $I(Y) \subseteq A(X)$. And the rational map

$$\varphi : X \dashrightarrow \mathbb{P}^n \quad x \mapsto [f_0(x), \dots, f_n(x)]$$

Then, φ is regular on $X - Y$, but not generally on Y . (Image $[0 : \dots, : 0]$ not well-defined)

The graph Γ_φ together with projection $\pi : \Gamma_\varphi \rightarrow X$ called the **blow-up** of X along Y , and denoted by

$$Bl_Y(X) = \tilde{X} = (\Gamma_\varphi, \pi)$$

Inverse image $E = \pi^{-1}(Y) \subseteq \tilde{X}$ is called **exceptional divisor**.

♠ Proposition 116.6.7.2

- Construction $Bl_Y(X)$ is local. i.e. for X any variety, Y a subvariety and $U \subseteq X$ an affine open subset. The inverse image of U in $Bl_Y(X)$ is

$$Bl_{Y \cap U}(U)$$

- The construction does not depend on the choice of generators of the ideal of Y in X .

◦ **Proof ...**

□

♠ Proposition 116.6.7.3

- Naturality property of blow-ups. So $BL_N(M)$ makes sense;
- If complex manifold M is a subvariety of $\mathbb{C}^n = \mathbb{A}_{\mathbb{C}}^n$, the two notions of $BL_N(M)$ coincide.

◦ **Proof ...**

□

► **Theorem** Let X be any variety and

-
-

◦ **Proof** ...

□

A well-know result by Hironaka: [He proved that it was possible to resolve singularities of varieties over fields of characteristic 0 by repeatedly blowing up along non-singular subvarieties, using a very complicated argument by induction on the dimension.]

More information can be read here: https://en.wikipedia.org/wiki/Resolution_of_singularities#

♣ **Example 116.6.7.4** Let $r \in X$ be any point. Then,

$$\pi_r : X \dashrightarrow \mathbb{P}^3$$

the projection from r . And $Q \subseteq \mathbb{P}^3$ the image of π_r .

116.6.8 Unrationality

Definition 116.6.8.1 An irreducible variety X is **unirational** if there is a dominant rational map

$$\varphi : \mathbb{P}^n \rightarrow X$$

Equivalently in the language of field, if the function field $K(X)$ can be embedded in a purely transcendental extension

$$K(z_1, \dots, z_n)$$

116.6.9 Appendix: Lüroth Problem

Question: (Lüroth) Is every unirational curve is rational?

Dimension 1

► **Theorem** (Lüroth, 1876) Suppose $k \subseteq L \subseteq k(t)$ is a field extension. k is algebraically closed (mostly $k = \mathbb{C}$). If L is a intermediate field which containing F . Then, $L = F(u)$ for some $u =$ transcendental over F .

◦ **Proof** ...

□

★ **Corollary 116.6.9.1** Every dominant irreducible algebraic curve is birational to $\mathbb{P}^1$.

◦ **Proof** As above.

□

Dimension 2(Castelnuovo, Enriques)

Higher Dimension

116.7 Blow-Ups, Nash Blow-Ups and the Resolution of Singularities

† **Problem 116.7.1** *Given any arbitrary variety X , does there exist a smooth variety Y birational to it?*

► **Theorem** (Hironaka) Let X be any variety(of zero characteristic). Then, there exists a smooth variety Y and a regular birational map

$$\pi : Y \rightarrow X$$

Definition 116.7.0.1 Such a map $\pi : X \rightarrow Y$ is called a **resolution of singularities** of X .

Chapter 117 Parameterizations

117.1 Parameter Spaces

Definition 117.1.0.1 Let B be a variety. Then, given a collection of projective varieties $\{V_b\}_{b \in B}$, whose elements is the projective space with **base** B ,

$$\mathcal{V} = B \times V_b \subseteq \mathbb{P}^n \quad V_b := \pi_1^{-1}(b)$$

The space $\mathcal{V}$ is called the **total space**. V_b is member. The pair $(B, \mathcal{V})$ is parametrized by B .

Suppose B be projective variety. Then it can be described as the zero locus of homogeneous polynomials $F_\alpha(Z, W)$;

Suppose B be affine variety/ Then it can be described as the zero locus of polynomials $F_\alpha(z, W)$.

◇ **Lemma 117.1.0.2**

- Let $X \subseteq \mathbb{P}^n$ be any projective variety and $\{V_b\}$ be any family of projective varieties in $\mathbb{P}^n$ with base B . Then, the set

$$\{b \in B \mid X \cap V_b \neq \emptyset\}$$

is closed in B .

- Let $\{W_b\}$ be another family, then

$$\{b \in B \mid V_b \cap W_b \neq \emptyset\}$$

is a closed subvariety of B

◦ **Proof**

•

□

◇ **Lemma 117.1.0.3** ...

◦ **Proof** ...

□

♣ **Example 117.1.0.4** (The Universal Hyperplane) If we think of the projective space $\mathbb{P}^{n*}$ as a collection of hyperplanes $H \subseteq \mathbb{P}^n$: (Here: 115.1.2.10)

Then, define a subset of product $\mathbb{P}^{n*} \times \mathbb{P}^n$ simply as

$$\Gamma = \{(H, p) \mid p \in H\}$$

This is a subvariety of $\mathbb{P}^{n*} \times \mathbb{P}^n$

In particular, it may be realized as a hyperplane section of the Segre variety

$$\mathbb{P}^{n*} \times \mathbb{P}^n \hookrightarrow \mathbb{P}^{n^2+2n}$$

♣ **Example 117.1.0.5** (The Universal Hyperplane Section)

$$\Omega_X = \{(H, p) | p \in H \cap X\} = \pi_2^{-1}(X)$$

♣ **Example 117.1.0.6** (Parameter Spaces of Hypersurfaces)

♣ **Example 117.1.0.7** (Universal Families of Hypersurfaces)

♣ **Example 117.1.0.8** (A Family of Lines)

117.2 Parameter Spaces and Moduli Spaces

117.3 Grassmannians and Related Varieties

Definition 117.3.0.1

- Let $G(k, n)$ denote the set of k -dimensional linear subspaces of vector space K^n ,

$$G(k, n) := \{V \subseteq K^n | V \in Vect_k, \dim V = k\}$$

Especially, for an abstract vector space, take the notation of $G(k, V)$.

- Of course: A k -dimensional linear subspace of a vector space K^n is the same thing as a $(k-1)$ plane of $\mathbf{P}^{n-1}$.

In most contexts, Grassmannians are defined initially via coordinate patches or as a quotient of groups:

- (Local Chart)
- Quotient: We could describe the Grassmanian as a homogeneous space.

If W is a k -dimensional subspace of V . We can associate the subspace W to a vector

$$\lambda = v_1 \wedge \cdots \wedge v_k, W = \text{span}(v_1, \dots, v_k)$$

Then: λ is determined up to scalars by W . If we choose a different basis, then

$$\lambda' = \det(A)\lambda$$

So: The following map is well-defined.

Definition 117.3.0.2 In algebraic geometry, the Grassmannian is specified by the **Plücker embedding**

$$\Psi : G(k, V) \rightarrow \mathbb{P}(\wedge^k V)$$

Inverse Process Given any $[\omega] = \psi(W)$, we can recover the corresponding subspace W as the space of vector spaces

$$W := \{v \in V | v \wedge \omega = 0 \in \wedge^{k+1} V\}$$

Definition 117.3.0.3 The homogeneous coordinates on $\mathbb{P}^N = \mathbb{P}(\wedge^k V)$ are called **Plücker coordinate** on $G(k, V)$.

Special Case: $k = 2$

117.3.1 Subvarieties of Grassmannians

Grassmannians $G(1, 3)$

Fano Varieties

A fundamental type of subvariety of the Grassmannian $G(k, n)$ is the **Fano variety**

Definition 117.3.1.1 Let $X \subseteq \mathbb{P}^n$ be a projective variety. Then, the **Fano variety** is the collection

$$F_k(X) := \{\Lambda \subseteq X \mid \Lambda \text{ is the variety of } k\text{-plane}\} \hookrightarrow G(k, n)$$

Firstly, we could observe the fano variety of a hypersurface $X = G(Z)$. Then,

$$F_k(X) = \{\}$$

117.3.2 Tangent Spaces to Grassmannians

117.4 Gauss Map, Tangential and Dual Varieties

Chapter 118 Rational Maps

Chapter 119 Plane Curves

119.1 Plane Curves

119.1.1 Intersection Multiplicities

Definition 119.1.1.1 The intersection multiplicity of C and C' at a point $P \in \mathbb{P}_k^2$ is defined as:

$$I_P(C, C') := \dim_k \mathcal{O}_{\mathbb{P}_k^2, P} / (f, g)$$

◇ Lemma 119.1.1.2 ...

◦ Proof ...

□

119.2 Local Properties

119.2.1 Multiplicities

119.2.2 Intersection Numbers

119.3 Projective Plane Curves

119.4 Divisors

Definition 119.4.0.1 Let C be a smooth projective curve over algebraically closed.(i.e. A smooth irreducible projective variety of dimension 1)

A **divisor** D on C is a finite formal sum given by

$$D = \sum_{i=1}^n n_i P_i; n_i \in \mathbb{Z}, P_i \in C$$

The **divisor group**

$$\text{Div}(C) := \{ \sum n_i P_i | n_i \in \mathbb{Z}, P_i \in C \} = \mathbb{Z}(C)$$

The **degree of divisor**

$$\deg(D) := \sum n_i$$

◇ Lemma 119.4.0.2 ...

◦ Proof ...

□

119.5 Bezout's Theorem

119.5.1 Degree

Definition 119.5.1.1 ...

119.5.2 Join of Varieties

Definition 119.5.2.1 Define:

$$J(X, Y) := \cup_{x \in X, y \in Y} \bar{x}y$$

◇ **Lemma 119.5.2.2** For varieties X, Y , we have

$$\deg J(X, Y) = \deg(X) \cdot \deg(Y)$$

◦ **Proof** ...

□

119.5.3 Multiplicities

Consider two varieties X, Y intersects at point $p \in X \cap Y$ and smooth. Then, we have

$$T_p X + T_p Y = T_p P^n$$

Definition 119.5.3.1

-
-
-
-

- $m_Z(X, Y) \geq 1$

-
-

Completions Operation

119.5.4 Bezout's Theorem

► **Theorem** (Bezout)

◦ **Proof** ...

□

♣ **Example 119.5.4.1** (Rational Curves)

The Bezout's theorem gives a tool for the research of Elliptic curve group structure:

Chapter 120 Normalization

Chapter 121 Quadrics

Suppose we work on algebraically closed field. Then, the quadric on $\mathbb{P}^n$ can be in the form of

$$F(X) = \sum \alpha_i X_i^2$$

Under the base transformation, we have $\text{rank}(F) = \text{card}(\text{supp}\{\alpha_i\})$. The rank

$$Q = \text{Im}(F), \text{rank}(Q) =$$

121.1 Tangent Spaces to Quadrics

♠ **Proposition 121.1.0.1** If $H \subseteq \mathbb{P}V$ is a hyperplane, and $Q' = Q \cap H$. Then,

$$\text{rank}(Q) - 2 \leq \text{rank}(Q') \leq \text{rank}(Q)$$

$$\text{rank}(Q) - 2k \leq \text{rank}(Q') \leq \text{rank}(Q)$$

◦ **Proof ...**

□

121.2 Quadrics on Plane and in Space

121.2.1 Quadrics on Plane

We have: $\text{rank}(Q) = 1, 2, 3$:

- $\text{rank}(Q) = 3$: It has a standard form $Q = V(X_1^2 + X_2^2 + X_3^2)$
Via veronese embedding, we have the isomorphism

$$v_2 : (\mathbb{P}^1) \times \mathbb{P}^1 \rightarrow Q \subseteq \mathbb{P}^2 \quad [X_0, X_1] \mapsto [X_0^2, X_0X_1, X_1^2] = [Z_0, Z_1, Z_2]$$

- $\text{rank}(Q) = 2$: $Q = V(X_1^2 + X_2^2) = V(Z_1Z_2)$, where $Z_1 = X_1 + iX_2, Z_2 = X_1 - iX_2$.
- $\text{rank}(Q) = 1$: $Q = V(X_1^2)$. It is a straight line with multiplicity 2;

121.2.2 Quadrics in Space

We have: $\text{rank}(Q) = 1, 2, 3, 4$:

- $\text{rank}(Q) = 4$: It is smooth. And $V(\sum X_i^2) = V((X_1 - iX_2)(X_1 + iX_2) - (X_3 + X_4)(X_3 - iX_4)) = V(Z_1Z_3 - Z_2Z_4)$.
- $\text{rank}(Q) = 3$: Quadric Cone.
- $\text{rank}(Q) = 2$: Union of two Planes
- $\text{rank}(Q) = 1$:

We have the following conclusion of birationality:

♠ **Proposition 121.2.2.1** (H& F,P84,Exercise 7.22) Quadrics in $\mathbb{P}^3$ is birational to $\mathbb{P}^2$.

◦ **Proof ...**

□

121.3 Linear Spaces on Quadrics

121.4 Families of Quadrics

121.5 Pencils of Quadrics

Chapter 122 Cubics

122.1 Plane Cubic Curve

122.1.1 Classification of Smooth Plane Cubics

122.2 Cubic Surfaces

Polar

Definition 122.2.0.1 For homogeneous cubic polynomial f in $x_0, \dots, x_3$, the polar of f is defined by

...

Then: The tangent space

...

Resultant

Definition 122.2.0.2

$$\begin{cases} r(u, v) = a_0u^2 + a_1uv + a_2v^2 \\ s(u, v) = b_0u^3 + b_1u^2v + b_1uv^2 + b_3v^3 \end{cases}$$

Then, $R()$

◇ **Lemma 122.2.0.3**

◦ **Proof ...** □

► *Theorem ...*

◦ **Proof ...** □

The Configuration of the 27 Lines

122.2.1 Rationality of Cubics

Chapter 123 Algebraic Groups

Remark We will discuss some of algebraic group in the context of classical algebraic geometry. More contents will be discussed in an independent part.

Definition 123.0.0.1

- (Algebraic Group) An **algebraic group** is defined to be simply a variety X with regular maps (X, m, i) , multiplication

$$m : X \times X \rightarrow X$$

and inverse

$$i : X \rightarrow X$$

satisfying the usual rules for multiplication and inverse in a group. e.g. There exists a point $e \in X$

$$\begin{aligned} m(e, p) &= m(p, e) = p; \\ m(p, i(p)) &= e \end{aligned}$$

- (Algebraic Group Homomorphism) A **morphism of algebraic groups** is a map $\varphi : G \rightarrow H$ such that: It is a
 - Group homomorphism;
 - Regular Map;

123.1 Matrix Algebraic Groups

123.1.1 The General Linear Group, $GL_n K$

Definition 123.1.1.1 Define the open subset of $\mathbb{A}^{n^2}$:

$$GL_n K := \{x \in \mathbb{A}^{n^2} | \det(x) \neq 0\}$$

With:

- Multiplication: $GL_n K \times GL_n K \rightarrow GL_n K$, Matrix Multiplication;
- Inverse: $GL_n K \rightarrow GL_n K$, Matrix Inverse;

123.1.2 Subgroup $SL_n K$

Definition 123.1.2.1 ...

123.2 Group Action

Definition 123.2.0.1

- By an action of an algebraic group G on an algebraic variety, we mean a regular map

...

- A projective action on a projective variety $X \subseteq \mathbb{P}^n$ mean the action of G on $\mathbb{P}^n$ such that

$$\varphi(g, X) = X, \forall g \in G$$

Stronger condition:

123.2.1 $PGL_{n+1}K$ on $\mathbb{P}^n$

123.3 Quotient

Chapter 124 Intersections and Degree Research

Chapter 125 Varieties of Arbitrary Dimension

125.1 Dimension of Arbitrary Varieties

Chapter 126 Hilbert Functions

This is a short note of Hilbert function, with the intuition from the background of algebraic geometry.

Problem Given a hypersurface $X \subseteq \mathbb{P}^n$ be an intersection of hypersurfaces. One of the basic problem, is to describe the hypersurfaces $X \subseteq \Gamma$.

126.1 Hilbert Function, Hilbert Polynomials

Definition 126.1.0.1 Define the function

$$h_X : \mathbb{N} \rightarrow \mathbb{N} \quad m \mapsto \dim(S(X)_m)$$

where $S(X) = K[Z_0, \dots, Z_n]/I(X)$ is the homogeneous coordinate ring and m denote the m graded piece.

126.2 Syzygies

Chapter 127 Resolutions

Chapter 128 Segre Varieties

Chapter 129 Appendix: From Classical Algebraic Geometry to Algebraic Geometry

Abstract The remaining chapters are mainly dependent on the book

Classical Algebraic Geometry: a modern view/Topics in Classical Algebraic Geometry

By Igor V. Dolgachev.

Remark 1 We will take $\mathbb{C}$ as the base field.

Chapter 130 Polarity

130.1 The Polar Paring

Take the following notations:

- E : A finite dimesnional vecot space; $E^\vee$ is its dual;
- $S^k E$: Symmetric k -th power;

Definition 130.1.0.1 Define the **bilinear paring**

$$S^k E \otimes S^d E^\vee \rightarrow S^{d-k} E^\vee; d \geq k$$

In coordinate, for the basis $E = \text{span} \langle \xi_0, \dots, \xi_{n-1} \rangle$, $E^\vee = \text{span} \langle t_0, \dots, t_{n-1} \rangle$, then the elements in

- $S^d E^\vee$ can be seen as a homogeneous polynomial f of degree d , in variables $t_0, \dots, t_{n-1}$.
- $S^k E$ can be seen as a homogeneous polynomial ψ of degree k in variables $\xi_1, \dots, \xi_{n-1}$

We have $\langle \xi_i, t_j \rangle = \delta_{ij}$. We can view each ξ_i as the operator

$$\xi_i = \frac{\partial}{\partial t_i}$$

Then, $D_\psi = \psi(\partial_0, \dots, \partial_{n-1})$, where ψ is the corresponding polynomial of $S^k E$. Then, the paring becomes:

$$\langle \psi(\xi_0, \dots, \xi_{n-1}), f(t_0, \dots, t_{n-1}) \rangle = D_\psi(f)$$

For monomials $\partial^i = \partial_0^{i_0} \dots \partial_{n-1}^{i_{n-1}}$, monomials $t^j = \partial_0^{j_0} \dots \partial_{n-1}^{j_{n-1}}$ we have

$$\partial^i(t^j) = \begin{cases} \frac{j!}{(j-i)!} t^{j-i} & j \geq i \\ 0 & \text{Otherwise} \end{cases}$$

Here, take the following notation

$$i! = \prod i_k! \quad i = (i_0, \dots, i_{n-1}) \geq 0 \Leftrightarrow i_k \geq 0 \quad |i| = \sum i_k$$

Consider the following cases:

$$\begin{aligned} \psi &= (a_0 \partial_0 + \dots + a_{n-1} \partial_{n-1})^k = \sum_{|i|=k} \frac{k!}{i!} a^i \partial^i \\ \Rightarrow D_\psi(f) &= \langle \sum_{|i|=k} \frac{k!}{i!} a^i \partial^i, f \rangle = \sum_{|i|=k} \frac{k!}{i!} a^i \partial^i(f) \end{aligned}$$

We have the following canonical isomorphisms

$$S^k E^\vee \cong (S^k E)^\vee \quad S^k E = (S^k E^\vee)^\vee$$

130.1.1 Hypersurface

Take the following notations:

- $|E| = \mathbb{P}_{sub}(E) := \mathbb{P}E$;
- If $E \cong \mathbb{C}^{n+1}$, then $|E| \cong (\mathbb{C}^{n+1} - \{0\})/\mathbb{C}^* = \mathbb{P}^n\mathbb{C}$;
- There exists a canonical projection $E \rightarrow \mathbb{P}_{sub}(E), v \mapsto [v]$;

Structure Sheaf on $|E|$

Definition 130.1.1.1

- Take the notation $\mathcal{O}_{|E|}(1)$: With global section

$$[\mathcal{O}_{|E|}(1)](|E|) = E^\vee$$

The sheaf is called **tautological invertible sheaf**.

- Take the notation $\mathcal{O}_{|E|}(d)$: With global section

$$[\mathcal{O}_{|E|}(d)](|E|) = S^d E^\vee$$

- For any $f \in S^d E^\vee$ take the notation

$$V(f) = \{x \in |E| \mid f(x) = 0\}$$

Such collection is also called **hypersurface of degree d** in $|E|$.

Especially, a hypersurface of degree 1 is a hyperplane.

Definition 130.1.1.2 Let $X = V(f)$ be a hypersurface of degree d in $|E|$. Let $a = [v] \in |E|$ for some $v \in E$, the hypersurface

$$P_{a^k}(X) = V(D_{v^k}(f))$$

♣ **Example 130.1.1.3** ...

Chapter 131 Conics and Quadric Surfaces

Part LVI

Curve and Surfaces-A Classical View

Part LVII

Rational Points on Elliptic Curve

Part LVIII

Algebraic Group

Part LIX

Schemes

Part LX

Intersection Theory

Part LXI

Algebraic Curves

The first part is based on the algebraic curve in complex field.

Chapter 132 Fundamental Concepts

132.1 The Plane Curve in Projective Plane $\mathbb{P}^2\mathbb{C}$

The knowledge is collected here: 118

132.2 Riemann Surface

The knowledge is collected here: XXI

132.3 Holomorphic and Meromorphic Functions

132.4 Holomorphic and Meromorphic Differentials

132.5 Differential Forms

132.6 Poincare-Hopf Theorem

132.7 Complex Manifolds

132.8 Algebraic Varieties

132.9 Smooth Points, Tangent Spaces and Implicit Function Theorem

132.10 Holomorphic Mapping from Compact Riemann Surface into Complex Projective Space

Chapter 133 The Normalization Theorem

133.1 Singularities of Plane Algebraic Curves

Given here:

133.2 Connectedness

133.3 Normalization

Chapter 134 The Riemann-Roch Theorem

Part LXII

Algebraic Surfaces

Complex

Part LXIII

Abelian Varieties

Chapter 135 Analytic Theory

135.1 Complex Tori

Part LXIV

Moduli Space

Part LXV

Hodge Theory

Part LXVI

Resolution of Singularities

Chapter 136 Resolution for Curves

Chapter 137 Resolution for Surfaces

Chapter 138 Strong Resolution in Characteristic Zero

Part LXVII

Birational Algebraic Geometry

Part LXVIII

Theoretic Categorical Theory

Part LXIX

Etale Cohomology

Part LXX

**PRISMS AND PRISMATIC
COHOMOLOGY**